



MX2

Proof 2024

Made with LuaT_EX and **copious** amounts of *Tea Shop Express*. (#NotSponsored ☺)

Contents

1	Language of Proof	3
2	Direct Proof	9
3	Contradiction & Contrapositive	11
4	Algebraic Inequalities	17
5	Mathematical Induction	27
6	Calculus Proofs & The Squeeze Theorem	38
7	Solutions	46
7.1	Language of Proof	46
7.2	Direct Proof	50
7.3	Contradiction & Contrapositive	54
7.4	Algebraic Inequalities	66
7.5	Mathematical Induction	91
7.6	Calculus Proofs & The Squeeze Theorem	133
	Appendix	151



Preface

Abbreviations:

- “HY” – half yearly exam
- “AT” – assessment task
- “God folder” – don’t ask.

Note:

- This document is for **non commercial usage only** and was made to benefit students.
- Sources for questions have been attributed in the *right margin*, except for the multiple choice, where they are listed in the solutions (for formatting reasons). ¹*These sources are merely where I have found these questions. I do not know if they are truly original or come from elsewhere.*
- To quickly access solutions, click the question **number**, e.g for Question **37**, you would click **37**. Doing this in solutions will take you back to the original question. There are 530 marks in total. For further explanation on solutions, consider asking your teacher, tutor or online forums such as this one or nswmaths.com.
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- Questions² are only sorted by type (mcq vs normal) and topic, not by any other metric (e.g difficulty, chronological; sorry if you get jumpscared by a Q16c better organisation will come later or never)
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“*Individuals*” – who have helped me learn $\text{\LaTeX}$.

“*Individuals*” – who have provided me with more papers/resources to add to this document.

Enjoy your studies!



Language of Proof

- 1 Consider the following statement for positive integers:

“If 3 divides $a - b$ then 3 divides $a^3 - b^3$ ”

Which of the following is always true?

- A. The statement is correct and its converse is incorrect.
- B. The statement is incorrect and its converse is correct.
- C. The statement is incorrect and its converse is incorrect.
- D. The statement is correct and its converse is correct.

- 2 Consider the statement:

“Every day next week, Max will do at least one maths problem”

If this statement is **not** true, which of the following statements is certainly true?

- A. Some day next week, Max will do no maths problems
- B. Some day next week, Max will do more than one maths problems
- C. On no day next week will Max do more than one maths problem
- D. Every day next week, Max will do no maths problems

- 3 Which of the following is true?

- A. $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : b = a^3$
- B. $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : b = \sqrt[3]{a}$
- C. $\forall a \in \mathbb{Z}^+ \exists b \in \mathbb{Z}^+ : a = b^3$
- D. $\forall a \exists b \in \mathbb{Z}^+ : b = a^3$

- 4 Consider the following statement:

“If you have no treasure, I have no kingdom.”

Which of the following is logically equivalent to this statement?

- A. “If I have no kingdom then you have no treasure”
- B. “If you have treasure then I have a kingdom”
- C. “If you have no kingdom then I have no treasure”
- D. “If I have a kingdom then you have treasure”



- 5 A sufficient condition for $\triangle ABC$ to be right angled is that $a^2 + b^2 = c^2$. Which of the following is an equivalent statement?
- A. If $\triangle ABC$ is right angled, then $a^2 + b^2 = c^2$.
 - B. If $a^2 = b^2 = c^2$, then $\triangle ABC$ is right-angled.
 - C. If $a^2 + b^2 \neq c^2$, then $\triangle ABC$ is not right angled.
 - D. $\triangle ABC$ is right-angled if and only if $a^2 + b^2 = c^2$
- 6 Suppose John found a raven, and it was black. Which of the following **must** be false?
- A. "There exist non-black ravens as well as black ravens."
 - B. "There exist non-black ravens."
 - C. "All ravens are black."
 - D. None of the above.
- 7 Let A and B be true statements such that $A \Rightarrow B$. Which of the following statements is necessarily false?
- A. $\sim B \Rightarrow \sim A$
 - B. A and $\sim B$
 - C. A or $\sim B$
 - D. None of the above.
- 8 Which of the following is **NOT** equivalent to 'If n is even, then n^2 is a multiple of 4'?
- A. n is even $\therefore n^2$ is a multiple of 4;
 - B. n^2 is a multiple of 4 if n is even;
 - C. n^2 is a multiple of 4 $\therefore n$ is even;
 - D. n is even implies that n^2 is a multiple of 4.
- 9 The negation of the statement $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y = 12$ is:
- A. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$;
 - B. $\exists x \in \mathbb{R}$ such that $\forall y \in \mathbb{R} 2x + 3y \neq 12$
 - C. $\forall x \in \mathbb{R}, \forall y \in \mathbb{R} 2x + 3y \neq 12$
 - D. $\exists x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $2x + 3y \neq 12$



10 Which of the following statements is FALSE?

- A. $\forall x \in \mathbb{R}, x^2 \geq 0$
- B. $\forall n \in \mathbb{Z}, \exists m \in \mathbb{Z}, m = n + 5$
- C. $\exists a \in \mathbb{R}, \forall x \in \mathbb{R}, ax = x$
- D. $\exists m \in \mathbb{Z}, \forall n \in \mathbb{Z}, m = n + 5$

11 Which of the following statements is true?

- A. $\forall x > 0, \exists y < 0 : y > \frac{1}{x}$
- B. $\forall x > 0, \exists y < 0 : xy > 1$
- C. $\forall x < 0, \exists y > 0 : y < \frac{1}{x}$
- D. $\forall x < 0, \exists y > 0 : xy < 1$

12 Consider the following claim:

“If p is not prime then $(p - 1)! + 1$ is not divisible by p ”.

Which of the following is a counterexample to this claim?

- A. A prime number p such that $(p - 1)! + 1$ is divisible by p .
- B. A prime number p such that $(p - 1)! + 1$ is not divisible by p .
- C. A composite number p such that $(p - 1)! + 1$ is divisible by p .
- D. A composite number p such that $(p - 1)! + 1$ is not divisible by p .

13 Which of the following is the negation of $p \Rightarrow (q \wedge r)$

- A. $\sim p \Rightarrow (q \wedge r)$
- B. $\sim p \wedge (q \wedge r)$
- C. $(\sim q \vee \sim r) \Rightarrow \sim p$
- D. $p \wedge (\sim q \vee \sim r)$

14 Which of the following statements is FALSE?

- A. $\exists x \in \mathbb{Z}, \forall y \in \mathbb{R}$ such that $\sqrt{x} < e^y$
- B. $\exists x \in \mathbb{R}^+, \forall y \in \mathbb{R}$ such that $|y| < x$
- C. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $\cos x = \cos(-y)$
- D. $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}^+$ such that $x^4 = 2y^3$



- 15 Suppose that a, b, c and d are integers such that $a + b$ divides $ac - bc$. Which of the following must be divisible by $a + b$?
- A. $ab + cd$
 - B. $ab - cd$
 - C. $bc - ad$
 - D. $bc + ad$
- 16 Consider the following statement for real numbers a and b :
- “If $a < b$ then $a^3 < b^3$.”
- A. If $a < b$ then $a^3 \geq b^3$.
 - B. If $a \geq b$ then $a \geq b^3$.
 - C. If $a^3 \geq b^3$ then $a \geq b$.
 - D. If $a^3 < b^3$ then $a < b$.
- 17 Which of the following is not equivalent to $P \Rightarrow Q$?
- A. P is sufficient for Q
 - B. Q is necessary for P
 - C. If Q is false, then P is false
 - D. None of the above
- 18 A single die is rolled and the uppermost face noted. Let p represent the statement “the uppermost face is divisible by 3” and let q represent the statement “the uppermost face is divisible by 6”. Considering the implication $p \Rightarrow q$, which of the following is correct?
- A. The negation is true and the converse is false.
 - B. The negation is true and the converse is false.
 - C. The negation is false and the converse is true.
 - D. The negation is false and the converse is false.
- 19 Given that x and y are real numbers, which of the following is a true statement?
- A. $\forall y, \exists x$ such that $x^2 - y^2 = x$
 - B. $\forall y, \exists x$ such that $x^2 - y^2 = y$
 - C. $\forall y, \exists x$ such that $x^2 + y^2 = x$
 - D. $\forall y, \exists x$ such that $x^2 + y^2 = y$



20

Consider the following proof: For $a, b, c \in \mathbb{Z}^+$, $\exists p, q, r \in \mathbb{Z}^+$ such that $a = 3p + 1, b = 3q + 1, c = 3r + 1$.

$$\begin{aligned} abc &= (3p + 1)(3q + 1)(3r + 1) \\ &= 27pqr + 9pq + 9pr + 9qr + 3p + 3q + 3r + 1 \\ &= 3(9pqr + 3pq + 3pr + 3qr + p + q + r) + 1 \\ &= 3k + 1, \quad k \in \mathbb{Z}^+ \end{aligned}$$

Which of the following can be deduced from the proof above?

- A. If the product of three integers is one more than a multiple of 3, then the three integers must all be one more than a multiple of 3.
- B. If the product of three integers is one more than a multiple of 3, then at least one of the three integers must be one more than a multiple of 3.
- C. If the product of three integers is two more than a multiple of 3, then none of the three integers can be one more than a multiple of 3.
- D. If the product of three integers is two more than a multiple of 3, then at least one of the three integers must not be one more than a multiple of 3.



Question 21 (1 mark)

SGGHS '20 Q11d

Consider the statement $\forall x \in \mathbb{R}, \exists y \in \mathbb{R}$ such that $x^2 + y^2 = 2$. Is this true or false? Justify your answer.

Question 22 (2 marks)

Killara '24 Q13a

Prove or refute the following: For any list of primes $p_1, \dots, p_n$, the number $(p_1 p_2 \cdots p_n) + 1$ is prime.

Question 23 (3 marks)

NSB '24 Q11a

State whether each statement is true or false, justifying your answer.

(i) $\forall n \in \mathbb{N}, n^2 > n$

1

(ii) $\exists x \in \mathbb{R}$ such as $x^2 + 1 = 0$

1

(iii) $\forall x \in \mathbb{R}, \exists y \in \mathbb{N}$ such as $x + y = 10$

1

Question 24 (10 marks)

Ascham '22 Q12a

Consider the statement:

$$\forall A \in \mathbb{R}: \text{If } \sin A = k \text{ then } \sin^{-1} k = A$$

(i) Write the converse.

2

(ii) Write the contrapositive.

2

(iii) Write the negation.

2

(iv) Determine whether or not the statement is an equivalence. Give reasons.

2

(v) Determine whether or not the statement is true. If true, justify using logic, or if not, give a counter-example.

2



Direct Proof

Question 25 (2 marks)

BBHS '24 Q14b

Prove that the sum of the squares of 3 consecutive positive integers is always 1 less than a multiple of 3.

Question 26 (2 marks)

Caringbah '24 Q11c

Prove algebraically that the difference between two consecutive square numbers is odd.

Question 27 (3 marks)

Baulko '24 Q16c

Prove

$$\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \frac{1}{\sqrt{5} + \sqrt{6}} \cdots + \frac{1}{\sqrt{2023} + \sqrt{2024}} > 22.$$

Question 28 (2 marks)

HAHS '24 Q13a

Prove that a number is even if and only if its square is even.

Question 29 (5 marks)

Ascham '24 Q15a

(i) Prove that $\tan^{-1}(k+1) - \tan^{-1}k = \tan^{-1}\left(\frac{1}{k(k+1)+1}\right)$ for $k = 1, 2, 3 \dots$

2

(ii) Hence prove that

3

$$\lim_{n \rightarrow \infty} \left[\tan^{-1}\left(\frac{1}{1.2+1}\right) + \tan^{-1}\left(\frac{1}{2.3+1}\right) + \tan^{-1}\left(\frac{1}{3.4+1}\right) + \dots + \tan^{-1}\left(\frac{1}{n(n+1)+1}\right) \right] = \frac{\pi}{4}$$

Question 30 (5 marks)

Penrith '24 Q14b

On the curve $y = e^x$, P and Q are two points with x -coordinates a and b respectively, $a > b$

(i) Prove that $y = e^x$ is concave up for all x .

1

(ii) Explain why $\frac{1}{2}(e^a + e^b) > e^{\frac{1}{2}(a+b)}$.

2

(iii) Hence show that $e^a + e^b + e^c + e^d > 4e^{\frac{a+b+c+d}{4}}$ if $a > b > c > d$

2

Question 31 (3 marks)

NSB '21 Q14a

Show that $\frac{(x+k)^2}{x^2+x+1} \leq \frac{4}{3}(k^2-k+1)$ for all real x and k .



Question 32 (2 marks)

Normo '22 Q14a

Prove that if $a, b \in \mathbb{R}$,

$$\frac{5}{8}a^4 + \frac{2}{5}b^2 \geq a^2b$$

Question 33 (2 marks)

Knox '23 Q16a

Prove that the sum of the squares of three consecutive integers that are even and positive, always gives a remainder 8 when divided by 12

Question 34 (4 marks)

Knox '23 Q15a

It is given that $3^x > x + \ln(x)$ for $x > 0$. By forming a series of n inequalities, show that $3^{n+1} > n^2 + n + 3 + 2 \ln(n!)$, where n is a positive integer. Do **NOT** use induction.

Question 35 (2 marks)

Knox '22 Q16d

For arbitrary pronumerals a, b, c, d, e, f, g and h , if $ab - gh$ is a divisor of $abef + cdgh$, show that $ab - gh$ is a divisor of $abcd + efgh$.



Contradiction & Contrapositive

36 Consider the following statement for $n \in \mathbb{Z}$

“If $n^2 + 4n + 1$ is even, then n is odd.”

Which of the following statements is the contrapositive of this statement for $n \in \mathbb{Z}$?

- A. If n is even, then $n^2 + 4n + 1$ is odd.
- B. If $n^2 + 4n + 1$ is odd, then n is even.
- C. If n is odd, then $n^2 + 4n + 1$ is even.
- D. If $n^2 + 4n + 1$ is even, then n is even.

37 The contrapositive of “If it barks it’s a dog” is:

- A. “If it doesn’t bark it isn’t a dog”
- B. “If it’s a dog it will bark”
- C. “If it doesn’t bark it’s a dog”
- D. “If it isn’t a dog it doesn’t bark”

38 The converse of “if it miaows it’s a cat” is

- A. If it’s a cat it miaows.
- B. If it’s not a cat it doesn’t miaow.
- C. If it doesn’t miaow it isn’t a cat.
- D. If it’s a cat it doesn’t miaow.

39 The contrapositive of “if it miaows it’s a cat” is

- A. If it’s a cat it miaows.
- B. If it’s not a cat it doesn’t miaow.
- C. If it doesn’t miaow it isn’t a cat.
- D. If it’s a cat it doesn’t miaow.



40

Consider the statement

“For all quadrilaterals, if the four angles are right angles, then the quadrilateral is a rectangle”

Which of the following is the contrapositive of the statement?

- A. There exists a quadrilateral such that the four angles are not right angles and the quadrilateral is not a rectangle.
- B. There exists a quadrilateral such that the quadrilateral is not a rectangle and the four angles are not right angles.
- C. For all quadrilaterals, if the quadrilateral is not a rectangle, then the four angles are not right angles.
- D. For all quadrilaterals, if the four angles are right angles, then the quadrilateral is not a rectangle.

Question 41 (3 marks)

SGHS '21 Q12d

In the set of real numbers, let P be the proposition:

“If $a + b$ is irrational then at least one of a and b is irrational.”

- (i) State the contrapositive of the proposition P . 1
- (ii) Write the converse of the proposition P , and state, with reasons, whether this converse is true or false. 2

Question 42 (3 marks)

SGS '21 Q12a

Prove by contradiction that $\frac{p}{q} + \frac{q}{p} \geq 2$ for $p, q \in \mathbb{Z}^+$.

Question 43 (3 marks)

Baulko '23 Q11e

Consider the statement about a real number x “if $x^5 + 4x^3 + 5x \geq 8x^4 + 2x^2 + 17$ then $x > 0$ ”

- (i) Write down the contrapositive of the statement. 1
- (ii) Hence, or otherwise, prove the statement “ $\forall x \in \mathbb{R}$, if $x^5 + 4x^3 + 5x \geq 8x^4 + 2x^2 + 17$ then $x \geq 0$.” 2

Question 44 (3 marks)

NSG '20 Q16a

Prove by contradiction that $\forall a, b \in \mathbb{Z}, a^2 - 4b \neq 2$.



Question 45 (3 marks)

God folder

Consider the polynomial $p(x) = 4x^3 + 2x^2 - 6x + 1$. Prove by contradiction that $p(x)$ has no integer roots.

Question 46 (4 marks)

Ruse '20 AT2 Q10c

Given a, b, c are positive real numbers with $a > b$ and $c^2 > ab$, prove by contradiction that

$$\frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2-c^2}} > 0$$

Question 47 (3 marks)

Manly '24 Q12e

Let a, b and c be positive integers.

Prove the following statement by contraposition:

"If $a^3 + b^3 + c^3$ is divisible by 9, then at least one of a, b and c is divisible by 3."

Question 48 (3 marks)

SGGHS '24 Q14d

Prove by contradiction that for $a \geq 2$, $\sqrt{a} + \sqrt{a+2} > \sqrt{a+8}$

Question 49 (1 mark)

Copium '23 Q13 a) iv)

Prove by contradiction that if a, b, c, d are positive real numbers such that $a \geq c$ and $b \geq d$, then $ab + cd \geq ad + bc$.

Question 50 (4 marks)

Ruse '22 AT2 Q8c

Suppose $a, b \in \mathbb{Z}$. Consider the statement:

If $a^2(b^2 - 2b)$ is odd, then a and b are odd.

(i) Write the contrapositive of the statement.

1

(ii) Use the contrapositive to prove the statement is true.

3

Question 51 (3 marks)

Baulko '24 Q15c i

Prove that the cube root of any irrational number, is also an irrational number.

Question 52 (4 marks)

Baulko '23 Q14a

Prove, by contradiction, that there are no positive integers p and q such that $4p^2 - q^2 = 25$

Question 53 (2 marks)NSB '24 Q14c ³

Let a, b, c be positive integers. Prove that it is impossible for all three numbers $a^2 + b + c, b^2 + a + c, c^2 + a + b$ to be perfect squares.

Question 54 (3 marks)

Killara '24 Q14a

Prove that, for any integer greater than one, there is only one prime factorisation.



Question 55 (3 marks)

NSB '21 Q13c

“If both ab and $a + b$ are even then both a and b are even.”

Prove this statement by proving its contrapositive.

Question 56 (2 marks)

SGHS '22 Q16a

Let a and b be positive integers. Prove by contradiction that there are no solutions to $a^2 = 9b - 6$.

Question 57 (3 marks)

SGHS '23 Q16a

Let $f(x)$ be a concave down function on a given interval and let x_1, x_2 and x_3 lie in the given interval.

Jensen's inequality states that

$$\frac{f(x_1) + f(x_2) + f(x_3)}{3} \leq f\left(\frac{x_1 + x_2 + x_3}{3}\right).$$

(Do NOT prove this)

(i) Show algebraically that $f(x) = \sin x$ is concave down for $0 < x < \pi$.

1

(ii) Suppose that A, B and C are the angles of a triangle. By using part (i), or otherwise, show that

2

$$\sin A + \sin B + \sin C \leq \frac{3\sqrt{3}}{2}.$$

Question 58 (8 marks)Baulko '21 Q14c ⁴

Let $S_n = \sum_{r=1}^n \frac{1}{\sqrt{r}}$ where $n \in \mathbb{Z}^+$.

(i) Prove by induction that $S_n \leq 2\sqrt{n} - 1$.

4

(ii) Show that $(4k + 1)\sqrt{k + 1} > (4k + 3)\sqrt{k} \quad \forall k \geq 0$.

2

(iii) Determine the smallest number c such that:

2

$$S_n \geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - c$$

Question 59 (3 marks)

Normo '22 Q14d

Suppose that $a, b, c \in \mathbb{Z}$. Consider the following proposition:

“If all a, b, c are odd integers, then the equation $ax^2 + bx + c = 0$ has no integer solutions.”

(i) State the contrapositive to the proposition.

1

(ii) Hence prove the proposition by proving the contrapositive.



2

Question 60 (7 marks)

NSG '23 Q16a

- (i) Given ω is a non-real root of $x^3 - 1 = 0$, show that ω is also a root of $x^2 + x + 1 = 0$. 1
- (ii) Use a proof by contradiction to show that $(x + 1)^{2n} + x^{2n} + 1$ is not divisible by $(x^2 + x + 1)$ if n is divisible by 3. 3

Question 61 (3 marks)

NSG '23 Q13c

Consider the statement S below:

$$S: \forall x, y \in \mathbb{R} \text{ If } y^3 + yx^2 \leq x^3 + xy^2 \text{ then } y \leq x.$$

- (i) State the contrapositive of the statement S . 1
- (ii) Hence, prove the statement by proving the contrapositive. 2

Question 62 (4 marks)

NSG '22 Q16b

- (i) By finding the equation of the tangent to $y = \ln x$ at $x = 1$, explain why 1

$$\ln x \leq x - 1 \quad \text{for } x > 0$$

- (ii) If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are positive numbers such that $a_1 + a_2 + a_3 + \dots + a_n = 1$ and $a_1 b_1 + a_2 b_2 + \dots + a_n b_n = 1$, prove that $b_1^{a_1} b_2^{a_2} b_3^{a_3} \dots b_n^{a_n} \leq 1$. 3

Question 63 (7 marks)

Manly '20 Q13d

let n be a natural number.

- (i) Prove that if n is composite, at least one factor of n is less than $\sqrt{n}$. 2
- (ii) State the converse of the proposition in part (i) 1
- (iii) Write down the contrapositive of the proposition in part (i) 2

Question 64 (4 marks)

FSHS '23 Q16a

Consider the proposition:

“If the remainder is 2 or 3 when an integer n is divided by 4”.

- (i) State the contrapositive to the proposition. 1
- (ii) Hence, prove the proposition by proving the contrapositive. 3



Question 65 (5 marks)

NSG '24 Q14c

The integers 1 to 30 are to be arranged randomly around a circle so that each integer occurs exactly once.

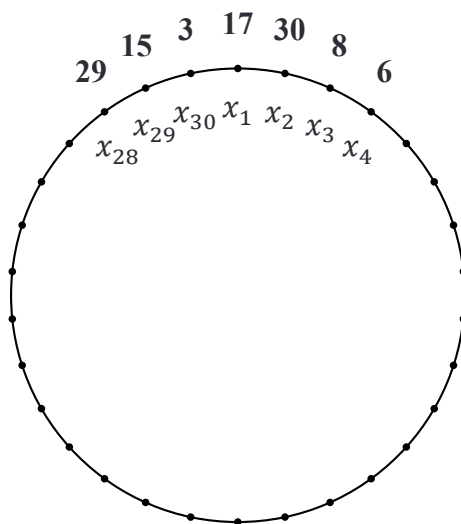
Consider the statement:

P : “There exists a set of three neighbouring integers around the circle which sum to at least 45.”

- (i) Without using the word “exist” or an equivalent, write down the negation of this statement. 1
An arbitrarily chosen integer on the circle is labelled x_1 , and the other integers are labelled consecutively around the circle $x_2, x_3, x_4, \dots, x_{30}$

The diagram below shows an example for the partial arrangement

$\dots, 29, 15, 3, 17, 30, 8, 6, \dots$



- (ii) Assume that the statement P is FALSE. 1

Hence write down an inequality (not specific to the example above) involving x_1, x_2 and x_3 , and another inequality involving x_{30}, x_1 and x_2 .

- (iii) Prove by contradiction that the statement P is true for any random arrangement. 3

(A proof by the pigeonhole principle will receive no credit.)



Algebraic Inequalities

- 66 Given that positive numbers a, b, c satisfy $a + 2b + 3c = 6$, which of the following expressions have the largest maximum value?

A. a^2bc
B. abc
C. ab^2c
D. ab^2c^2

- 67 Suppose that a and b are both positive real numbers with $a > b$. If m is any real number, how many of the following statements are always true?

$$\begin{aligned}\frac{a}{b} &< \frac{a+m}{b+m} \\ \frac{a}{b} &\geq \frac{a+m}{b+m} \\ \frac{a+m}{b+m} &\leq 1\end{aligned}$$

A. 0
B. 1
C. 2
D. 3

- 68 Let a, b, c be positive real numbers. Which of the following expressions has the smallest minimum value?

A. $\frac{(a+b)(b+c)(a+c)}{abc}$
B. $\frac{a+b}{c} + \frac{b+c}{a} + \frac{a+c}{b}$
C. $\left(a + \frac{1}{a}\right)\left(b + \frac{1}{b}\right)\left(c + \frac{1}{c}\right)$
D. $(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right)$



- 69 Let a and b be positive real numbers with $a < b$. Given that:

$$Q = \frac{2}{\frac{1}{a} + \frac{1}{b}}$$

$$M = \sqrt{ab}$$

$$N = \frac{a+b}{2}$$

$$R = \sqrt{\frac{a^2 + b^2}{2}}$$

which of the following statements is true?

- A. $Q < N < M < R$
- B. $Q < M < N < R$
- C. $M < N < Q < R$
- D. $Q < M < R < N$
- 70 If x, y and z are any real numbers and $x > y$, which of the following statements must be true?
- A. $x^2 > y^2$
- B. $\frac{1}{x} < \frac{1}{y}$
- C. $x + z > y + z$
- D. $xz > yz$
- 71 If a, b and c are real numbers such that $0 < a < b < c$, which of the following is NOT necessarily true?
- A. $\frac{1}{a} + \frac{1}{b} - \frac{1}{c} > 0$
- B. $\frac{1}{c+a} + \frac{1}{b+c} - \frac{1}{a+b} > 0$
- C. $\frac{1}{ab} - \frac{1}{bc} > 0$
- D. $\frac{c}{ab} + \frac{a}{bc} - \frac{b}{ac} > 0$



Question 72 (5 marks)

Ascham '22 Q14a

- (i) If
- a, b
- are real, show that
- $a^2 + b^2 \geq 2ab$
- .

1

Hence, or otherwise, show that if a, b, c are real then

(ii) $a^2 + b^2 + c^2 \geq ab + bc + ca$

2

(iii) $(a^2 + b^2 + c^2)^2 \geq 3abc(a + b + c)$

2

Question 73 (3 marks)Baulko '22 Q16c ⁵If $a, b, c > 0$, prove that $\frac{a}{b+c} + \frac{b}{a+c} + \frac{c}{a+b} \geq \frac{3}{2}$.**Question 74** (3 marks)

Ruse '22 Q11b

- (i) Prove that
- $x + y \geq 2\sqrt{xy}$

1

- (ii) Hence or otherwise, find the minimum value of the function

2

$$f(x) = \frac{9x^2 \sin^2 x + 4}{x \sin x}$$

in the domain $0 < x < \pi$.**Question 75** (4 marks)Sydney Tech '22 Q16c ⁶Prove the following, where x, a, b, c are real positive numbers.

(i) $x + \frac{1}{x} \geq 2$

2

(ii) Hence, show $(a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \geq 9$

2

Question 76 (4 marks)

Sydney Girls '22 Q15c

For all non-negative real numbers a and b , $(a + b)^2 \geq 4ab$ (Do not prove this)

- (i) Using this fact, show that for all non-negative real numbers
- a, b
- and
- c

2

$$(a + b + c)^2 \geq 8c\sqrt{ab}.$$

- (ii) Using part (i), or otherwise, show that for all non-negative real numbers,
- a, b
- and
- c

2

$$(a + b + c)^3 \geq 16c\sqrt{2}abc.$$



Question 77 (5 marks)

Penrith '22 Q12c

Suppose that a, b, c are real numbers.

(i) Prove that $a^4 + b^4 + c^4 \geq a^2b^2 + a^2c^2 + b^2c^2$ 2

(ii) Show that $a^2b^2 + a^2c^2 + b^2c^2 \geq a^2bc + b^2ac + c^2ab$ 2

(iii) Deduce that if $a + b + c = d$ then $a^4 + b^4 + c^4 \geq abcd$ 1

Question 78 (6 marks)

NSG '22 Q15c

(i) Show that $\frac{1}{(k-1)k(k+1)} > \frac{1}{k^3}$, where $k > 1$. 2

(ii) Given that $S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} = \sum_{k=3}^n \frac{1}{k^3}$, by using partial fractions on part (i) or 4
otherwise, prove that $S_n \leq \frac{1}{12} \quad \forall n \geq 3$.

Question 79 (5 marks)

Normo '22 AT3 Q3a

It is given that $a, b, c \in \mathbb{R}$ and $a + b + c = 1$.

(i) Prove that $\frac{a}{b} + \frac{b}{a} \geq 2$. 1

(ii) Prove that $\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq 9$. 2

(iii) Hence, or otherwise, prove $(1-a)(1-b)(1-c) \geq 8abc$. 2

Question 80 (3 marks)

God folder

Suppose that $a > b > c > d > 0$.

Show that:

$$ac(b+d) - bd(a+c) > c(ac+b^2) - b(bd+c^2)$$



Question 81 (6 marks)

NSB '23 AT2 Q3

(a) It is given that $\sqrt[n]{t_1 t_2 t_3 \dots t_n} \leq \frac{t_1 + t_2 + t_3 + \dots + t_n}{n}$ for any positive integer n and for any positive numbers $t_1, t_2, t_3, \dots$. Find the minimum value of $x^5 + y^5 + z^5 - 5xyz$ **2**

(b) (i) Use part (a) above to show that $n \geq \sqrt{(2n-1)}$ **1**

(ii) Without using Mathematical Induction, prove that $n^n \geq 1 \times 3 \times 5 \times \dots \times (2n-1)$ for all positive integers n . **3**

Question 82 (8 marks)

NSB '19 Q15c

Given that a, b and c are positive real numbers:

(i) Prove that $a^2 + (bc)^2 \geq 2abc$ **2**

(ii) Prove that $a^2 + b^2 + c^2 \geq ab + bc + ca$ **2**

(iii) Prove that $a^2(1+b^2) + b^2(1+c^2) + c^2(1+a^2) \geq 6abc$ **2**

(iv) Hence, or otherwise prove that $a^2(1+a^2) + b^2(1+b^2) + c^2(1+c^2) \geq 6abc$ **2**



Question 83 (6 marks)

ST '24 Q16c

It is known that the arithmetic mean of three numbers is greater than the geometric mean of those numbers, that is

$$\frac{a+b+c}{3} \geq \sqrt[3]{abc}$$

- (i) Given that $a+b+c=t$ and $(a,b,c>0)$, prove that

3

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \geq \frac{9}{t}$$

- (ii) Hence, given $a+b+c=1$ and $(a,b,c>0)$, prove that

3

$$\left(\frac{1}{a}-1\right)\left(\frac{1}{b}-1\right)\left(\frac{1}{c}-1\right) \geq 8$$

Question 84 (6 marks)

BBHS '24 Q14d

- (i) Let x, y be real numbers such that $x \geq 0$ and $y \geq 0$.

1

Prove that $\frac{x+y}{2} \geq \sqrt{xy}$

- (ii) Suppose that a, b, c are real numbers.

2

Prove that $a^8 + b^8 + c^8 \geq a^4b^4 + a^4c^4 + b^4c^4$.

- (iii) Show that $a^4b^4 + a^4c^4 + b^4c^4 \geq a^4b^2c^2 + b^4a^2c^2 + c^4a^2b^2$.

2

- (iv) Deduce that if $a^2 + b^2 + c^2 = d^2$, then $a^8 + b^8 + c^8 \geq a^2b^2c^2d^2$.

1

Question 85 (3 marks)Ruse '24 Q12d ⁷

- (i) Show that $a^2 + b^2 \geq 2ab$.

1

- (ii) Hence or otherwise, show that for all positive numbers a, b and c ,

2

$$2(a^3 + b^3 + c^3) \geq ab(a+b) + bc(b+c) + ca(c+a)$$



Question 86 (4 marks)

NSG '24 Q16b

- (i) Show that $\frac{2k-1}{2k} < \frac{2k}{2k+1}$ for all positive integers k 1
- (ii) By considering $p = \frac{1}{2} \cdot \frac{3}{4} \cdot \frac{5}{6} \cdots \frac{2n-1}{2n}$ and $q = \frac{2}{3} \cdot \frac{4}{5} \cdot \frac{6}{7} \cdots \frac{2n}{2n+1}$, show that $\frac{1}{2} \cdot \frac{3}{4} \cdots \frac{2n-1}{2n} < \frac{1}{\sqrt{2n}}$ 3

Question 87 (2 marks)

Baulko '23 Q16b

Given p, q are positive real numbers:

- (i) Prove that $\frac{1}{2}(p+q) \geq \sqrt{pq}$ 1
- (ii) Hence, deduce that $\sqrt{p} \leq \frac{1}{2} \left(\frac{p}{\sqrt{p}} + \sqrt{q} \right)$ 1

Question 88 (5 marks)

Manly '24 Q16c

- (i) Prove for all $a, b > 0$ that $\frac{a+b}{2} \geq \sqrt{ab}$ 1
- (ii) Prove that if $x > 1$ then $\frac{x}{\sqrt{x-1}} \geq 2$ 2
- (iii) Hence or otherwise prove that for $a, b > 1$, the following inequality holds: 2

$$\frac{a^2}{b-1} + \frac{b^2}{a-1} \geq 8$$

Question 89 (5 marks)Baulko '23 Q14b ⁸Three numbers a, b and c are all positive.

- (i) Use the Arithmetic/Geometric mean inequality, or otherwise, to prove that $(a+b+c)^3 \geq 27abc$ 2
- (ii) Hence, or otherwise, if a, b and c multiply to give 1, prove that $a^5c^4 + b^5a^4 + c^5b^4 \geq 3$. 3

Question 90 (3 marks)

NSB '24 Q14a

Let $a_1, a_2, \dots, a_n$ be positive real numbers such that $a_1 a_2 \cdots a_n = 1$. Prove that

$$(a_1^2 + a_1)(a_2^2 + a_2) \cdots (a_n^2 + a_n) \geq 2^n$$



Question 91 (4 marks)

NSB '24 Q13b

Let p and q be positive real numbers with $\frac{1}{p} + \frac{1}{q} = 1$. Prove that

(i)

2

$$\frac{1}{3} \leq \frac{1}{p(p+1)} + \frac{1}{q(q+1)} \leq \frac{1}{2}$$

(ii)

2

$$\frac{1}{p(p-1)} + \frac{1}{q(q-1)} \geq 1$$

Question 92 (2 marks)

Killara '22 Q13b

Prove $|a^2 - ab| + |ab - b^2| \geq (a+b)(a-b)$ for $a > b > 0$.

Question 93 (5 marks)

Normo '13 Q16b

(i) By squaring, or otherwise, show that for $k \geq 0$

1

$$2k + 3 > 2\sqrt{k+2}\sqrt{k+1}$$

(ii) By decomposing $2k + 3$ and factorising $2\sqrt{k+2}\sqrt{k+1} - 2(k+1)$, show that for $k \geq 1$,

2

$$\frac{1}{\sqrt{k+1}} > 2(\sqrt{k+2} - \sqrt{k+1})$$

(iii) Hence, or otherwise, show for $n \geq 1$

2

$$1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \cdots + \frac{1}{\sqrt{n}} > 2(\sqrt{n+1} - 1)$$

Question 94 (4 marks)

God folder

(i) Prove that, for all positive real numbers a, b, c ,

2

$$\frac{abc}{a^3 + b^3 + abc} \leq \frac{c}{a + b + c}$$

(ii) Hence prove that

2

$$\frac{abc}{a^3 + b^3 + abc} + \frac{abc}{b^3 + c^3 + abc} + \frac{abc}{c^3 + a^3 + abc} \leq 1$$



Question 95 (4 marks)

God folder

Let a, b and c be positive real numbers such that $a + b + c = 1$. Show that

$$\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \leq 3 + \frac{2(a^3 + b^3 + c^3)}{abc}.$$

Question 96 (6 marks)NSB '22 Q16b v) ⁹

Given that $a + b + c + d = 0$ and $a^2 + b^2 + c^2 + d^2 = 12$, find:

(i) The maximum value of $abcd$.

2

(ii) The minimum value of $abcd$.

4

Question 97 (4 marks)

SGHS '22 Q15c

For all non-negative real numbers a and b , $(a + b)^2 \geq 4ab$. (Do not prove this.)

(i) Using this fact, show that for all non-negative real numbers a, b and c

2

$$(a + b + c)^2 \geq 8c\sqrt{ab}$$

(ii) Using part (i), or otherwise, show that for all non-negative real numbers a, b and c

2

$$(a + b + c)^3 \geq 16\sqrt{2}abc.$$

Question 98 (4 marks)

Random Question

Let a, b, c be reals with $a, b, c > 1$ such that

$$\frac{1}{a^2 - 1} + \frac{1}{b^2 - 1} + \frac{1}{c^2 - 1} = 1$$

Prove that

$$\frac{1}{a + 1} + \frac{1}{b + 1} + \frac{1}{c + 1} \leq 1$$

Question 99 (5 marks)

Ruse '20 Q15b

Given that a, b and c are positive real numbers:

(i) Prove that $a^3 + b^3 \geq a^2b + ab^2$.

2

(ii) Hence prove

3

$$abc \leq \left(\frac{a + b + c}{3} \right)^3 \leq \frac{a^3 + b^3 + c^3}{3}$$

You may assume that $a + b + c \geq 3\sqrt[3]{abc}$ and

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 6abc.$$



Question 100 (3 marks)

Ruse '22 AT3 Q9a

Let $a + b + c = 1$, where a, b and c are positive real numbers. Find the smallest possible value of $\left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right)$

Question 101

Ruse '20 AT2 Q10d

Given $\frac{x + y + z}{3} \geq \sqrt[3]{xyz}$, show that $\frac{1}{A} + \frac{1}{B} + \frac{1}{C} \geq \frac{9}{\pi}$ if A, B, C are three interior angles (in radians), of $\triangle ABC$.

Question 102 (4 marks)

NSB '21 AT2 Q3 a) i

Prove $\frac{x^2}{y^2} + \frac{y^2}{z^2} + \frac{z^2}{x^2} \geq \frac{1}{yz}\sqrt{x^2y^2 + z^4} + \frac{1}{xz}\sqrt{z^2y^2 + x^4}$ for $x, y, z > 0$.

Question 103 (3 marks)

Knox '22 Q15d

If a, b and $c \in \mathbb{Z}^+$, prove that $\frac{b+c}{a} + \frac{c+a}{b} + \frac{a+b}{c} \geq 6$.



Mathematical Induction

Question 104 (3 marks)

God folder

Sigma rolls n dice and adds up all the numbers rolled. Prove by mathematical induction that the expected sum of the numbers is $\frac{7n}{2}$

Question 105 (3 marks)

SGHS '21 Q13b

Consider the recurrence relation defined by $T_1 = 1$ and

$$T_{n+1} = 3T_n + 4 \text{ for } n = 1, 2, \dots$$

Prove by mathematical induction that

$$T_n = 3^n - 2 \text{ for } n = 1, 2, \dots$$

Question 106 (3 marks)

Abbotsleigh '24 Q15c i¹⁰

Given $P_n(x) = (x - a_1)(x - a_2) \dots (x - a_n)$, prove by mathematical induction that

$$\frac{P'_n(x)}{P_n(x)} = \frac{1}{(x - a_1)} + \frac{1}{x - a_2} + \dots + \frac{1}{x - a_n} \quad (\text{for } n \geq 2)$$

Question 107 (5 marks)

Caringbah '24 Q16d

Let $f(x) = 1 + \frac{1}{x}$ and $\varphi = \frac{1 + \sqrt{5}}{2}$.

Define $f \circ f(x)$ to be the composition of $f(x)$ with itself twice, that is $f \circ f(x) = f(f(x))$.

Define $f^n(x)$ to be the function $f(x)$ composed within itself n times, for integers $n \geq 0$, that is,

$$f^n(x) = \begin{cases} 1 & n = 0 \\ \underbrace{f \circ f \circ \dots \circ f(x)}_{n \text{ times}} & n \geq 1 \end{cases}$$

(i) Show that $1 + \frac{1}{\varphi}$ and $1 + \frac{1}{1 - \varphi} = 1 - \varphi$ 1

(ii) Show, using mathematical induction, that for all integers $n \geq 0$, 3

$$f^n(1) = \frac{\varphi^{n+2} - (1 - \varphi)^{n+2}}{\varphi^{n+1} - (1 - \varphi)^{n+1}}$$

(iii) Hence, or otherwise, find the value of the infinite composition $\lim_{n \rightarrow \infty} f^n(1)$. 1



Question 108 (3 marks)

Ruse 2020 AT3 Q1e

Prove by mathematical induction that for $n \geq 3$, the sum of the interior angles of a convex polygon with n vertices is $180(n - 2)^\circ$

Question 109 (4 marks)

Ruse '24 Q14b

Consider the following number sequence given by

$$a_1 = 0$$

$$a_n = \frac{1 + a_{n-1}}{2 + a_{n-1}} \quad n \in \mathbb{Z}, n \geq 2$$

Prove by mathematical induction that $a_{n-1} < a_n$ for all integers $n \geq 2$.

Question 110 (3 marks)

ST '24 Q14b

Use mathematical induction to prove that

$$\frac{d^n}{dx^n} (xe^{2x}) = 2^{n-1}(2x + n)e^{2x}$$

for $n \geq 1$

Question 111 (3 marks)

SG '24 Q15d

Prove by mathematical induction that for all positive integers n ,

$$\frac{1^2}{\sqrt{2}} + \frac{2^2}{\sqrt{3}} + \cdots + \frac{n^2}{\sqrt{n+1}} < n^2\sqrt{n}$$

Question 112 (4 marks)

BBHS '24 Q16 c

Given that $x_1 + x_2 > \sqrt{x_1 x_2}$, use the method of mathematical induction to show that for all positive integers $n \geq 2$, if $x_j > 1, j = 1, 2, 3, \dots, n$, then

$$\ln(x_1 + x_2 + \cdots + x_n) > \frac{1}{2^{n-1}} (\ln x_1 + \ln x_2 + \cdots + \ln x_n)$$

Question 113 (4 marks)

Ruse '24 Q13 d

Prove by mathematical induction that

$$\frac{1}{2} + \cos \theta + \cos 2\theta + \cdots + \cos(n-1)\theta = \frac{\sin\left(\frac{2n-1}{2}\theta\right)}{2 \sin \frac{\theta}{2}}$$

for all $n \geq 2, n \in \mathbb{Z}$ and $\theta \neq 2k\pi$ for all $k \in \mathbb{Z}$.

Question 114 (4 marks)

Manly '24 Q16d

Prove by mathematical induction that

$$\int_0^{\frac{\pi}{2}} \sin^{2n} x \, dx = \frac{2\pi(2n)!}{4^{n+1}(n!)^2}$$



for all integers $n \geq 1$

Question 115 (4 marks)

SBHS '24 Q13b

A sequence $u_1, u_2, u_3, \dots$ is such that

$$u_1 = \frac{1}{4}$$
$$u_{n+1} = u_n + \frac{1}{n(n+1)} + 2^{-n}, n \in \mathbb{Z}^+$$

(i) Prove by mathematical induction that for $n \in \mathbb{Z}^+$,

3

$$u_n = \frac{9}{4} - \frac{1}{n} - 2^{-n+1}$$

(ii) Show that $u_n < \frac{9}{4}$ for $n \in \mathbb{Z}^+$.

1

Question 116 (3 marks)

NSG '24 Q16a

In the country of Discretia, the currency consists of \$3 and \$8 coins.

Use mathematical induction to prove that any whole dollar amount of \$14 or more can be achieved by taking a combination of \$3 and \$8 coins.

Hint: You may wish to consider two cases in the inductive step – no \$8 coins and at least one \$8 coin.

Question 117 (5 marks)

SGGHS '24 Q15a

(i) Show $\frac{2k}{k+2} < \frac{2k+2}{k+3}$

2

(ii) Use mathematical induction to prove for $n \in \mathbb{Z}, n \geq 1$

3

$$\frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \dots + \frac{n}{(n+2)!} < \frac{2n}{n+2} - \frac{1}{(n+2)!}$$

Question 118 (4 marks)

SGGHS '24 Q16a ¹¹

A sequence is defined by the following formula for $n \in \mathbb{Z}^+$:

$$T_0 = 0$$
$$T_n = \sqrt{T_{n-1} + 2}$$

Prove by mathematical induction that $T_n = 2 \cos\left(\frac{\pi}{2^{n+1}}\right)$ for integer $n \geq 0$



Question 119 (6 marks)Baulko '23 Q16c ¹²

Given $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are positive real numbers, where $0 < b_1 \leq b_2 \leq b_3 \leq \dots \leq b_n$, and $A_n = a_1 + a_2 + a_3 + \dots + a_n$, $B_n = b_1 + b_2 + b_3 + \dots + b_n$, where $A_r \leq B_r$ for $r = 1, 2, 3, \dots, n$.

(i) Prove, by mathematical induction for all positive integers n , that

3

$$\begin{aligned} & \frac{1}{\sqrt{b_n}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) B_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) B_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \\ &= \sqrt{b_1} + \sqrt{b_2} + \sqrt{b_3} + \dots + \sqrt{b_n}. \end{aligned}$$

(ii) Hence, given that

1

$$\begin{aligned} & \frac{a_1}{\sqrt{b_1}} + \frac{a_2}{\sqrt{b_2}} + \frac{a_3}{\sqrt{b_3}} + \dots + \frac{a_n}{\sqrt{b_n}} \\ &= \frac{1}{\sqrt{b_n}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) A_{n-1} + \left(\frac{1}{\sqrt{b_{n-2}}} - \frac{1}{\sqrt{b_{n-1}}} \right) A_{n-2} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) A_1, \end{aligned}$$

Show that $\sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \leq \sum_{r=1}^n \sqrt{b_r}$.

(iii) Using the result from part (b), or otherwise, deduce that

2

$$\sum_{r=1}^n \sqrt{a_r} \leq \sum_{r=1}^n \sqrt{b_r}.$$

Question 120 (3 marks)

Killara '24 Q13b

Prove by mathematical induction that $x^n - y^n$ is divisible by $x + y$ when n is even.

Question 121 (5 marks)

Girraween '24 Q14a

(i) Solve $k^2 - k - 2 \geq 0$

2

(ii) Hence prove by induction that $3^n > 2n^2 - 3 \forall n \in \mathbb{Z}^+ \geq 2$

3

Question 122 (4 marks)NSB '24 Q13a ¹³

(i) If k is an integer where $k \geq 3$ and $(k-1)(k+1) < k^2$, show that

1

$$\frac{1}{(k-1)k(k+1)} > \frac{1}{k^3}$$

(ii) Given that $S_n = \frac{1}{3^3} + \frac{1}{4^3} + \frac{1}{5^3} + \dots + \frac{1}{n^3} = \sum_{k=3}^n \frac{1}{k^3}$, use partial fractions from previous part

3

or otherwise to prove that $S_n < \frac{1}{12}$



Question 123 (4 marks)

Normo '21 AT2 Q2c

Suppose that p is a prime number and that a is a positive integer.

- (i) Use the expansion of $(1 + a)^p$ to prove that $(1 + a)^p - 1 - a^p$ is divisible by p . (You may use the fact that $\binom{p}{k}$ is divisible by p for $k = 1, 2, \dots, p - 1$) 1
- (ii) Hence or otherwise, prove by mathematical induction that $a^p - a$ is divisible by p for all positive integers a . 3

Question 124 (3 marks)

Normo '22 Q16a

Let $a \in \mathbb{R}^+$ and $n \in \mathbb{Z}^+$.

- (i) Show that $a + \frac{1}{a} \geq 2$. It is given that 1

$$a^{n+1} + \frac{1}{a^{n+1}} + a^{n-1} + \frac{1}{a^{n-1}} = \left(a + \frac{1}{a}\right) \left(a^n + \frac{1}{a^n}\right) \quad (\text{Do NOT prove this})$$

- (ii) Use the principle of mathematical induction to prove that 2

$$a^n + \frac{1}{a^n} \geq a^{n-1} + \frac{1}{a^{n-1}}$$

for all positive integers $n \geq 1$.

Question 125 (5 marks)

NSG '20 Q15a

- (i) Use mathematical induction to prove that $\frac{1}{n!} < \frac{1}{e^n}$ for $n \geq 6, n \in \mathbb{Z}^+$ 3

- (ii) Hence or otherwise, show that $\frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \dots < \frac{1}{e^6 - e^5}$. 2

Question 126 (3 marks)

Knox '22 Q16a

Use mathematical induction to prove that $3^n \geq n^2 + 3n - 1$, for any integer $n \geq 1$.



Question 127 (10 marks)

SGHS '22 Q16c

Let $I_n = \int_0^1 \frac{x^{2n-1}}{(2n-1)!} \sin x \, dx$ where n is an integer, $n \geq 1$.

(i) Show that

3

$$I_n = \frac{\sin 1}{(2n-2)!} - \frac{\cos 1}{(2n-1)!} - I_{n-1}, \quad \text{for } n \geq 2.$$

(ii) Using part (i), prove by mathematical induction that for every positive integer n there exists integers a_n and b_n such that

3

$$I_n = \frac{a_n \sin 1 + b_n \cos 1}{(2n-1)!}.$$

(iii) It is given that $0 < I_n < \frac{1}{(2n)!}$ and $\cos 1 > \frac{1}{2}$ (Do not prove this.) Using these results and part (ii), show that for every positive integer n there exist integers a_n and b_n such that

2

$$0 < a_n \tan 1 + b_n < \frac{1}{n}$$

(iv) Using part (iii), prove by contradiction that $\tan 1$ is irrational.

2

Question 128 (3 marks)NSG '13 Q16c¹⁴

Using the fact that $\tan^{-1} x + \tan^{-1} y = \tan^{-1} \frac{x+y}{1-xy}$ for $0 \leq x < 1$ and $0 \leq y < 1$, prove by mathematical induction that for all positive integers n

$$\tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2 \times n^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2n+1}$$

Question 129 (3 marks)

HAHS '24 Q13d

Use mathematical induction to prove that, for $n \geq 1$,

$$x^{(3^n)} - 1 = (x-1)(x^2+x+1)(x^6+x^3+1)\dots(x^{(2 \times 3^{n-1})} + x^{(3^{n-1})} + 1)$$

Question 130 (3 marks)

SGS '21 Q15a

Bernoulli's inequality famously states that,

$$\text{"if } x \geq -1 \text{ then } (1+x)^n \geq 1+nx, \text{ for all positive integers } n.$$

Prove this inequality using mathematical induction.



Question 131 (5 marks)

Ruse '22 AT3 Q10

The function f is defined by $f(x) = \frac{1}{2}(1 - \sqrt{1-x})$ for $0 < x < 1$.

Given that $f_1 = f$ and $f_n = \underbrace{f \circ f \circ \dots \circ f}_{n \text{ compositions}}$, prove by Mathematical Induction that for $0 < \theta < \frac{\pi}{2}$ and positive integers n

$$f_n(\sin^2 \theta) = \sin^2 \left(\frac{\theta}{2^n} \right)$$

Question 132 (4 marks)

SBHS '21 AT2 Q11d

(i) Prove, using induction that $\frac{1}{n+1} + \frac{1}{n+2} + \frac{1}{n+3} + \dots + \frac{1}{2n+1} < 1$ for $n > 1$, where n is an integer. 3

(ii) Hence show that $\sum_{k=n+1}^{2n} \frac{1}{k} < 1$. 1

Question 133 (3 marks)

NSB '19 Q15b

A sequence x_n is defined by the following rules: $x_0 = 2a$, $x_1 = -a^2$ and

$$x_{n+1} = -ax_n + a^2x_{n-1}, \quad \text{for } n \geq 1.$$

Prove by mathematical induction

$$x_n = a^{n+1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^n + \left(\frac{-1 - \sqrt{5}}{2} \right)^n \right], \quad \text{for } n \geq 0.$$

Question 134 (6 marks)Normo '22 Q13a ¹⁵

For integer values of n , $I_n = \int_1^e (1 - \ln x)^n dx$ for $n \geq 0$.

(i) Show that $I_n = -1 + nI_{n-1}$ for $n \geq 1$. 2

(ii) Use mathematical induction to show that: 4

$$I_n = n!e - 1 - \sum_{r=1}^n {}^nP_r \quad \text{for all integers } n \geq 1$$



Question 135 (4 marks)

BoS '24 Q16a

For positive integer n and $a_k \in \{0, 1, 2, 3, 4, 5, 6, 7, 8, 9\}$ with $a_{2n-1} \neq 0$, let

$$u_{2n} = \sum_{k=0}^{2n-1} 10^k a_k \quad \text{and} \quad w_{2n} = \sum_{k=0}^{2n-1} (-1)^k a_k.$$

Note that u_{2n} represents a $2n$ digit number with leading digit a_{2n-1} and final digit a_0 .

(i) Suppose that w_{2n} is divisible by 11 for all positive integer n . Prove by mathematical induction that u_{2n} is also divisible by 11. 3

(ii) A *palindromic* number is a number which is unchanged when it's digits are in reverse order. Using the result in part (i), deduce that all palindromic numbers with an even number of digits is divisible by 11. 1

Question 136 (4 marks)

HAHS '17 Q16c

The n th Fermat number, F_n , is defined by $F_n = 2^{2^n} + 1$ for $n = 0, 1, 2, 3, \dots$. Prove by mathematical induction, that for all positive integers:

$$F_0 \times F_1 \times F_2 \times \dots \times F_{n-1} = F_n - 2$$

Question 137 (4 marks)Normo '21 AT2 Q2b ¹⁶

The *towers of Hanoi* is a classical puzzle. You have three pillars, with n gold plates on the first pillar, all in different sizes with the smaller plates above the larger plates. Your task is to move one plate at a time, such that you end up with all n plates at the last pillar. During this procedure, a larger plate is never allowed to be on top of a smaller plate.

Use mathematical induction to prove that this can be done using $2^n - 1$ moves.

Question 138 (5 marks)

Ruse '23 AT3 Q8a

A function $f(x)$ is defined such that $f(x) > 0$ for all real numbers x and $f(a + b) = f(a) \times f(b)$ for any real numbers a and b .

(i) Show that $f(0) = 1$ and deduce that $f(-x) = \frac{1}{f(x)}$ 3

(ii) By using mathematical induction, show that $f(nx) = [f(x)]^n$ for $n \in \mathbb{Z}^+$. 2

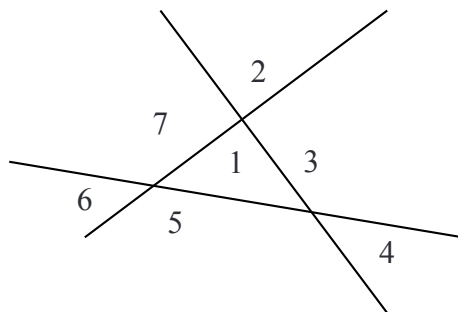


Question 139 (5 marks)

In howard textbook and Lambertnet trial

Suppose that n lines are drawn in a plane such that no three lines intersect at the same point i.e. not concurrent, and no two lines are drawn in parallel.

Let R_n be the number of regions into which these lines divide the plane. The diagram below shows that for $n = 3$ that $R_3 = 7$.



(i) By drawing diagrams, find R_1, R_2, R_4 and R_5 .

2

(ii) From these results, find the values of a, b and c in the recurrence relationship below that describes the numbers of lines and regions, where $a, b, c \in \mathbb{Z}$.

$$R_1 = a$$

$$R_n = bR_{n-1} + cn, n \geq 2.$$

(iii) Prove that $R_n = \frac{n^2 + n + 2}{2}, n \geq 1$.

2



Question 140 (8 marks)

Copium '23 Q14a ¹⁷

A tetromino tile is a tile that can be formed by gluing together four unit square tiles, edge to edge. For each positive integer n , consider a bathroom whose floor is in the shape of a $2 \times 2n$ rectangle. Let T_n be the number of ways to tile this bathroom floor with tetromino tiles. For example, $T_2 = 4$ since there are four ways to tile a 2×4 rectangular bathroom floor with tetromino tiles, as shown below:



- (i) Let F_n be the Fibonacci sequence, defined as $F_{n+2} = F_{n+1} + F_n$ for $n \geq 0$, and $F_0 = 1, F_1 = 1$. 1
Prove for positive integers n :

$$F_n^2 + F_{n-3}^2 = 2(F_{n-1}^2 + F_{n-2}^2)$$

- (ii) If the tetromino covering the leftmost 2×1 column of the $2 \times 2n$ rectangle is a 2×2 square, 1
explain why the number of ways to tile the rectangle is now T_{n-1}
- (iii) If the leftmost 2×1 column is covered by two horizontal 1×4 rectangles instead, explain why 1
the number of ways to tile the rectangle is T_{n-2} .
- (iv) It is given that if the leftmost column of the $2 \times 2n$ rectangle is covered by an L-shaped tetromino, 3
the number of ways to tile the rectangle is

$$2(T_0 + T_1 + \dots + T_{n-2})$$

(Do NOT prove this)

Hence prove that for positive integers $n \geq 3$,

$$T_n - 2T_{n-1} - 2T_{n-2} + T_{n-3} = 0$$

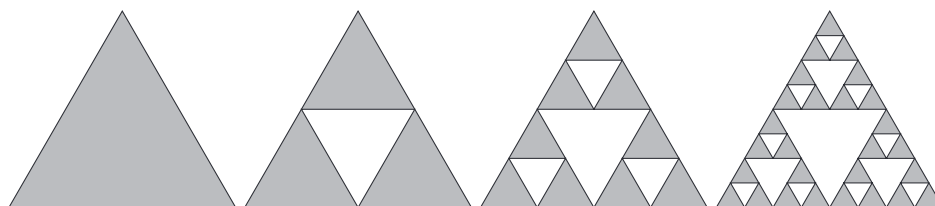
- (v) Hence prove by mathematical induction that $T_n = F_n^2$ for all positive integers n . 2



Question 141 (3 marks)

”The –C ult Q2a”

The Sierpinski triangle is a fractal that starts with a shaded equilateral triangle. Then, a triangle is hollowed out from the middle of each shaded triangle such that the vertices of that triangle is exactly in the middle of each of the black triangle’s edges.



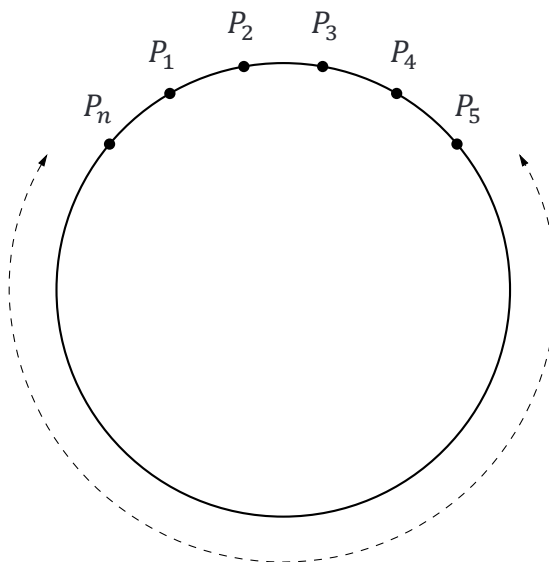
- (i) Prove by mathematical induction that there are a total of $\frac{3^n - 1}{2}$ shaded or white triangles in the n th iteration. 2

- (ii) Find the shaded area of the n th iteration, if the original triangle has side length one. 1

Question 142 (5 marks)

BoS '18 Q15c

There are n people sitting in a circle labelled P_1 to P_n as shown in the diagram below.



Every second person is removed from the table in a clockwise direction, initially removing P_2 , until eventually there is only one person remaining on the table.

- (i) Suppose there are $2m$ people in the circle labelled P_1 to P_{2m} , where m is a positive integer. Use mathematical induction to prove that P_1 always wins. 3
- (ii) Hence, find the labelled position of the remaining person if the circle contains 2018 people. 2



Calculus Proofs & The Squeeze Theorem

Question 143 (12 marks)

God folder

The harmonic series is defined by:

$$H_n = \sum_{k=1}^n \frac{1}{k} = 1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \cdots + \frac{1}{n}$$

The alternating harmonic series is defined by:

$$A_n = \sum_{k=1}^n \frac{(-1)^{k+1}}{k} = 1 - \frac{1}{2} + \frac{1}{3} - \frac{1}{4} + \cdots + \frac{(-1)^{n+1}}{n}$$

Denote H_∞ and A_∞ as the infinite sum of the above series.

(i) By considering upper rectangles for the area under $y = \frac{1}{x}$, show that:

2

$$H_n > \ln n$$

(ii) Hence, show that the harmonic series diverges i.e H_∞ grows without bound

1

(iii) Show that for $x \geq 0$:

1

$$\frac{x^n}{1+x} \leq x^{n-1}$$

(iv) Prove that for $x \neq -1$:

3

$$\frac{1}{1+x} = \frac{(-1)^n x^n}{1+x} + \sum_{k=0}^{n-1} (-x)^k$$

(v) Hence, show that:

3

$$A_n = \ln 2 - (-1)^n \int_0^1 \frac{x^n}{1+x} dx$$

(vi) Hence, by showing that:

2

$$\ln 2 - \int_0^1 \frac{x^n}{1+x} dx \leq A_n \leq \ln 2 + \int_0^1 \frac{x^n}{1+x} dx$$

prove that $A_\infty = \ln 2$



Question 144 (2 marks)

BoS '20 Q15b

Show that for $x > 0$

$$0 < \ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}$$

Question 145 (7 marks)

Abbotsleigh '24 Q16a

Given $x \leq 0$

(i) Prove that $1 - x \leq \frac{1}{1+x} \leq 1$ 2

(ii) For $x > 0$, show that $1 - \frac{1}{2x} \leq x \ln x \left(1 + \frac{1}{x}\right) \leq 1$. 3

(iii) Hence deduce that $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x}\right)^x = e$. 2

Question 146 (5 marks)

BBHS '24 Q16 c

Consider the function $f(x) = \frac{(n+1+x)^{n+1}}{(n+x)^n}$, $x > 0$, where $n \geq 1$ is a fixed positive integer.

(i) Show that for $x > 0$, $f(x)$ is an increasing function. 2

(ii) Hence show that $\left(1 + \frac{x}{n+1}\right)^{n+1} > \left(1 + \frac{x}{n}\right)^n$ 2

(iii) Deduce that $(n+2)^{n+1}n^n > (n+1)^{2n+1}$ 1

Question 147 (5 marks)

Ruse '24 Q14d

(i) Show that, for $x \in \mathbb{R}$ such that $|x| < 1$, 1

$$\sum_{k=0}^n (-1)^k x^k = \frac{1}{1+x} - \frac{(-1)^{n+1} x^{n+1}}{1+x}$$

(ii) Prove that 2

$$0 \leq \int_0^1 \frac{x^{n+1}}{1+x} dx \leq \frac{1}{n+2}$$

(iii) By integrating part (i) and using the result in part (ii), show that 2

$$\lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^k}{k+1} = \ln 2$$



Question 148 (5 marks)

Baulko '22 Q13d

Consider the geometric series

$$S_{2n} = 1 - h + h^2 - \dots + h^{2n}, \quad 0 < h < 1$$

(i) Show that $S_{2n-1} < \frac{1}{1+h} < S_{2n}$ 2

(ii) Integrate the previous result between $h = 0$ and $h = x$, where $0 < x < 1$, and hence write down a polynomial for $\ln(1+x)$ 2

(iii) Use $n = 3$ to estimate the value of $\ln \frac{5}{4}$ to 4 significant figures. Show full working. 1

Question 149 (4 marks)

BBHS '24 Q16 b

If a function $f(x)$ is continuous for $a \leq x \leq b$ and differentiable:

(i) Show that $\left| \int_a^b f(x) dx \right| \leq \int_a^b |f(x)| dx$ 1

(ii) Hence prove that $\left| \int_0^\pi 27^x \cos x dx \right| \leq \frac{3^{3\pi} - 1}{3 \ln 3}$ 3

Question 150 (5 marks)

NSG '13 Q16d

Consider $f(x) = \log x - x + 1$.

(i) Show that $f(x) \leq 0$ for all $x > 0$. 2

(ii) Consider the set of n positive numbers $p_1, p_2, p_3, \dots, p_n$ such that 1

$$p_1 + p_2 + p_3 + \dots + p_n = 1$$

By using the result in part (i), deduce that

$$\sum_{r=1}^n \log(np_r) \leq np_1 + np_2 + np_3 + \dots + np_n - n$$

(iii) Show that $\sum_{r=1}^n \log np_r \leq 0$. 1

(iv) Hence deduce that $0 < n^n p_1 p_2 p_3 \dots p_n \leq 1$. 1



Question 151 (6 marks)

Knox '15 Q16c

Suppose that x is a positive real number.

(i) Show that $\frac{1}{1+t^2} < 1 - t^2 + t^4 - t^6 + \dots + t^{4n} < \frac{1}{1+t^2} + t^{4n+2}$ for $0 < t < x$ 3

(ii) Hence show that 1

$$\tan^{-1} x < x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots + \frac{x^{4n+1}}{4n+1} < \tan^{-1} x + \frac{x^{4n+3}}{4n+3}$$

(iii) Explain why $\tan^{-1} x = x - \frac{x^3}{3} + \frac{x^5}{5} - \frac{x^7}{7} + \dots$ for $0 \leq x \leq 1$. 1

(iv) Deduce that $\pi = 4 \times \left(1 - \frac{1}{3} + \frac{1}{5} - \frac{1}{7} + \dots\right)$ 1

Question 152 (8 marks)

Ruse '22 AT3 Q8b

Suppose $0 \leq t \leq \frac{1}{\sqrt{2}}$.

(i) Show that $0 \leq \frac{2t^2}{1-t^2} \leq 4t^2$. 2

(ii) Hence show that $0 \leq \frac{1}{1+t} + \frac{1}{1-t} - 2 \leq 4t^2$. 2

(iii) By integrating the result in part (ii) from $t = 0$ to $t = x$ (where $0 \leq x \leq \frac{1}{\sqrt{2}}$), show that $0 \leq$ 2

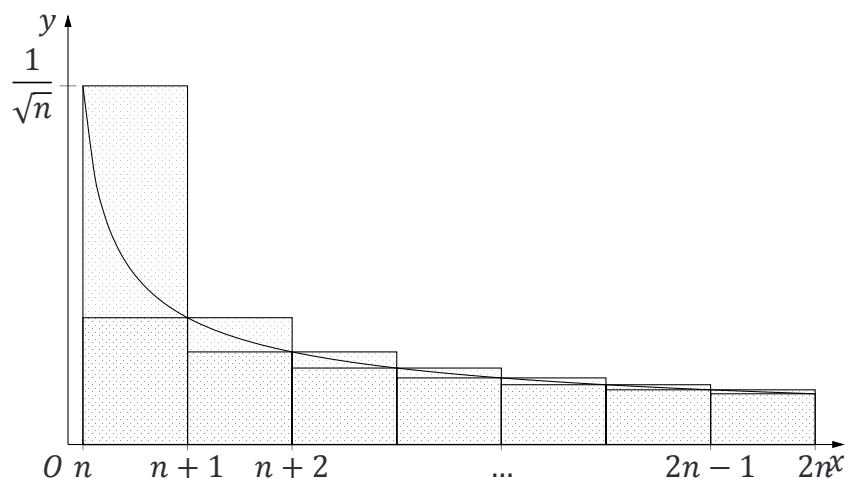
$$\ln\left(\frac{1+x}{1-x}\right) - 2x \leq \frac{4x^3}{3}$$

(iv) Hence show that for $0 \leq x \leq \frac{1}{\sqrt{2}}$, $1 \leq \left(\frac{1+x}{1-x}\right)e^{-2x} \leq e^{\frac{4x^3}{3}}$. 2



Question 153 (7 marks)Baulko '20 Q16a¹⁸

In the diagram below, the diagram of $y = \frac{1}{\sqrt{x}}$ has been drawn, and n rectangles have been constructed between $x = n$ and $x = 2n$, each of width 1 unit.



(i) Show that $\int_n^{2n} \frac{dx}{\sqrt{x}} = 2\sqrt{n}(\sqrt{2} - 1)$

1

(ii) Let $S_n = \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n}}$, show that

4

$$2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} < S_n < 2\sqrt{n}(\sqrt{2} - 1)$$

(iii) Hence find, correct to three decimal places:

2

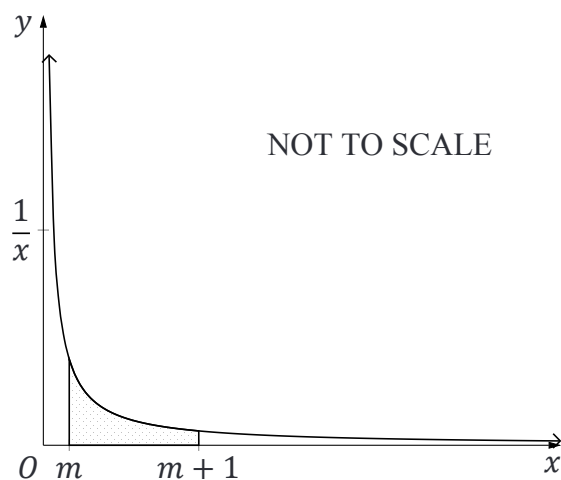
$$\frac{1}{\sqrt{1000000}} + \frac{1}{\sqrt{1000001}} + \frac{1}{\sqrt{1000002}} + \dots + \frac{1}{\sqrt{2000000}}$$



Question 154 (6 marks)

Ruse '20 Q13a

Consider the function $f(x) = \frac{1}{x}$ for $x > 0$.



- (i) Let m be a positive real number, $m \geq 1$. By integrating $f(x)$ between m and $m + 1$, show that

2

$$\frac{1}{m+1} < \ln(m+1) - \ln m < \frac{1}{m}$$

- (ii) Let m take on successively the values $1, 2, \dots, n-1$. Hence show that

4

$$\ln(k+1) < S_k < 1 + \ln k$$

where

$$S_k = 1 + \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$$



Question 155 (8 marks)

Normo '23 AT3 Q2

- (a) (i) Show that, for
- $x > 1$

1

$$0 < \frac{1}{x\sqrt{x}} < \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}}$$

- (ii) Hence show that, for
- $t \geq 1$
- ,

2

$$\frac{2(\sqrt{t} - 1)}{\sqrt{t}} \leq \ln t \leq 3(\sqrt[3]{t} - 1)$$

- (iii) Hence find
- $\lim_{t \rightarrow \infty} \frac{\ln t}{\sqrt{t}}$
- .

2

- (b) Consider the function
- $f(x) = \tan^{-1} x$
- .

- (i) Show that
- $f''(x) < 0$
- for
- $x > 0$
- .

1

- (ii) Hence, by considering the concavity of
- $f(x)$
- , or otherwise, show that for
- $a > b > 0$
- :

2

$$\tan^{-1}(a - b) + \tan^{-1}(a + b) < 2 \tan^{-1} a$$

Question 156 (10 marks)

HGHS '24 Q16b

Let $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx$ and let $J_n = (-1)^n I_{2n}$

- (i) Show that
- $I_n + I_{n+2} = \frac{1}{n+1}$
- .

2

- (ii) Find the value of
- $I_4 = \int_0^{\frac{\pi}{4}} \tan^4 x \, dx$
- .

2

- (iii) Deduce that
- $J_n - J_{n-1} = \frac{(-1)^n}{2n-1}$
- for
- $n \geq 1$
- .

1

- (iv) By considering

2

$$J_m = (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3} + \cdots) + (J_1 - J_0) + J_0$$

, show that

$$J_m = \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1} \quad \text{for } n \geq 1$$



(v) Use the substitution $u = \tan x$ to show that $I_n = \int_0^1 \frac{u^n}{1+u^2} du$.

1

(vi) Deduce that $0 \leq I_n \leq \frac{1}{n+1}$ and conclude that $I_n \rightarrow 0$ as $n \rightarrow \infty$.

2

Question 157 (4 marks)

NSG '22 Q16b ii

(i) By finding the equation of the tangent to $y = \log_e x$ at $x = 1$, explain why

1

$$\log_e x \leq x - 1 \quad \text{for } x > 0.$$

(ii) If $a_1, a_2, a_3, \dots, a_n$ and $b_1, b_2, b_3, \dots, b_n$ are positive numbers such that $a_1 + a_2 + a_3 + \dots + a_n = 1$ and $a_1 b_1 + a_2 b_2 + a_3 b_3 + \dots + a_n b_n = 1$, prove that

3

$$b_1^{a_1} b_2^{a_2} b_3^{a_3} \dots b_n^{a_n} \leq 1.$$



Solutions

Language of Proof

1 (Copium): **B.**

If 3 divides $a - b$, let $a - b = 3k$. Then

$$a^3 - b^3 = (a - b)(a^2 + ab + b^2) = 3[k(a^2 + ab + b^2)]$$

which is divisible by 3 because $k(a^2 + ab + b^2) \in \mathbb{Z}$. Therefore the statement is correct.

If 3 divides $a^3 - b^3$, let it be $3m$:

$$3m = a^3 - b^3 = (a - b)((a - b)^2 + 3ab) = (a - b)^3 + 3ab(a - b)$$

so $(a - b)^3 = 3[m - ab(a - b)]$ which is divisible by 3 because $m - ab(a - b) \in \mathbb{Z}$. Thus $a - b$ is divisible by 3, so the converse is also correct.

2 (Baulko '20 Q6): **A.**

If the statement is false, then there must exist a day where it is not the case that Max does at least one problem. The complementary event to doing at least one problem, is to do no problems at all. So it must certainly be true that there exists a day next week the Max does the complementary event

3 (Girra '20 Q6): **A.**

- A. For all positive integers a , a^3 is indeed positive.
- B. $\sqrt[3]{a}$ exists for all a , unless a is a perfect cube, $\sqrt[3]{a}$ is not an integer.
- C. Obviously false.
- D. Domain of a unspecified; a can be negative, etc, so this is false.

4 (Ruse '20 Q2): **D.**

Let T (for treasure) be the statement “you have treasure” and let K (for kingdom) be the statement “I have a kingdom”. The original statement is now $\neg T \rightarrow \neg K$. The contrapositive of $\neg T \rightarrow \neg K$ is $K \rightarrow T$, which is D.

5 (Baulko '24 Q9): **B.**

In $P \Rightarrow Q$, P is a sufficient condition for Q . Q is a necessary condition for P . $\therefore P : a^2 + b^2 = c^2$
 $Q : \triangle ABC$ is right angled, hence B is correct.

6 (Ruse '24 Q3): **C.**

“All ravens are black.” must be false because the existence of even one non-black raven disproves this statement.



7 (Ruse '24 Q6): **B**.

- A. $\sim B \Rightarrow \sim A$ is the contrapositive and is always true.
- B. A and $\sim B$ is false since $A \Rightarrow B$ implies B is true if A is true.

8 (NSB '24 Q1): **C**.

- A. Equivalent
- B. Rewording
- C. The converse of the statement, which is not equivalent
- D. The original implication

9 (NSB '24 Q4): **B**.

- A. This keeps the same quantifier order as the original statement, so it is not correct.
- B. a
- C. This says that for every pair (x, y) the equation fails, which is much stronger than the correct negation

10 (NSG '24 Q9): **D**.

- A. Clearly true.
- B. Translated: For every integer n , I can find an integer m larger than it.
- C. Translated: I can find an a [$a \neq 0$] such that every number times a is zero.
- D. Translated: I can find an integer m such that **every** integer is 5 more than m .

11 (BoS '21 Q3): **D**.

- A. implies that a negative value is greater than a positive value, which is false.
- B. implies that a negative value (xy) is greater than 1, which is false.
- C. implies that a positive value is less than a negative value, which is false.
- D. implies that a negative value (xy) is less than 1, which is true.



12 (SGHS '21 Q9): C.

“‘If P then Q ’ is equivalent to ‘If not Q then not P ’. So the claim is equivalent to ‘If $(p - 1)! + 1$ is divisible by p , then p is prime’. A counter-example is therefore a composite number such that $(p - 1)! + 1$ is divisible by p .

13 (Normo '22 Q6): D.

14 (Normo '22 Q9): B.

15 (SGHS '22 Q9): C.

We want $a + b$ multiplied by something to give $ac - bd$. This motivates the expansion

$$(a + b)(c - d) = ac - bd + bc - ad$$

We know that $ac - bd$ is divisible by $a + b$, and we can see that $(a + b)(c - d)$ is divisible by $a + b$. It follows that $bc - ad$ must also be divisible by $a + b$

16 (SGHS '22 Q1): D.

The converse of ‘If P then Q ’ is ‘If Q then P ’. Therefore the converse of ‘If $a < b$ then $a^3 < b^3$ ’ is ‘If $a^3 < b^3$ then $a < b$ ’.

18 (SGS '24 Q8): A.

$p \Rightarrow q$ is false (3 uppermost). Negation: $p \wedge \sim q$ is true $\sim (p \Rightarrow q)$ Converse: $q \Rightarrow p$ is true as 6 is divisible by 3.

19 (SGS '24 Q10): A.

$$\begin{aligned}\forall y, \exists x : x^2 - y^2 &= x \\ x^2 - x &= y \\ \left(x - \frac{1}{2}\right)^2 &= y^2 + \frac{1}{4}\end{aligned}$$

RHS $\geq \frac{1}{4} \forall y$, so always solutions for x . Not the case for others.

20 (SGS '22 Q10): B.

The proof states:

“If a, b, c are 1 more than a multiple of 3, then abc is 1 more than a multiple of 3.”

Contrapositive: If abc is *not* 1 more than a multiple of 3, then *not all* of a, b, c are 1 more than a multiple of 3.

D fits this structure.

Question 21 (1 mark) (SGGHS '20 Q11d)

If the set of all counter examples $x \in \mathbb{R}, x^2 > 2$, then $x > 3$. If $x = 3$, then there does not exist a **real** y such that $x^2 + y^2 = 2$, i.e no $y \in \mathbb{R}$ exists that satisfies $x^2 + y^2 = 2$. $\therefore$ false statement.



Question 22 (2 marks) (Killara '24 Q13a)

Consider the set $\{3, 5\}$.

$$\begin{aligned} 3 \times 5 + 1 &= 16 \\ &= 4 \times 4 \end{aligned}$$

$\therefore$ The statement is refuted by a counter example.

Question 23 (3 marks) (NSB '24 Q11a)

Consider the set $\{3, 5\}$.

$$\begin{aligned} 3 \times 5 + 1 &= 16 \\ &= 4 \times 4 \end{aligned}$$

which is not prime. The statement is refuted by counter example.

Question 24 (10 marks) (Ascham '22 Q12a)

(i) If $\sin A = k$ then $\sin^{-1} k = A$.

(ii) If $\sin^{-1} k \neq A$ then $\sin A \neq k$.

(iii) Not all A such that $\sin A = k$ means $\sin^{-1} k = A$ or $\exists \sin A = k$ such that $\sin^{-1} k \neq A$.

(iv) Equivalence if $A \Rightarrow B$ and $B \Rightarrow A$ are both true. $A \Rightarrow B$ is NOT true ($\sin \frac{5\pi}{6} = \frac{1}{2}$ but $\sin^{-1} \frac{1}{2} = \frac{\pi}{6}$) but $B \Rightarrow A$ is true. $\therefore$ Not an equivalence.



Direct Proof

Question 25 (2 marks) (BBHS '24 Q14b)

Let the 3 consecutive numbers be $n, n + 1, n + 2$.

$$\begin{aligned}n^2 + (n + 1)^2 + (n + 2)^2 &= n^2 + n^2 + 2n + 1 + n^2 + 4n + 4 \\&= 3n^2 + 6n + 5 \\&= 3n^2 + 6n + 6 - 1 \\&= 3(n^2 + 2n + 3) - 1\end{aligned}$$

$\therefore$ This is one less than a multiple of 3.

Question 26 (2 marks) (Caringbah '24 Q11c)

Let two consecutive numbers be $k, (k + 1)^2, k \in \mathbb{Z}$. Then,

$$\begin{aligned}(k + 1)^2 - k^2 &= k^2 + 2k + 1 - k^2 \\&= 2k + 1\end{aligned}\quad (\text{which is odd})$$

$\therefore$ the difference between two consecutive square numbers is odd.

Question 27 (3 marks) (Baulko '24 Q16c)

$$\begin{aligned}&\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \frac{1}{\sqrt{5} + \sqrt{6}} + \dots + \frac{1}{\sqrt{2023} + \sqrt{2024}} \\&> \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{4} + \sqrt{5}} + \frac{1}{\sqrt{6} + \sqrt{7}} + \dots + \frac{1}{\sqrt{2024} + \sqrt{2025}} \\&\therefore 2 \left(\frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \frac{1}{\sqrt{4} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{6}} + \dots + \frac{1}{\sqrt{2023} + \sqrt{2024}} \right) \\&> \frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{2} + \sqrt{3}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \frac{1}{\sqrt{4} + \sqrt{5}} + \dots + \frac{1}{\sqrt{2023} + \sqrt{2024}} + \frac{1}{\sqrt{2024} + \sqrt{2025}} \\&= \frac{\sqrt{2} - 1}{1} + \frac{\sqrt{3} - \sqrt{2}}{1} + \frac{\sqrt{4} - \sqrt{3}}{1} + \frac{\sqrt{5} - \sqrt{4}}{1} + \dots + \frac{\sqrt{2024} - \sqrt{2023}}{1} + \frac{\sqrt{2025} - \sqrt{2024}}{1} \\&= \sqrt{2025} - 1 \\&= 45 - 1 \\&= 44 \\&\therefore \frac{1}{1 + \sqrt{2}} + \frac{1}{\sqrt{3} + \sqrt{4}} + \frac{1}{\sqrt{4} + \sqrt{5}} + \frac{1}{\sqrt{5} + \sqrt{6}} + \dots + \frac{1}{\sqrt{2023} + \sqrt{2024}} > 22\end{aligned}$$



Question 28 (2 marks) (HAHS '24 Q13a)

Let p be even, so that $p = 2k$ where $k \in \mathbb{Z}$. Then,

$$\begin{aligned} p^2 &= (2k)^2 \\ &= 4k^2 \\ &= 2(2k^2) \end{aligned}$$

Which is even. Hence, if p is even, then p^2 is even. Proving the converse: Let q^2 be even, such that $q^2 = 2k$ where $k \in \mathbb{Z}$. Then,

$$\frac{q \times q}{2} = k$$

So 2 divides either q or q , that is 2 divides q as k is an integer. Therefore, a number is even if and only if the number squared is even.

Question 29 (5 marks) (Ascham '24 Q15a)

(i) Consider

$$\begin{aligned} \tan^{-1}(k+1) &= \alpha \\ \tan^{-1}(k) &= \beta \\ \text{then, } \tan(\alpha - \beta) &= \frac{\tan \alpha - \tan \beta}{1 + \tan \alpha \tan \beta} \\ &= \frac{k+1 - k}{1 + (k+1)k} \\ \therefore \alpha - \beta &= \tan^{-1}\left(\frac{1}{k(k+1)}\right) \end{aligned}$$

(ii) Consider LHS:

$$\lim_{n \rightarrow \infty} \left[\tan^{-1}\left(\frac{1}{1.2+1}\right) + \tan^{-1}\left(\frac{1}{2.3+1}\right) + \tan^{-1}\left(\frac{1}{3.4+1}\right) + \dots + \tan^{-1}\left(\frac{1}{n(n+1)+1}\right) \right] =$$

Question 30 (5 marks) (Penrith '24 Q14b)

(i) $y, y', y'' = e^x$, which is positive for $x \in \mathbb{R}$

(ii) $\left(e^{\frac{a}{2}} - e^{\frac{b}{2}}\right)^2 > 0$ as e^x is always positive. For $a > b$, $e^a > e^b$ from (i). Therefore, $e^a - 2e^{\frac{a}{2}}e^{\frac{b}{2}} + e^b > 0$. Rearranging, $\frac{1}{2}(e^a + e^b) > e^{\frac{a}{2}}e^{\frac{b}{2}}$.



(iii) Just cycle lmao

$$\begin{aligned}\frac{1}{2}(e^a + e^b) &> e^{\frac{a}{2}}e^{\frac{b}{2}} && \text{(from (i))} \\ \frac{1}{2}(e^c + e^d) &> e^{\frac{c}{2}}e^{\frac{d}{2}} && (a > b > c > d) \\ \frac{1}{2}(e^a + e^b) + \frac{1}{2}(e^c + e^d) &> 2\left(e^{\frac{1}{2}(a+b)} + e^{\frac{1}{2}(c+d)}\right) \\ &> 2\left(2e^{\frac{1}{2}\frac{(a+b+c+d)}{2}}\right)\end{aligned}$$

by re-applying part (i), where new $a = \frac{1}{2}(a+b)$ and new $b = \frac{1}{2}(c+d)$

Question 31 (3 marks) (NSB '21 Q14a)

Let $y = \frac{(x+k)^2}{x^2+x+1}$. Rearranging it gives

$$(y-1)x^2 + (y-2k)x + (y-k^2) = 0$$

This is a quadratic equation in x , $\forall x \in \mathbb{R}, \Delta \geq 0$

$$\begin{aligned}(y-2)^2 - 4(y-1)(y-k^2) &\geq 0 \\ 3y^2 - 4(k^2 - k + 11y) &\leq 0 \\ y &\leq \frac{4}{3}(k^2 - k + 1)\end{aligned}$$

Question 32 (2 marks) (Normo '22 Q14a)

$$\begin{aligned}(5a^2 - 4b)^2 &\geq 0 \\ 25a^4 - 40a^2b + 16b^2 &\geq 0 \\ 25a^4 + 16b^2 &\geq 40a^2b \\ \therefore \frac{5}{8}a^4 + \frac{2}{5}b^2 &\geq a^2b\end{aligned}$$

Question 33 (2 marks) (Knox '23 Q16a)

Let the three consecutive integers be $2n, 2n+2, 2n+4$ where n is a positive integer, and let S be the sum of these three consecutive integers.

$$\begin{aligned}S &= (2n)^2 + (2n+2)^2 + (2n+4)^2 \\ &= 4n^2 + 4n^2 + 8n + 4 + 4n^2 + 16n + 16 \\ &= 12n^2 + 24n + 20 \\ &= 12n^2 + 24n + 12 + 8 \\ &= 12(n^2 + 2n + 1) + 8 \\ &= 12(n+1)^2 + 8 \\ &= 12M + 8 \\ M &= (n+1)^2 \in \mathbb{N}\end{aligned}$$



From the result above we can see that the sum of any three consecutive even integers can be divided by 12 and leave a remainder of 8.

Question 34 (4 marks) (Knox '23 Q15a)

From the statement $3^x > x + \ln x$:

$$\begin{aligned} x = 1 &\rightarrow 3^1 > 1 + \ln 1 \\ x = 2 &\rightarrow 3^2 > 2 + \ln 2 \\ x = 3 &\rightarrow 3^3 > 3 + \ln 3 \\ &\vdots \\ x = n &\rightarrow 3^n > n + \ln n \end{aligned}$$

Summing LHS,

$$3^1 + 3^2 + \dots + 3^n = \frac{3(3^n - 1)}{3 - 1} = \frac{3^{n+1} - 3}{2}$$

Summing RHS,

$$\begin{aligned} 1 + 2 + 3 + \dots + n &= \frac{n}{2}(n + 1) = \frac{1}{2}(n^2 + n) \\ \ln 1 + \ln 2 + \dots + \ln n &= \ln(1 \times 2 \times 3 \times \dots \times n) = \ln n! \end{aligned}$$

Hence by summing all the statements for $x = 1$ to $x = n$:

$$\begin{aligned} \frac{3^{n+1} - 3}{2} &> \frac{1}{2}(n^2 + n) + \ln n! \\ 3^{n+1} - 3 &> n^2 + n + 2 \ln n! \\ 3^{n+1} &> n^2 + n + 3 + 2 \ln n! \end{aligned}$$

Question 35 (2 marks) (Knox '22 Q16d)

$$\begin{aligned} abcd + efgh &= abcd - cdgh + cdgh + efgh - abef + abef \\ &= cd(ab - gh) + cdgh - ef(ab - gh) + abef \\ &= (ab - gh)(cd - ef) + abef + cdgh \end{aligned} \tag{1}$$

Now, as $ab - gh$ is a divisor of $abef + cdgh$ then

$$abef + cdgh = M(ab - gh) \tag{2}$$

, where M is a pronumeral; though possibly an integer.

$$\begin{aligned} (2) \rightarrow (1) : abcd + efgh &= (ab - gh)(cd - ef) + M(ab - gh) \\ &= (ab - gh)(cd - ef + M) \end{aligned}$$

Hence, $ab - gh$ is a divisor of $abcd + efgh$.



Contradiction & Contrapositive

36 (HGHS '20): A.

37 (Girra '20 Q5): D.

40 (Knox '23 Q5): C.

For a statement “For all, if P then Q ”, its contrapositive statement is “For all, if not Q then not P ”.

So as the statement is “For all quadrilaterals, if the four angles are right angles, then the quadrilateral is a rectangle”, the contrapositive statement would be “For all quadrilaterals, if the quadrilateral is not a rectangle, then the four angles are not right angles”.

Question 41 (3 marks) (SGHS '21 Q12d)

(i) If a and b are rational, then $a + b$ is rational.

(ii) **Converse of P :**

If at least one of a and b is irrational, then $a + b$ is irrational.

This converse is false. For a counter-example, consider $a = \sqrt{2}$ and $b = -\sqrt{2}$. Then, at least one of a and b is irrational, while $a + b = \sqrt{2} - \sqrt{2} = 0$ is rational. Other counter examples are $a = \pi, b = -\pi$ or $a = e, b = -e$, etc.

Question 42 (3 marks) (SGS '21 Q12a)

Assume $\frac{p}{q} + \frac{q}{p} < 2$

$$\begin{aligned}p^2 + q^2 &< 2pq \\p^2 - 2pq + q^2 &< 0 \\(p - q)^2 &< 0\end{aligned}$$

This is a contradiction, since a square is non-negative by definition.

Hence $\frac{p}{q} + \frac{q}{p} \geq 2$ using proof by contradiction.

Question 43 (3 marks) (Baulko '23 Q11e)

(i) If $x < 0$, then $x^5 + 4x^3 + 5x < 8x^4 + 2x^2 + 17$.

(ii) If $x < 0$ then $x^5 + 4x^3 + 5x < 0$.

If $x < 0$ then $8x^4 + 2x^2 + 17 > 0$. $\therefore 8x^4 + 2x^2 + 16 > x^5 + 4x^3 + 5x$.



Question 44 (3 marks) (NSG '20 Q16a)

Assume $\exists a, b \in \mathbb{Z}^+ : a^2 - 4b = 2$.

$$\begin{aligned}\text{Then } a^2 &= 4b + 2 \\ &= 2(2b + 1)\end{aligned}$$

$\therefore a^2$ is even.

So, $a = 2k$ for some $k \in \mathbb{Z}^+$.

$$\begin{aligned}(2k)^2 &= 2(2b + 1) \\ 2k^2 &= 2b + 1\end{aligned}$$

LHS= $2k^2$ is even and RHS= $2b + 1$ is odd. There is a contradiction, hence the assumption is incorrect.
So $\forall a, b \in \mathbb{Z}, a^2 - 4b \neq 2$.

Question 45 (3 marks) (God folder)

Suppose for a contradiction that p has an integer root k . We have $4k^3 + 2k^2 - 6k + 1 = 0$.

Then $2(2k^3 + k^2) = 2(-3k) - 1$. We have an even equal to an odd.

Question 46 (4 marks) (Ruse '20 AT2 Q10c)

Assume that $\frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2-c^2}} \leq 0$

$$\begin{aligned}\frac{a+c}{\sqrt{a^2+c^2}} &\leq \frac{b+c}{\sqrt{b^2-c^2}} \\ (a+c)(\sqrt{b^2-c^2}) &\leq (b+c)(\sqrt{a^2+c^2}) \\ (a+c)^2(b^2+c^2) &\leq (b+c)^2(a^2+c^2) \\ (a^2+2ac+c^2)(b^2+c^2) &\leq (b^2+2bc+c^2)(a^2+c^2) \\ a^2b^2+2ab^2c+c^2b^2+a^2c^2+2ac^3+c^4 &\leq a^2b^2+2a^2bc+a^2c^2+b^2c^2+2bc^3+c^4 \\ 2ab^2c+2ac^3 &\leq 2a^2bc+2bc^3 \\ ab^2c+2ac^3 &\leq 2a^2bc+2bc^3 \\ ab^2+ac^2 &\leq a^2b+bc^2 \\ ab^2+ac^2-a^2b-bc^2 &\leq 0 \\ ab(b-a)+c^2(a-b) &\leq 0 \\ (a-b)(c^2-ab) &\leq 0\end{aligned}$$

As $a > b \therefore (a-b) > 0$ and $c^2 > ab \therefore (c^2-ab) > 0$.

Hence

$$(a-b)(c^2-ab) > 0.$$

$\therefore$ This is a contradiction, hence $\frac{a+c}{\sqrt{a^2+c^2}} - \frac{b+c}{\sqrt{b^2-c^2}} > 0$



Question 47 (3 marks) (Manly '24 Q12e)

The contrapositive statement is:

“If **none** of a, b, c are divisible by 3 then $a^3 + b^3 + c^3$ is not divisible by 9”.

If each of a, b and c are not divisible by 3 then each one is of the form $3k + 1$ or $3k + 2$.

Hence, a^3, b^3 and c^3 end in either 1 or 8.

$\therefore a^3 + b^3 + c^3$ ends in either 3, 10, 17 or 24 $\Rightarrow$ no common factor of 9 exists.

$\therefore a^3 + b^3 + c^3$ is not divisible by 9.

$\therefore$ The statement is true by contradiction.

Question 48 (3 marks) (SGGHS '24 Q14d)

Assume $\exists a \geq 2$ such that $\sqrt{a} + \sqrt{a+2} \leq \sqrt{a+8}$

$$\begin{aligned}\therefore (a + \sqrt{a+2})^2 &\leq (\sqrt{a+8})^2 \\ a + 2\sqrt{a(a+2)} + a + 2 &\leq a + 8 \\ 2a + 2\sqrt{a^2 + 2a} + 2 &\leq a + 8 \\ a + 2\sqrt{a^2 + 2a} - 6 &\leq 0 \\ (2\sqrt{a^2 + 2a})^2 &\leq (6 - a)^2 \\ 4(a^2 + 2a) &\leq 36 - 12a + a^2 \\ 4a^2 + 8a &\leq 36 - 12a + a^2 \\ 3a^2 + 20a - 36 &\leq 0 \\ a &= 1.47, -8.14\end{aligned}$$

So, the negation is false.

$\therefore$ By contradiction, $\sqrt{a} + \sqrt{a+2} > \sqrt{a+8}$

Question 49 (1 mark) (Copium '23 Q13 a) iv))

Assume $ab + cd < ad + bc$. Then $ab + cd - ad - bc < 0$ so

$$a(b - d) - c(b - d) < 0 \Rightarrow (a - c)(b - d) < 0$$

But $a \geq c$ and $b \geq d$ implies that $a - c, b - d \geq 0$ giving $(a - c)(b - d) \geq 0$, a contradiction. Thus the original statement is true.

- **1 mark** Writing assumption and obtaining contradiction.
- **Comment:** It is good practice to restate the original statement is true after obtaining a contradiction.



Question 50 (4 marks) (Ruse '22 AT2 Q8c)

- (i) Let $P(x)$ mean ' x is P '.

Then for $a, b \in \mathbb{Z}$, the statement as given can be expressed as

$$\text{odd}(a^2(b^2 - 2b)) \Rightarrow \text{odd}(a) \wedge \text{odd}(b)$$

The contrapositive is

$$\begin{aligned} \neg(\text{odd}(a) \wedge \text{odd}(b)) &\Rightarrow \neg \text{odd}(a^2(b^2 - 2b)) \\ \equiv \neg \text{odd}(a) \vee \neg \text{odd}(b) &\Rightarrow \neg \text{odd}(a^2(b^2 - 2b)) \end{aligned}$$

which is equivalent to

$$\text{even}(a) \vee \text{even}(b) \Rightarrow \text{even}(a^2(b^2 - 2b))$$

"If either a or b is even, then $a^2(b^2 - 2b)$ is even."

So if at least one of a or b is even, then the given expression is true.

- (ii) Suppose a is even and b is any integer. Then

$$a = 2k \quad \exists k \in \mathbb{Z}$$

We have,

$$\begin{aligned} a^2(b^2 - 2b) &= 4k^2(b^2 - 2b) \\ &= 2(2k^2(b^2 - 2b)) \end{aligned}$$

Since $\mathbb{Z}$ is closed under $+$, $\times$, $-$, it follows that $2k^2(b^2 - 2b) \in \mathbb{Z}$.
Hence $2|a^2(b^2 - 2b)$ if a is even.

Now suppose that $b = 2k$ ($\exists k \in \mathbb{Z}$) and a is an integer. Then

$$\begin{aligned} a^2(b^2 - 2b) &= a^2(4k^2 - 4k) \\ &= 2(a^2(2k^2 - 2k)) \end{aligned}$$

Again, since $\mathbb{Z}$ is closed under the previously stated operations,

$$a^2(2k^2 - 2k) \in \mathbb{Z}$$

So

$$2|a^2(b^2 - 2b)$$

Hence, if at least one of a or b is even, then $a^2(b^2 - 2b)$ is even.
Therefore, if $a^2(b^2 - 2b)$ is odd, then both a is odd and b is odd.



Question 51 (3 marks) (Baulko '24 Q15c i)

Let r be an irrational number and that $\sqrt[3]{r} = \frac{p}{q}$ where $(p \wedge q \in \mathbb{Z}^+) \wedge q \neq 0$, i.e $\sqrt[3]{r}$ is rational and also p and q are coprime.

$$r = \frac{p^3}{q^3}$$
$$rq^3 = p^3$$

$\therefore r$ is a factor of p^3 . However, rq^3 is irrational as r is irrational, yet p^3 is rational ($p \in \mathbb{Z}$).
 $\therefore \sqrt[3]{r}$ is irrational, by contradiction.

Question 52 (4 marks) (Baulko '23 Q14a)

Suppose, by contradiction, that positive integers p and q do exist such that $4p^2 - q^2 = 25$. Then, $(2p + q)(2p - q) = 25$.

The factors of 25 are 1,5,25.

Since $q \neq 0$ (q is a positive integer), then the factors must be 1 and 25 as they are distinct.

$$(2) - (1) : 2q = 24$$

$$q = 12$$

$$2p - q = 1 \text{ (1) and } 2p + q = 25 \text{ (2).} \quad \Rightarrow \quad 2p + q = 25 \quad \therefore \text{a contradiction exists since } p \text{ is a positive}$$
$$2p = 13$$

$$p = 6\frac{1}{2}$$

integer. $\therefore$ there are no positive integers p and q such that $4p^2 - q^2 = 25$.

Question 53 (2 marks) (NSB '24 Q14c)

Assume $a \leq b \leq c$.

$$c^2 < c^2 + a + b \leq c^2 + 2c < (c + 1)^2$$
$$c^2 + a + b \leq (c + 1)^2$$

$c^2 + a + b$ cannot be perfect square, similarly for the other two.

Question 54 (3 marks) (Killara '24 Q14a)

Assume $\exists A \in \mathbb{Z}$ with 2 unique prime factorisations, that is:

$$A = p_1 p_2 p_3 \dots p_i \quad \text{where } p_i \text{ is prime}$$

$$A = q_1 q_2 q_3 \dots q_i \quad \text{where } q_i \text{ is prime}$$

$\therefore p_1 p_2 p_3 \dots p_i = q_1 q_2 q_3 \dots q_i$ are equal and A . As both sides are divisible by p_1 , and all q_i are prime, then one of q_i must be q_1 .

$\therefore p_1 p_3 \dots p_i = q_2 q_3 \dots q_i$, since $p_2 = q_2$ and so on, so each p_i must match q_i , and therefore the prime factorisations are the same, a contradiction, so any integer A must only have one prime factorisation.



Question 55 (3 marks) (NSB '21 Q13c)

The contrapositive of the statement is:

“If a and b are not both even, then ab and $(a + b)$ are not both even.”

i.e, if either a or b is odd, then either ab or $a + b$ is odd.

Suppose a is odd. Then $a = 2k + 1$ for some integer k . Then, $ab = (2k + 1)b = 2kb + b$, which is **odd** if b is **odd**, even otherwise. And $a + b = 2k + 1 + b$ which is odd if b is even, and even otherwise. So regardless of whether b is even or odd, one of ab or $a + b$ is necessarily odd.

Question 56 (2 marks) (SGHS '22 Q16a)

Assume that there is at least one solution to

$$a^2 = 9b - 6$$

where a and b are both positive integers.

$$a^2 = 3(3b - 2)$$

Since a is a positive integer, a^2 must have prime factors in pairs. Hence, $(3b - 2)$ must have 3 as a factor so that a has 3^2 as a factor. Since b is a positive integer, $(3b - 2)$ is two less than a multiple of 3, and hence $(3b - 2)$ is NOT a multiple of 3. Hence, 3 is a factor of a^2 but 3^2 is not and this contradicts the assumption that a is a positive integer. $\therefore$ By contradiction, there is no solution to

$$a^2 = 9b - 6$$

where a and b are both positive integers.

Question 57 (3 marks) (SGHS '23 Q16a)

(i)

$$\begin{aligned} f'(x) &= \cos x \\ f''(x) &= -\sin x < 0 \quad (0 < x < \pi) \end{aligned}$$

Hence $f''(x)$ is concave down for $0 < x < \pi$.

(ii) Let $f(x) = \sin x, x_1 = A, x_2 = B, x_3 = C$:

$$\begin{aligned} \frac{\sin A + \sin B + \sin C}{3} &\leq \sin\left(\frac{A + B + C}{3}\right) \\ &= \sin\left(\frac{\pi}{3}\right) \\ &= \frac{\sqrt{3}}{2} \\ \therefore \sin A + \sin B + \sin C &\leq \frac{3\sqrt{3}}{2} \end{aligned}$$



Question 58 (8 marks) (Baulko '21 Q14c)

(i) When $n = 1$:

$$\begin{aligned} LHS &= S_1 & RHS &= 2 - 1 \\ &= \frac{1}{\sqrt{1}} & &= 1 \\ &= 1 \end{aligned} \tag{1}$$

$\therefore$ true for $n = 1$.

Assume true for $n = k$, $k \in \mathbb{Z}^+$, i.e $S_k \leq 2\sqrt{k} - 1$.

Prove true for $n = k + 1$, i.e $S_{k+1} \leq 2\sqrt{k+1} - 1$. Proof:

$$\begin{aligned} S_{k+1} &= S_k + \frac{1}{\sqrt{k+1}} \\ &\leq 2\sqrt{k} - 1 + \frac{1}{\sqrt{k+1}} && \text{(by assumption)} \\ &= \frac{2\sqrt{k(k+1)} + 1}{\sqrt{k+1}} - 1 \\ &= \frac{\sqrt{4k^2 + 4k + 1}}{\sqrt{k+1}} - 1 \\ &< \frac{\sqrt{4k^2 + 4k + 1 + 1}}{\sqrt{k+1}} - 1 && (4k^2 + 4k < 4k^2 - 4k + 1) \\ &= \frac{\sqrt{(2k+1)^2 + 1}}{\sqrt{k+1}} - 1 \\ &= \frac{2k+1+1}{\sqrt{k+1}} - 1 \\ &= \sqrt{2(k+1)} - 1 \end{aligned}$$

Hence, the result is true for $n = k + 1$, if it is true for $n = k$. Since the result is true for $n = 1$, then it is true $\forall n$ where $n \in \mathbb{Z}^+$ by induction.

(ii)

$$\begin{aligned} &(4k+1)\sqrt{k+1} - (4k+3)\sqrt{k} \\ &= \sqrt{(4k+1)^2(k+1)} - \sqrt{(4k+3)^2k} \\ &= \sqrt{16k^3 + 8k^2 + k + 16k^2 + 8k + 1} - \sqrt{16k^3 + 24k^2 + 9k} \\ &= \sqrt{16k^3 + 24k^2 + 9k + 1} - \sqrt{16k^3 + 24k^2 + 9k} \\ &> 0 && \text{(as } 1 > 0) \\ &\therefore (4k+1)\sqrt{k+1} > (4k+3)\sqrt{k} \forall k > 0 \end{aligned}$$



(iii)

$$\begin{aligned} S_n &\geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - c \\ c &\geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - S_n \\ &\geq 2\sqrt{n} + \frac{1}{2\sqrt{n}} - 2\sqrt{n} + 1 \quad (S_n \leq 2\sqrt{n} - 1) \\ &= \frac{1}{2\sqrt{n}} + 1 \end{aligned}$$

As n increases, $\frac{1}{2\sqrt{n}}$ will decrease, thus its largest value will occur when $n = 1$.

$$\begin{aligned} \therefore c &\geq \frac{1}{2} + 1 \\ &= \frac{3}{2} \end{aligned}$$

Question 59 (3 marks) (Normo '22 Q14d)

(i) If the equation $ax^2 + bx + c = 0$ has at least one integer solution, then at least one of a, b, c is an even integer.

(ii) If $ax^2 + bx + c = 0$ then at least one solution is either odd or even.

Case 1: Assume the equation has at least one odd integer solution: Let $x = 2m + 1$, where $m \in \mathbb{Z}$.

$$\begin{aligned} a(2m + 1)^2 + b(2m + 1) + c &= 0 \\ 4am^2 + 4am + a + 2bm + b + c &= 0 \\ a + b + c &= 2k \quad (\text{where } k = -2am^2 - 2am - 2bm) \end{aligned}$$

i.e. the sum of a, b, c is even since $k \in \mathbb{Z}$. Assume a, b, c are all odd: Let $a = 2A + 1, b = 2B + 1, c = 2C + 1$, where $A, B, C \in \mathbb{Z}$.

$$\begin{aligned} a + b + c &= (2A + 1) + (2B + 1) + (2C + 1) \\ &= 2p + 1 \quad (\text{where } p = A + B + C + 1) \end{aligned}$$

i.e. the sum of a, b, c is odd since $p \in \mathbb{Z}$. But this contradicts the previous result, and so a, b, c cannot all be odd. $\therefore$ By contradiction, if the equation has at least one odd integer solution, then at least one of a, b, c is even.

Case 2: Assume the integer has at least one even integer solution: Let $x = 2n$, where $n \in \mathbb{Z}$.

$$\begin{aligned} a(2n)^2 + b(2n) + c &= 0 \\ 4an^2 + 2bn + c &= 0 \\ &= 2q \quad (\text{where } q = -2an^2 - bn) \end{aligned}$$

c is even since $q \in \mathbb{Z}$. $\therefore$ If the equation has at least one even integer solution, then at least one of a, b, c is even. Hence by the contrapositive, if all a, b, c are odd integers, then the equation $ax^2 + bx + c = 0$ has no integer solutions.



Question 60 (7 marks) (NSG '23 Q16a)

(i)

$$\begin{aligned}\omega - 1^3 &= 0 \\ (\omega - 1)(\omega^2 + \omega + 1) &= 0 \\ \omega &= 1 \quad \text{or} \quad \omega^2 + \omega + 1 = 0\end{aligned}$$

As ω is non-real, $\omega^2 + \omega + 1 = 0$.

(ii) RTP:

“If n is divisible by 3 then $(x + 1)^{2n} + x^{2n} + 1$ is not divisible by $x^2 + x + 1$.”

For the sake of contradiction, assume that n is divisible by 3 and $(x + 1)^{2n} + x^{2n} + 1$ is divisible by $x^2 + x + 1$.

Then,

$$\begin{aligned}(x + 1)^{2n} + x^{2n} + 1 &= (x^2 + x + 1) Q(x) && (Q(x) \text{ is polynomial}) \\ (\omega + 1)^{2n} + \omega^{2n} + 1 &= (\omega^2 + \omega + 1) Q(\omega) && (\text{substitute } x = \omega) \\ &= 0 && (\omega^2 + \omega + 1 = 0) \\ (-\omega^2)^{2n} + \omega^{2n} + 1 &= 0 && (\text{Now } n = 3k, k \in \mathbb{Z}) \\ (-\omega)^{6k} + \omega^{6k} + 1 &= 0 \\ (-1)^{6k}(\omega^3)^{4k} + (\omega^3)^{2k} + 1 &= 0 \\ 1 + 1 + 1 &= 0\end{aligned}$$

Hence our assumption is incorrect.

$\therefore$ if n is divisible by 3, $(x + 1)^{2n} + x^{2n} + 1$ is not divisible by $x^2 + x + 1$.

Question 61 (3 marks) (NSG '23 Q13c)

(i) $\forall x, y \in \mathbb{R}^+$, if $y > x$ then $y^3 - x^2y > xy^3 - x^3$.

(ii) Consider

$$\begin{aligned}y^3 - x^2y - xy^2 - x^3 &= y(y^2 - x^2) - x(y^2 - x^2) \\ &= (y - x)(y^2 - x^2) \\ &= (y - x)^2(y + x) \\ &> 0\end{aligned}$$



Question 62 (4 marks) (NSG '22 Q16b)

(i)

$$\begin{aligned}y &= \ln x \\y' &= \frac{1}{x} \\x &= 1, m_1 = 1 \\y - \ln 1 &= 1(x - 1) \\y &= x - 1\end{aligned}$$

As $\ln x$ is concave down $\forall x$, it never lies above any tangent, $\therefore \ln x \leq x - 1$

(ii)

$$\begin{aligned}\ln(b_1^{a_1} b_2^{a_2} \dots b_n^{a_n}) &= \ln(b_1^{a_1}) + \ln(b_2^{a_2}) + \dots + \ln(b_n^{a_n}) \\&= a_1 \ln b_1 + a_2 \ln b_2 + \dots + a_n \ln b_n \\&\leq a_1(b_1 - 1) + a_2(b_2 - 1) + \dots + a_n(b_n - 1) && \text{(from (i))} \\&= a_1 b_1 + a_2 b_2 + \dots + a_n b_n - (a_1 + a_2 + \dots + a_n) \\&= 1 - 1 \\&= 0 \\b_1^{a_1} b_2^{a_2} b_3^{a_3} \dots b_n^{a_n} &\leq e^0 \\&\leq 1\end{aligned}$$

Question 63 (7 marks) (Manly '20 Q13d)

(i) Let $n = pq$ where $p < n, q < n$ for $n > 1$.

If $p \leq \sqrt{n}$ then the proposition is true.

If $p < \sqrt{n}$ then $pq = n \Rightarrow q < \sqrt{n}$

$\therefore$ the proposition is true.

(ii) n has a factor less than $\sqrt{n} \Rightarrow n$ is composite.

(iii) If a factor exists greater than $\sqrt{n}$ then n is NOT composite.



Question 64 (4 marks) (FSHS '23 Q16a)

(i) If $n = k^2$, where $k \in \mathbb{Z}$, then the remainder is 0 or 1 when an integer, n , is divided by 4.

(ii) **Case 1:** $n = k^2$

$$\begin{aligned}k &= 4q \quad (q \in \mathbb{Z}) \\k^2 &= 16q^2 \\n &= 16q^2 \quad \text{remainder is 0 when divided by 4}\end{aligned}$$

Case 2: $k = 4q + 1$

$$\begin{aligned}k^2 &= (4q + 1)^2 \\&= 16q^2 + 8q + 1 \\n &= 4(4q^2 + 2q) + 1 \quad \text{remainder is 1 when divided by 4}\end{aligned}$$

Case 3: $k = 4q + 2$

$$\begin{aligned}k^2 &= (4q + 2)^2 \\&= 16q^2 + 16q + 4 \\n &= 4(4q^2 + 4q + 1) \quad \text{remainder is 0 when divided by 4}\end{aligned}$$

Case 4: $k = 4q + 3$

$$\begin{aligned}k^2 &= (4q + 3)^2 \\&= 16q^2 + 24q + 9 \\n &= 4(4q^2 + 6q + 2) + 1 \quad \text{remainder is 1 when divided by 4}\end{aligned}$$

$\therefore$ Since the contrapositive statement is true, then the original statement is true.



Question 65 (5 marks) (NSG '24 Q14c)

(i) Every set of three neighbouring integers adds to less than 45.

(ii)

$$\begin{aligned}x_1 + x_2 + x_3 &< 45 \\x_{30} + x_1 + x_2 &< 45\end{aligned}$$

(iii)

$$\begin{aligned}x_1 + x_2 + x_3 &< 45 \\x_2 + x_3 + x_4 &< 45 \\&\vdots \\x_{28} + x_{29} + x_{30} &< 45 \\x_{29} + x_{30} + x_1 &< 45 \\x_{30} + x_1 + x_2 &< 45\end{aligned}$$

Adding:

$$\begin{aligned}3(x_1 + x_2 + \dots + x_{30}) &< 30 \times 45 \\x_1 + x_2 + \dots + x_{30} &< 450\end{aligned}$$

But

$$\begin{aligned}x_1 + x_2 + \dots + x_{30} &= 1 + 2 + \dots + 30 \\&= \frac{30}{2} (1 + 30) \\&= 465\end{aligned}$$

Which is a contradiction, $\therefore P$ is true.



Algebraic Inequalities

66 (Copium '23): A.

By the AM-GM inequality, we have that:

$$\begin{aligned}\frac{\frac{1}{2}a + \frac{1}{2}a + 2b + 3c}{4} &\geq \sqrt[4]{\frac{1}{2}a \times \frac{1}{2}a \times 2b \times 3c} \Rightarrow a^2bc \leq \left(\frac{6}{4\sqrt[4]{\frac{1}{2} \times \frac{1}{2} \times 2 \times 3}}\right)^4 \\ \frac{a + 2b + 3c}{3} &\geq \sqrt[3]{a \times 2b \times 3c} \Rightarrow abc \leq \left(\frac{6}{3\sqrt[3]{1 \times 2 \times 3}}\right)^3 \\ \frac{a + b + b + 3c}{4} &\geq \sqrt[4]{a \times b \times b \times 3c} \Rightarrow ab^2c \leq \left(\frac{6}{4\sqrt[4]{1 \times 1 \times 1 \times 3}}\right)^4 \\ \frac{a + b + b + \frac{3}{2}c + \frac{3}{2}c}{5} &\geq \sqrt[5]{a \times b \times b \times \frac{3}{2}c \times \frac{3}{2}c} \Rightarrow ab^2c^2 \leq \left(\frac{6}{5\sqrt[5]{1 \times 1 \times 1 \times \frac{3}{2} \times \frac{3}{2}}}\right)^5\end{aligned}$$

where equality can be obtained from:

$$(a, b, c) = \left(3, \frac{3}{4}, \frac{1}{2}\right)$$

$$(a, b, c) = \left(2, 1, \frac{2}{3}\right)$$

$$(a, b, c) = \left(\frac{3}{2}, \frac{3}{2}, \frac{1}{2}\right)$$

$$(a, b, c) = \left(\frac{6}{5}, \frac{6}{5}, \frac{4}{5}\right)$$

and A gives the maximum value.

67 (Ruse '21 AT3 Q4): A.

$$\begin{aligned}\frac{a}{b} &< \frac{a+m}{b+m} && \text{is false } (m = 0) \\ \frac{a}{b} &\geq \frac{a+m}{b+m} && \text{is false } (m < 0) \\ \frac{a+m}{b+m} &\leq 1 && \text{is false } (m > 0)\end{aligned}$$

So A.



68 (BoS '16 Q10): **B**.

A.

$$\begin{aligned}\frac{(a+b)(b+c)(a+c)}{abc} &\geq \frac{2\sqrt{ab} \ 2\sqrt{bc} \ 2\sqrt{ac}}{abc} \\ &\geq \frac{8\sqrt{a^2b^2c^2}}{abc} \\ &= 8\end{aligned}$$

B.

$$\begin{aligned}\frac{a+b}{c} + \frac{b+c}{a} + \frac{a+c}{b} &= \left(\frac{a}{c} + \frac{c}{a}\right) + \left(\frac{b}{c} + \frac{c}{b}\right) + \left(\frac{b}{a} + \frac{a}{b}\right) \\ &\geq 2 + 2 + 2 \\ &= 6\end{aligned}$$

C.

$$\begin{aligned}\left(a + \frac{1}{a}\right)\left(b + \frac{1}{b}\right)\left(c + \frac{1}{c}\right) &\geq 2 \times 2 \times 2 \\ &= 8\end{aligned}$$

D.

$$\begin{aligned}(a+b+c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) &\geq 3\sqrt[3]{abc} \cdot 3\sqrt[3]{\frac{1}{abc}} \\ &= 9\end{aligned}$$

69 (Ruse '21 AT3 Q5): **B**.

This can be determined by substituting in a value of a and b , e.g $a = 5, b = 3$.

70 (Ruse '23 AT3 Q3): **C**.

A. Consider $x = -1, y = -2 \Rightarrow (-1)^2 > (-2)^2 \Rightarrow -1 > 4. \therefore$ not A.

B. Consider $x = 1, y = -2$, then the statement implies $\frac{1}{1} < \frac{1}{-2}, \therefore$ not B.

C.

$$\begin{aligned}x &> y \\ \therefore x + z &> y + z\end{aligned}$$

D. Consider $z \leq 0$: equality or the inequality flips.



71 (NSG '20 Q10): **B.**

A. $b < c \Rightarrow \frac{1}{b} > \frac{1}{c}$ so $\frac{1}{b} - \frac{1}{c} > 0$ so $\frac{1}{a} + \frac{1}{b} - \frac{1}{c} > 0 \therefore$ not A.

B.

$$\begin{array}{cc} c > b & c > a \\ c + a > a + b & b + c > a + b \\ \frac{1}{c + a} < \frac{1}{a + b} & \frac{1}{b + c} < \frac{1}{a + b} \end{array}$$

e.g $a = 1, b = 2, c = 5$

$$\frac{1}{6} + \frac{1}{7} - \frac{1}{3} < 0$$

C.

$$\begin{array}{l} a > c \\ ab > bc \end{array}$$

so $\frac{1}{ab} > \frac{1}{bc}$ so $\frac{1}{ab} - \frac{1}{bc} > 0 \therefore$ not C

D. $\frac{c}{ab} + \frac{a}{bc} - \frac{b}{ac} = \frac{c^2 + a^2 - b^2}{abc}$ as $\begin{array}{l} c^2 > b^2 \\ c^2 - b^2 > 0 \\ c^2 + a^2 - b^2 > 0 \end{array} \therefore$ not D

Question 72 (5 marks) (Ascham '22 Q14a)

(i) Consider:

$$\begin{array}{l} a^2 - 2ab + b^2 = (a - b)^2 \\ \geq 0 \\ \therefore a^2 + b^2 \geq 2ab \end{array}$$

(ii) RTP: $a^2 + b^2 + c^2 \geq ab + bc + ca$ **Proof:** From (i),

$$a^2 + b^2 \geq 2ab \quad (1)$$

$$b^2 + c^2 \geq 2bc \quad (2)$$

$$a^2 + c^2 \geq 2ac \quad (3)$$

$$(1) + (2) + (3) :$$

$$2a^2 + 2b^2 + 2c^2 \geq 2ab + 2bc + 2ca$$

$$\therefore a^2 + b^2 + c^2 \geq ab + bc + ca$$



Question 73 (3 marks) (Baulko '22 Q16c)

$$\begin{aligned} LHS &= \left(\frac{a}{b+c} + 1\right) + \left(\frac{b}{a+c} + 1\right) + \left(\frac{c}{a+b} + 1\right) - 3 \\ &= \frac{a+b+c}{b+c} + \frac{a+b+c}{a+c} + \frac{a+b+c}{a+b} - 3 \\ &= (a+b+c) \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b}\right) - 3 \\ &= \frac{1}{2} \times ((b+c) + (a+c) + (a+b)) \times \left(\frac{1}{b+c} + \frac{1}{a+c} + \frac{1}{a+b}\right) - 3 \\ &\geq \frac{1}{2} \times (3 \times \sqrt[3]{(b+c)(a+c)(a+b)}) \times \left(3 \times \sqrt[3]{\frac{1}{(b+c)(a+c)(a+b)}}\right) - 3 \\ &= \frac{9}{2} \times \sqrt[3]{1} - 3 \\ &= \frac{3}{2} \end{aligned}$$

Question 74 (3 marks) (Ruse '22 Q11b)

(i)

$$\begin{aligned} (\sqrt{x} - \sqrt{y})^2 &\geq 0 \\ x - 2\sqrt{x}\sqrt{y} + y &\geq 0 \\ x + y &\geq 2\sqrt{xy} \end{aligned}$$

(ii)

$$\begin{aligned} f(x) &= \frac{9x^2 \sin^2 x + 4}{x \sin x} \\ &= 9x \sin x + \frac{4}{x \sin x} \\ &\geq 2\sqrt{9x \sin x + \frac{4}{x \sin x}} \\ &= 12 \end{aligned} \tag{i}$$

$\therefore$ minimum value of $f(x)$ is 12.



Question 75 (4 marks) (Sydney Tech '22 Q16c)

(i)

$$\begin{aligned}\left(\sqrt{x} + \frac{1}{\sqrt{x}}\right)^2 &\geq 0 \\ x - 2 + \frac{1}{x} &\geq 0 \\ x + \frac{1}{x} &\geq 2\end{aligned}$$

(ii)

$$\begin{aligned}(a + b + c)\left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c}\right) &= 1 + 1 + 1 + \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} \\ &= 3 + \frac{a}{b} + \frac{1}{\left(\frac{a}{b}\right)} + \frac{a}{c} + \frac{1}{\left(\frac{a}{c}\right)} + \frac{b}{c} + \frac{1}{\left(\frac{b}{c}\right)} \\ &\geq 3 + 2 + 2 + 2 && \text{(by (i))} \\ &= 9\end{aligned}$$

Question 76 (4 marks) (Sydney Girls '22 Q15c)

(i)

$$\begin{aligned}(a + b + c)^2 &= ((a + b) + c)^2 \\ &\geq 4(a + b)c \\ &\geq 4 \times 2\sqrt{abc} \\ &= 8c\sqrt{ab}\end{aligned}\tag{1}$$

(ii) Similarly,

$$(a + b + c)^2 \geq 8a\sqrt{bc}\tag{2}$$

$$(a + b + c)^2 \geq 8b\sqrt{ac}\tag{3}$$

(1) $\times$ (2) $\times$ (3) :

$$\begin{aligned}(a + b + c)^6 &\geq 8^3 abc \sqrt{a^2 b^2 c^2} \\ &= 8^2 a^2 b^2 c^2 \\ \therefore (a + b + c)^3 &= 8\sqrt{8abc} \\ &= 16\sqrt{2abc}\end{aligned}$$



Question 77 (5 marks) (Penrith '22 Q12c)

(i)

$$\begin{aligned}(\sqrt{x} - \sqrt{y})^2 &\geq 0 \\x + y &\geq 2\sqrt{xy} \\ \frac{x + y}{2} &\geq \sqrt{xy}\end{aligned}$$

Cycle, let $x \rightarrow a^4, y = b^4$, then:

$$\begin{aligned}\frac{a^4 + b^4}{2} &\geq \sqrt{a^4 b^4} = a^2 b^2 \\ \frac{b^4 + c^4}{2} &\geq \sqrt{b^4 c^4} = b^2 c^2 \\ \frac{c^4 + a^4}{2} &\geq \sqrt{c^4 a^4} = c^2 a^2 \\ \frac{(a^4 + b^4) + (b^4 + c^4) + (c^4 + a^4)}{2} &\geq a^2 b^2 + b^2 c^2 + c^2 a^2 \\ \text{i.e., } \frac{2(a^4 + b^4 + c^4)}{2} &\geq a^2 b^2 + b^2 c^2 + c^2 a^2 \\ \text{i.e., } a^4 + b^4 + c^4 &\geq a^2 b^2 + b^2 c^2 + c^2 a^2\end{aligned}$$

(ii) From (i),

$$\begin{aligned}\frac{a^2 b^2 + a^2 c^2}{2} &= \frac{a^2 (b^2 + c^2)}{2} \geq \sqrt{a^2 b^2 \cdot a^2 c^2} = a^2 bc \\ \frac{b^2 c^2 + b^2 a^2}{2} &= \frac{b^2 (c^2 + a^2)}{2} \geq \sqrt{b^2 c^2 \cdot b^2 a^2} = b^2 ca \\ \frac{c^2 a^2 + c^2 b^2}{2} &= \frac{c^2 (a^2 + b^2)}{2} \geq \sqrt{c^2 a^2 \cdot c^2 b^2} = c^2 ab\end{aligned}$$

By addition,

$$\begin{aligned}\frac{a^2 (b^2 + c^2) + b^2 (c^2 + a^2) + c^2 (a^2 + b^2)}{2} &\geq a^2 bc + b^2 ca + c^2 ab \\ \text{i.e., } \frac{2(a^2 b^2 + b^2 c^2 + c^2 a^2)}{2} &\geq a^2 bc + b^2 ca + c^2 ab \\ \text{i.e., } a^2 b^2 + b^2 c^2 + c^2 a^2 &\geq a^2 bc + b^2 ca + c^2 ab\end{aligned}$$

(iii) From (ii), (iii),

$$\begin{aligned}a^4 + b^4 + c^4 &\geq a^2 b^2 + b^2 c^2 + c^2 a^2 \\ a^2 b^2 + b^2 c^2 + c^2 a^2 &\geq a^2 bc + b^2 ca + c^2 ab\end{aligned}$$

$$a^4 + b^4 + c^4 \geq a^2 bc + b^2 ca + c^2 ab$$

Thus

$$\begin{aligned}&= abc(a + b + c), \text{ factorising} \\ &= abcd, \text{ since } a + b + c = d\end{aligned}$$

$$a^4 + b^4 + c^4 \geq abcd.$$



Question 78 (6 marks) (NSG '22 Q15c)

(i)

$$\begin{aligned}\frac{1}{(k-1)k(k+1)} - \frac{1}{k^3} &= \frac{k^2 - (k-1)(k+1)}{k^3(k-1)(k+1)} \\ &= \frac{1}{k^3(k-1)(k+1)} \\ &> 0 \quad (\text{as } k > 1) \\ \therefore \frac{1}{(k-1)k(k+1)} &> \frac{1}{k^2} \quad (\forall k > 1)\end{aligned}$$

(ii) Let $\frac{1}{(k-1)k(k+1)} = \frac{A}{k-1} + \frac{B}{k} + \frac{C}{k+1}$.

$$\begin{aligned}1 &= Ak(k+1) + B(k-1)(k+1) + Ck(k-1) \\ \therefore C &= -\frac{1}{2}, B = -1, A = \frac{1}{2} \\ \therefore S_n &= \sum_{k=3}^n \frac{1}{k^3} \\ &< \sum_{k=3}^n \left(\frac{1/2}{k-1} - \frac{1}{k} + \frac{1/2}{k+1} \right) \\ &= \frac{1}{2} \sum_{k=3}^{n-1} \frac{1}{k} - \sum_{k=3}^n \frac{1}{k} + \frac{1}{2} \sum_{k=4}^{n+1} \frac{1}{k} \\ &= \left[\frac{1}{2} \sum_{k=4}^{n-1} \frac{1}{k} + \frac{1}{2} + \frac{1}{3} \right] - \left[\sum_{k=4}^{n-1} \frac{1}{k} + \frac{1}{2} + \frac{1}{n} \right] + \frac{1}{2} \left[\sum_{k=4}^{n-1} \frac{1}{k} + \frac{1}{n} + \frac{1}{n+1} \right] \\ &= \frac{1}{4} + \frac{1}{6} - \frac{1}{3} - \frac{1}{n} + \frac{1}{2n} + \frac{1}{2(n+1)} \\ &= \frac{1}{12} - \frac{1}{2} \left(\frac{1}{n} - \frac{1}{n+1} \right) \\ &< 12 \quad \left(\text{as } \frac{1}{n} > \frac{1}{n+1} \right)\end{aligned}$$



Question 79 (5 marks) (Normo '22 AT3 Q3a)

(i)

$$\begin{aligned}\left(\sqrt{\frac{a}{b}} - \sqrt{\frac{b}{a}}\right)^2 &\geq 0 \\ \frac{a}{b} - 2 + \frac{b}{a} &\geq 0 \\ \therefore \frac{a}{b} + \frac{b}{a} &\geq 2\end{aligned}$$

(ii)

$$\begin{aligned}\frac{1}{a} + \frac{1}{b} + \frac{1}{c} &= (a + b + c) \left(\frac{1}{a} + \frac{1}{b} + \frac{1}{c} \right) \\ &= 1 + 1 + 1 + \frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} \\ &\geq 3 + 2 + 2 + 2 \\ &= 9\end{aligned}$$

(iii)

$$\begin{aligned}(1 - a)(1 - b)(1 - c) &= (1 - b - a + ab)(1 - c) \\ &= 1 - c - b + bc - a + ac + ab - abc \\ &= 1 - (a + b + c) + ab + ac + bc - abc \\ &= 0 + ab + ac + bc - abc - abc \\ &\geq 9abc - abc \\ &= 8abc\end{aligned}$$



Question 81 (6 marks) (NSB '23 AT2 Q3)

(a) Let $n = 5, t_1 = x^5, t_2 = y^5, t_3 = z^5, t_4 = 1^5$ and $t_5 = 15$.

$$\begin{aligned}\sqrt[5]{x^5 y^5 z^5 1^5} &\leq \frac{x^5 + y^5 + z^5 + 1^5 + 15}{5} \\ xyz &\leq \frac{x^5 + y^5 + z^5 + 1^5}{5} \\ x^5 + y^5 + z^5 - 5xyz &\leq -2\end{aligned}$$

i.e, the minimum value of $x^5 + y^5 + z^5 - 5xyz$ is -2.

(b) (i) Let $n = 2, t_1 = 1$ and $t_2 = (2n - 1)$

$$\begin{aligned}\sqrt{1 \cdot (2n - 1)} &\leq \frac{1 + (2n - 1)}{2} \\ n &\geq \sqrt{2n - 1}\end{aligned}$$

(ii) Similarly,

$$\begin{aligned}n &\geq \sqrt{3(2n - 3)} \\ n &\geq \sqrt{5(2n - 5)} \\ &\vdots \\ n &\geq \sqrt{(2n - 5) \cdot 5} \\ n &\geq \sqrt{(2n - 3) \cdot 3} \\ n &\geq \sqrt{(2n - 1) \cdot 1}\end{aligned}$$

Multiplying these inequalities:

$$\begin{aligned}n^n &\geq \sqrt{1^2 \cdot 3^2 \cdot 5^2 \dots (2n - 1)^2} \\ &= 1 \times 3 \times 5 \times \dots (2n - 1)\end{aligned}$$



Question 82 (8 marks) (NSB '19 Q15c)

(i)

$$x^2 + y^2 \geq 2xy$$
$$a^2 + (bc)^2 \geq 2a(bc) = 2abc$$

(ii)

$$a^2 + b^2 \geq 2ab$$
$$a^2 + c^2 \geq 2ac$$
$$b^2 + c^2 \geq 2bc$$
$$2a^2 + 2b^2 + 2c^2 \geq 2(ab + ac + bc)$$
$$a^2b^2 + c^2 \geq ab + ac + bc.$$

(iii)

$$a^2(1 + b^2) + b^2(1 + c^2) + c^2(1 + a^2) = a^2 + b^2 + c^2 + a^2b^2 + a^2c^2 + b^2c^2$$
$$a^2 + b^2c^2 \geq 2abc$$
$$b^2 + a^2c^2 \geq 2abc$$
$$a^2 + c^2c^2 \geq 2abc$$
$$\geq 6abc$$

(iv)

$$a^4 + b^4 + c^4 + a^2 + b^2 + c^2 \geq a^2b^2 + b^2c^2 + a^2c^2 + a^2 + b^2 + c^2$$
$$\geq 6abc$$



Question 83 (6 marks) (ST '24 Q16c)

(i)

$$\begin{aligned}
 \frac{a+b+c}{3} &\geq \sqrt[3]{abc} \\
 \frac{t}{3} &\geq (abc)^{\frac{1}{3}} \\
 \frac{3}{t} &\leq \frac{1}{(abc)^{\frac{1}{3}}} \\
 \frac{1}{(abc)^{\frac{1}{3}}} &\geq \frac{3}{t} \\
 \frac{\frac{1}{a} + \frac{1}{b} + \frac{1}{c}}{3} &\geq \sqrt[3]{\frac{1}{a} \cdot \frac{1}{b} \cdot \frac{1}{c}} \\
 \frac{1}{a} + \frac{1}{b} + \frac{1}{c} &\geq 3 \left(\frac{1}{abc} \right)^{\frac{1}{3}} \\
 &\geq 3 \left(\frac{3}{t} \right) \\
 &\geq \frac{9}{t}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 \left(\frac{1}{a} - 1 \right) \left(\frac{1}{b} - 1 \right) \left(\frac{1}{c} - 1 \right) &= \left(\frac{1-a}{a} \right) \left(\frac{1-b}{b} \right) \left(\frac{1-c}{c} \right) \\
 &= \frac{(1-a)(1-b)(1-c)}{abc} \\
 &= \frac{1 - (a+b+c) + (ab+ac+bc) - abc}{abc} \\
 &= \frac{1 - 1 + ab + ac + bc - abc}{abc} \\
 &= \frac{1}{c} + \frac{1}{b} + \frac{1}{a} - 1 \\
 &= \frac{9}{1} - 1 \\
 &= 8
 \end{aligned}$$



Question 84 (6 marks) (BBHS '24 Q14d)

(i)

$$\begin{aligned}(a+b)^2 &\geq 0 \\ a^2 + 2ab + b^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab \\ \frac{a+b}{2} &\geq \sqrt{ab}\end{aligned}$$

(ii)

$$\begin{aligned}a^8 + b^8 + c^8 &= \frac{1}{2}(a^8 + b^8) + \frac{1}{2}(b^8 + c^8) + \frac{1}{2}(a^8 + c^8) \\ \frac{1}{2}(a^8 + b^8) &\geq \sqrt{a^8 b^8} \\ \frac{1}{2}(b^8 + c^8) &\geq \sqrt{b^8 c^8} \\ \frac{1}{2}(a^8 + c^8) &\geq \sqrt{a^8 c^8} \\ a^8 + b^8 + c^8 &\geq \sqrt{a^8 b^8} + \sqrt{b^8 c^8} + \sqrt{a^8 c^8} \\ &\geq a^4 b^4 + b^4 c^4 + a^4 c^4\end{aligned}$$

(iii)

$$a^4 + b^4 \geq 2a^2 b^2 \quad (1)$$

$$b^4 + c^4 \geq 2b^2 c^2 \quad (2)$$

$$a^4 + c^4 \geq 2a^2 c^2 \quad (3)$$

$$(1) + (2) + (3)$$

$$2(a^4 + b^4 + c^4) \geq 2(a^2 b^2 + b^2 c^2 + a^2 c^2)$$

$$(a^4 + b^4 + c^4) \geq (a^2 b^2 + b^2 c^2 + a^2 c^2)$$

Replacing a with ab , b by bc and c by ac :

$$(ab)^4 + (bc)^4$$



Question 85 (3 marks) (Ruse '24 Q12d)

(i)

$$\begin{aligned}(a + b)^2 &\geq 0 \\ a^2 + 2ab + b^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab\end{aligned}$$

(ii)

$$\begin{aligned}a^2 + b^2 &\geq 2ab && \text{(i)} \\ (a^2 + b^2)(a + b) &\geq 2ab(a + b) \\ a^3 + a^2b + b^2a + b^3 &\geq 2a^2b + 2ab^2 \\ a^3 + b^3 &\geq a^2b + ab^2 && \text{(1)}\end{aligned}$$

Cycling,

$$b^3 + c^3 \geq b^2c + c^2b \quad (2)$$

$$a^3 + c^3 \geq a^2c + c^2a \quad (3)$$

(1) + (3) :

$$\begin{aligned}2(a^3 + b^3 + c^3) &\geq a^2b + ab^2 + b^2c + bc^2 + c^2a + ca^2 \\ &\geq ab(a + b) + bc(b + c) + ca(c + a)\end{aligned}$$



Question 86 (4 marks) (NSG '24 Q16b)

(i)

$$\begin{aligned}\frac{2k}{2k+1} - \frac{2k-1}{2k} &= \frac{4k^2 - (2k+1)(2k-1)}{2k(2k+1)} \\ &= \frac{1}{2k(2k+1)} \\ &> 0 \\ \therefore \frac{2k-1}{2k} &< \frac{2k}{2k+1}\end{aligned}$$

(ii) From (i), $\frac{1}{2} < \frac{2}{3}, \frac{3}{4} < \frac{4}{5}, \frac{5}{6} < \frac{6}{7}, \dots, \frac{2n-1}{2n} < \frac{2n}{2n+1}$. Multiplying:

$$\begin{aligned}p &< q \\ p^2 &< pq && \text{(as } p > 0\text{)} \\ p^2 &< \frac{1}{2} \cdot \frac{2}{3} \cdot \frac{3}{4} \cdot \frac{4}{5} \cdot \frac{5}{6} \cdot \frac{6}{7} \cdot \dots \cdot \frac{2n-1}{2n} \cdot \frac{2n}{2n+1} \\ &= \frac{1}{2n+1} \\ p &< \frac{1}{\sqrt{2n+1}}\end{aligned}$$

Question 87 (2 marks) (Baulko '23 Q16b)

(i)

$$\begin{aligned}(\sqrt{p} + \sqrt{q})^2 &\geq 0 \\ p + 2\sqrt{pq} + q &\geq 0 \\ \frac{1}{2}(p + q) &\geq \sqrt{pq}\end{aligned}$$

(ii) Dividing both sides in (i) by $\sqrt{q}$:

$$\begin{aligned}\sqrt{p} &\leq \frac{1}{2} \left(\frac{p}{\sqrt{q}} + \sqrt{q} \right) \\ &\leq \frac{1}{2} \left(\frac{p}{\sqrt{q}} + \sqrt{q} \right)\end{aligned}$$



Question 88 (5 marks) (Manly '24 Q16c)

(i)

$$\begin{aligned}(\sqrt{a} - \sqrt{b})^2 &\geq 0 \\ a - 2\sqrt{ab} + b &\geq 0 \\ a + b &\geq 2\sqrt{ab} \\ \frac{a+b}{2} &\geq \sqrt{ab}\end{aligned}$$

(ii)

$$\begin{aligned}(\sqrt{x-1} - 1)^2 &\geq 0 \\ x - 1 - 2\sqrt{x-1} + 1 &\geq 0 \\ x - 2\sqrt{x-1} &\geq 0 \\ x &\geq 2\sqrt{x-1} \\ \frac{x}{\sqrt{x-1}} &\geq 2\end{aligned}$$

(iii) From (i)

$$\begin{aligned}\frac{a^2}{b-1} + \frac{b^2}{a-1} &\geq 2\sqrt{\frac{a^2}{b-1} \times \frac{b^2}{a-1}} \\ &= 2\sqrt{\frac{a^2}{b-1}}\sqrt{\frac{b^2}{a-1}} \\ &= 2 \times \frac{a}{\sqrt{b-1}} \times \frac{b}{\sqrt{a-1}} \\ &\geq 2 \times 2 \times 2 \\ &= 8\end{aligned}$$



Question 89 (5 marks) (Baulko '23 Q14b)

(i)

$$\begin{aligned}\frac{a+b+c}{3} &\geq \sqrt[3]{abc} \\ a+b+c &\geq 3\sqrt[3]{abc} \\ (a+b+c)^3 &\geq 27abc\end{aligned}$$

(ii) Let $a \rightarrow \frac{a}{b^4}, b \rightarrow \frac{b}{c^4}, c \rightarrow \frac{c}{a^4}$:

$$\begin{aligned}\frac{\frac{a}{b^4} + \frac{b}{c^4} + \frac{c}{a^4}}{3} &\geq \sqrt[3]{\frac{abc}{a^4b^4c^4}} \\ \frac{a}{b^4} + \frac{b}{c^4} + \frac{c}{a^4} &\geq 3\sqrt[3]{\frac{1}{(abc)^4}} \\ a^4b^4c^4 \left(\frac{a}{b^4} + \frac{b}{c^4} + \frac{c}{a^4} \right) &\geq 3\sqrt[3]{\frac{1}{1}} (abc)^4 \\ a^5c^4 + b^5a^4 + c^5b^4 &\geq 3\end{aligned}$$

Question 90 (3 marks) (NSB '24 Q14a)

$$\begin{aligned}a_1 &\geq \sqrt{a_1} \\ (a_1 - \sqrt{a_1}) &\geq 0 \\ a_1^2 + 2a_1\sqrt{a_1} + a_1 &\geq 0 \\ a_1^2 + a_1 &\geq 2a_1\sqrt{a_1}\end{aligned} \tag{1}$$

Similarly, for a_2 to a_n

$$\begin{aligned}a_2^2 + a_2 &\geq 2a_2\sqrt{a_2} \\ &\vdots \\ a_n^2 + a_n &\geq 2a_n\sqrt{a_n}\end{aligned} \tag{2} \tag{3} \tag{n}$$

Now $(1) \times (2) \times \dots \times (n)$:

$$\begin{aligned}(a_1^2 + a_1)(a_2^2 + a_2)(a_3^2 + a_3) \dots (a_n^2 + a_n) &\geq 2 \times a_1\sqrt{a_1} \times 2a_2\sqrt{a_2} \times \dots \times 2a_n\sqrt{a_n} \\ &\geq 2^n (a_1a_2 \dots a_n) \sqrt{a_1a_2 \dots a_n} \\ &\geq 2^n\end{aligned}$$



Question 91 (4 marks) (NSB '24 Q13b)

(i) If $\frac{1}{p} + \frac{1}{q} = 1$,

$$p + q = pq = s$$

$$(p + q)^2 = p^2 + q^2 + 2pq \quad (1)$$

$$(p + q)^2 = (pq)^2 = s^2 \quad (2)$$

$$(1) + (2) \Rightarrow (p + q)^2 \geq 4pq \Rightarrow s \geq 4$$

$$\begin{aligned} \frac{1}{p(p+1)} + \frac{1}{g(g+1)} &= \frac{1}{p} - \frac{1}{p+1} + \frac{1}{g} - \frac{1}{g(g+1)} \\ &= 1 - \frac{p+q+2}{(p+1)(g+1)} \\ &= 1 - \frac{s+2}{2s+1} \\ &= \frac{s-1}{2s+1} \end{aligned}$$

$\therefore$ we have to prove

$$\frac{1}{3} \leq \frac{s-1}{2s+1} \leq \frac{1}{2}$$

But $2s + 1 \leq 3s - 3 \leftrightarrow 4 \leq s$ and

$$2s - 2 \leq 2s + 1 \leftrightarrow -2 \leq 1$$

(ii)

$$\begin{aligned} \frac{1}{p(p-1)} + \frac{1}{g(g-1)} &= \frac{1}{p-1} - \frac{1}{p} + \frac{1}{g-1} - \frac{1}{g} \\ &= \frac{p+g-2}{(p-1)(g-1)} - 1 \\ &= \frac{s-2}{s-s+1} - 1 \\ &= s-3 \quad \text{and as } s \geq 4 \end{aligned} \quad (i)$$

$$\begin{aligned} s-3 &\geq 1 \\ \therefore \frac{1}{p(p-1)} + \frac{1}{g(g-1)} &\geq 1 \end{aligned}$$



Question 92 (2 marks) (Killara '22 Q13b)

$$\begin{aligned}|a^2 - ab| + |ab - b^2| &\geq |a^2 - ab + ab - b^2| \\|a^2 - ab| + |ab - b^2| &\geq |a^2 - b^2| \\|a^2 - ab| + |ab - b^2| &\geq |(a + b)(a - b)|\end{aligned}$$

And as $a > b > 0$ we know:

$$\begin{aligned}a + b &> 0 \text{ and } a - b > 0, \text{ so} \\|a^2 - ab| + |ab - b^2| &\geq (a + b)(a - b)\end{aligned}$$

Question 93 (5 marks) (Normo '13 Q16b)

(i) Square both sides.

$$\begin{aligned}LHS &= (2x + 3)^2 = 4k^2 + 12k + 9 \\RHS &= 4(k^2 + 3k + 2) = 4k^2 + 12k + 8 \\LHS &> RHS\end{aligned}$$

(ii)

$$\begin{aligned}2k + 3 &> 2\sqrt{k + 2}\sqrt{k + 1} & (i) \\\therefore 2k + 3 &= 2k + 1 + 2 \\\therefore 1 &> 2\sqrt{k + 2}\sqrt{k + 1} + 2(k + 1) \\1 &> 2\sqrt{k + 1}(\sqrt{k + 2} - \sqrt{k + 1}) \\\therefore \frac{1}{\sqrt{k + 1}} &> 2(\sqrt{k + 2} - \sqrt{k + 1})\end{aligned}$$

(iii)

$$\begin{aligned}1 + \frac{1}{\sqrt{2}} + \frac{1}{\sqrt{3}} + \dots + \frac{1}{\sqrt{n}} &= \sum_{k=0}^{n-1} \frac{1}{\sqrt{k + 1}} \\\geq 2 \sum_{k=0}^{n-1} (\sqrt{k + 2} - \sqrt{k + 1}) \\&= 2(\sqrt{2} - \sqrt{1} + \sqrt{3} - \sqrt{2} + \dots + \sqrt{n + 1} - \sqrt{n}) \\&= 2(\sqrt{n + 1} - 1)\end{aligned}$$



Question 94 (4 marks) (God folder)

- (i) When dealing with inequalities, it is useful to work backwards (on a piece of scrap paper) to find an equivalent form of the inequality which is easier to solve.

$$\begin{aligned}\frac{abc}{a^3 + b^3 + abc} &\leq \frac{c}{a + b + c} \Leftrightarrow abc(a + b + c) \leq c(a^3 + b^3 + abc) \\ &\Leftrightarrow a^2bc + b^2ca + c^2ab \leq ca^3 + cb^3 + c^2ab\end{aligned}$$

This inequality is easier to deal with as we have removed the fraction and expanded terms. To prove this last inequality, we can consider the difference:

$$\begin{aligned}ca^3 + cb^3 + c^2ab - (a^2bc + b^2ca + c^2ab) &= c(a^3 + b^3 + \cancel{abc} - a^2b - ab^2 - \cancel{abc}) \\ &= c((a + b)^3 - 4a^2b - 4ab^2) \\ &= c((a + b)^3 - 4ab(a + b)) \\ &= c(a + b)((a + b)^2 - 4ab) \\ &= c(a + b)(a - b)^2 \\ &\geq 0\end{aligned}$$

Now reversing this process will give the required inequality.

- (ii) We can generate similar inequalities by cycling through the variables in (i)

- Replacing (a, b, c) with (b, c, a) gives

$$\frac{abc}{b^3 + c^3 + abc} \leq \frac{a}{a + b + c}$$

- Replacing (a, b, c) with (c, a, b) gives

$$\frac{abc}{c^3 + a^3 + abc} \leq \frac{b}{a + b + c}$$

Adding these two inequalities, along with the original inequality from (i) gives:

$$\begin{aligned}\frac{abc}{a^3 + b^3 + abc} + \frac{abc}{b^3 + c^3 + abc} + \frac{abc}{c^3 + a^3 + abc} &\leq \frac{c}{a + b + c} + \frac{a}{a + b + c} + \frac{b}{a + b + c} \\ &= \frac{a + b + c}{a + b + c} \\ &= 1\end{aligned}$$



Question 95 (4 marks) (God folder)

First common denominate so we can clear the abc denominator so we get a polynomial inequality:

$$\frac{ab + bc + ca}{abc} \leq \frac{3abc + 2(a^3 + b^3 + c^3)}{abc}$$
$$ab + bc + ca \leq 3abc + 2(a^3 + b^3 + c^3)$$

The LHS is a degree 2 and the RHS is also a degree 3.

so to make this homogenous, we will multiply the LHS by $a + b + c = 1$. So now we want to prove

$$(a + b + c)(ab + bc + ca) \leq 3abc + 2(a^3 + b^3 + c^3).$$

The $3abc$ term is annoying because it breaks the symmetry in the problem. Good thing is expanding the LHS also has a $3abc$ term so this cancels out.

$$a^2b + abc + a^2c + ab^2 + b^2c + abc + abc + bc^2 + ac^2 \leq 3abc + 2(a^3 + b^3 + c^3)$$
$$a^2b + a^2c + b^2c + b^2a + c^2a + c^2b \leq 2(a^3 + b^3 + c^3)$$

This is a symmetric homogenous polynomial inequality so the rest of this should be standard technique. The usual way to deal with three variable inequalities is to find a suitable two variable inequality, substitute variables and add them together.

A good choice would be to guess $a^3 + b^3 \geq a^2b + ab^2$. This is a two variable inequality in a and b and indeed, summing these inequalities cyclically gives us the answer.

That is, $\sum_{\text{cyc}} a^3 + b^3 \geq \sum_{\text{cyc}} a^2b + ab^2$.

All that is left is to verify $a^3 + b^3 \geq a^2b + ab^2$ by considering:

$$\begin{aligned} a^3 + b^3 - a^2b - ab^2 &= (a + b)(a^2 - ab + b^2) - ab(a + b) \\ &= (a + b)(a^2 - 2ab + b^2) \\ &= (a + b)(a - b)^2 \\ &\geq 0 \end{aligned}$$



Question 96 (6 marks) (NSB '22 Q16b v))

(i)

$$\begin{aligned}\sqrt{|a||b||c||d|} &= \frac{|a||b|}{2} + \frac{|c||d|}{2} \\ &= \frac{\frac{a^2+b^2}{2}}{2} + \frac{\frac{c^2+d^2}{2}}{2} \\ &= \frac{a^2 + b^2 + c^2 + d^2}{4} \\ &= \frac{12}{4} \\ &= 3\end{aligned}$$

Hence, the maximum value of $abcd$ is 9 (3^2).

(ii)

$$\begin{aligned}a + b + c + d &= 0 \\ \therefore d &= -(a + b + c) \\ \therefore abcd &= -abc(a + b + c)\end{aligned}$$

Minimising $-abc(a + b + c)$ is the same as maximising $abc(a + b + c)$.

$$\begin{aligned}a^2 + b^2 + c^2 + d^2 &= 12 \\ a^2 + b^2 + c^2 + [-(a + b + c)]^2 &= 12 \\ a^2 + b^2 + c^2 + a^2 + b^2 + c^2 + 2ab + 2ac + 2bc &= 12 \\ 2(a^2 + b^2 + c^2 + ab + ac + bc) &= 12 \\ (a^2 + b^2 + c^2 + ab + ac + bc) &= 6 \tag{1} \\ (a + b + c)^2 &= a^2 + b^2 + c^2 + 2ab + 2ac + 2bc \\ &= a^2 + b^2 + c^2 + ab + ac + bc + ab + ac + bc \\ &= 6 + ab + ac + bc \\ &= 6 + \frac{ab + ac + bc + ab + ac + bc}{2} \\ &\leq 6 + \frac{a^2 + b^2 + c^2 + ab + ac + bc}{2} \\ &= 6 + \frac{6}{2} \\ &= 9 \\ \therefore (a + b + c)^2 &\leq 9 \tag{2}\end{aligned}$$



Question 97 (4 marks) (SGHS '22 Q15c)

(i)

$$\begin{aligned}(a + b + c)^2 &= ((a + b) + c)^2 \\ &\geq 4(a + b)c \\ &\geq 4 \times 2\sqrt{abc} \\ &= 8c\sqrt{ab}\end{aligned}\tag{1}$$

(ii) Cycling from part (i):

$$(a + b + c)^2 \geq 8a\sqrt{bc}\tag{2}$$

$$(a + b + c)^2 \geq 8b\sqrt{ac}\tag{3}$$

Now, (1) $\times$ (2) $\times$ (3)

$$\begin{aligned}(a + b + c)^6 &\geq 8^3 abc \sqrt{a^2 b^2 c^2} \\ &= 8^3 a^2 b^2 c^2 \\ \therefore (a + b + c)^3 &\geq 8\sqrt{8abc} \\ &= 16\sqrt{2abc}\end{aligned}$$

Question 98 (4 marks) (Random Question)

this solution was made by a state ranker lmao

$$\begin{aligned}\frac{1}{a^2 - 1} + \frac{1}{b^2 - 1} + \frac{1}{c^2 - 1} &= 1 \\ \frac{a^2}{a^2 - 1} + \frac{b^2}{b^2 - 1} + \frac{c^2}{c^2 - 1} &= 4 \\ \left(\frac{a^2}{a^2 - 1} + \frac{b^2}{b^2 - 1} + \frac{c^2}{c^2 - 1}\right) \left(\frac{1}{a^2 - 1} + \frac{1}{b^2 - 1} + \frac{1}{c^2 - 1}\right) &\geq \left(\frac{a^2}{a^2 - 1} + \frac{b^2}{b^2 - 1} + \frac{c^2}{c^2 - 1}\right)^2 \\ &\quad \text{(CS)} \\ \therefore \left(\frac{a^2}{a^2 - 1} + \frac{b^2}{b^2 - 1} + \frac{c^2}{c^2 - 1}\right) &\leq 2 \\ \frac{a - 1}{a^2 - 1} + \frac{b - 1}{b^2 - 1} + \frac{c - 1}{c^2 - 1} + \left(\frac{1}{a^2 - 1} + \frac{1}{b^2 - 1} + \frac{1}{c^2 - 1}\right) &\leq 2 \\ \frac{1}{a + 1} + \frac{1}{b + 1} + \frac{1}{c + 1} + 1 &\leq 2 \\ \frac{1}{a^2 - 1} + \frac{1}{b^2 - 1} + \frac{1}{c^2 - 1} &\leq 1\end{aligned}$$



Question 99 (5 marks) (Ruse '20 Q15b)

(i)

$$\begin{aligned}
 (a-b)^2 &\geq 0 \\
 a^2 + 2ab + b^2 &\geq 0 \\
 a^2 + b^2 &\geq -2ab \\
 a^3 + b^3 &= (a+b)(a^2 - ab + b^2) \\
 a, b &> 0 \\
 a^3 &\geq 0, b^3 \geq 0 \\
 \therefore a^3 + b^3 &\geq 0 \\
 (a+b)(a^2 - ab + b^2) &\geq (a+b)(2ab - ab) \\
 (a+b)(a^2 - ab + b^2) &\geq (a+b)ab \\
 a^3 + b^3 &\geq a^2b + ab^2
 \end{aligned}$$

(ii) Given $a + b + c \geq 3\sqrt[3]{abc}$ and

$$(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a^2b + a^2c + b^2a + b^2c + c^2a + c^2b) + 6abc,$$

consider left, middle and right.

$$\begin{aligned}
 a + b + c &\geq 3\sqrt[3]{abc} \\
 \text{From } \frac{a + b + c}{3} &\leq \sqrt[3]{abc} \\
 \left(\frac{a + b + c}{3}\right)^3 &\geq abc \quad (1)
 \end{aligned}$$

$$\begin{aligned}
 (a + b + c)^3 &\leq (a^3 + b^3 + c^3) + 3(a^2 \dots) + 6abc \\
 &\leq 7(a^3 + b^3 + c^3) + 6abc
 \end{aligned}$$

but using $a + b + c \leq 3\sqrt[3]{abc}$

$$a^3 + b^3 + c^3 \leq 3abc$$

$$6abc \leq 2(a^3 + b^3 + c^3)$$

Hence, $(a + b + c)^3 \leq 9(a^3 + b^3 + c^3)$

$$\left(\frac{a + b + c}{3}\right)^3 \leq \frac{a^3 + b^3 + c^3}{3} \quad (2)$$

Combining (1) and (2),

$$abc \leq \left(\frac{a + b + c}{3}\right)^3 \leq \frac{a^3 + b^3 + c^3}{3}$$



Question 100 (3 marks) (Ruse '22 AT3 Q9a)

$$\begin{aligned}(\sqrt{x} + \sqrt{y})^2 &\geq x + 2\sqrt{xy} + y \\x + y &\geq 2\sqrt{xy} \\ \left(\frac{1}{a} - 1\right)\left(\frac{1}{b} - 1\right)\left(\frac{1}{c} - 1\right) &= \frac{(1-a)(1-b)(1-c)}{abc} \\ &= \frac{(b+c)(a+c)(a+b)}{abc} && \text{(since } a+b+c=1\text{)} \\ &\geq \frac{2\sqrt{bc} \times 2\sqrt{ac} \times 2\sqrt{ab}}{abc} && \text{(since } x+y \geq 2\sqrt{xy}\text{)} \\ &= \frac{8abc}{abc} \\ &= 8\end{aligned}$$

∴ the smallest possible value is 8.

Question 101 (Ruse '20 AT2 Q10d)

$$\begin{aligned}A + B + C &= \pi \quad (\text{interior angle sum of } \Delta) \\ A > 0, B > 0, C > 0 \\ \frac{A + B + C}{3} &\geq \sqrt[3]{ABC} \quad (\text{given}) \\ A + B + C &\geq 3\sqrt[3]{ABC} \\ \frac{\frac{1}{A} + \frac{1}{B} + \frac{1}{C}}{3} &\geq \sqrt[3]{\frac{1}{A} \times \frac{1}{B} \times \frac{1}{C}} \quad (\text{given}) \\ \therefore \frac{1}{A} + \frac{1}{B} + \frac{1}{C} &\geq \frac{3}{\sqrt[3]{ABC}} \\ (A + B + C) \left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C}\right) &\geq 3\sqrt[3]{ABC} \times \frac{3}{\sqrt[3]{ABC}} \\ \pi \left(\frac{1}{A} + \frac{1}{B} + \frac{1}{C}\right) &\geq 9 \\ \therefore \frac{1}{A} + \frac{1}{B} + \frac{1}{C} &\geq \frac{9}{\pi}\end{aligned}$$



Question 102 (4 marks) (NSB '21 AT2 Q3 a) i)

$$\begin{aligned}\frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} &\geq \sqrt{\frac{x^2}{y^2} \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)} \\ \frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} &\geq \sqrt{\frac{x^2}{y^2} \left(\frac{x^2 y^2 + z^4}{x^2 z^2}\right)} \\ \frac{\frac{x^2}{y^2} + \left(\frac{y^2}{z^2} + \frac{z^2}{x^2}\right)}{2} &\geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4}\end{aligned}\tag{1}$$

Similarly,

$$\begin{aligned}\frac{\frac{y^2}{z^2} + \left(\frac{x^2}{y^2} + \frac{z^2}{x^2}\right)}{2} &\geq \frac{1}{xz} \sqrt{y^2 z^2 + x^4} \\ (1) + (2) : \\ \frac{x^2}{y^2} + \frac{y^2}{z^2} + \frac{z^2}{x^2} &\geq \frac{1}{yz} \sqrt{x^2 y^2 + z^4} + \frac{1}{xz} \sqrt{y^2 z^2 + x^4}\end{aligned}\tag{1}$$

Question 103 (3 marks) (Knox '22 Q15d)

$$\begin{aligned}(a - b)^2 &\geq 0 \\ a^2 + b^2 &\geq 2ab \\ \frac{a^2}{ab} + \frac{b^2}{ab} &\geq 2 \\ \frac{a}{b} + \frac{b}{a} &\geq 2\end{aligned}\tag{1}$$

Cycling,

$$\frac{a}{c} + \frac{c}{a} \geq 2\tag{2}$$

$$\frac{b}{c} + \frac{c}{b} \geq 2\tag{3}$$

Now, (1)+(2)+(3):

$$\begin{aligned}\frac{a}{b} + \frac{b}{a} + \frac{a}{c} + \frac{c}{a} + \frac{b}{c} + \frac{c}{b} &\geq 6 \\ \frac{a+c}{b} + \frac{b+c}{a} + \frac{a+b}{c} &\geq 6\end{aligned}$$



Mathematical Induction

Question 104 (3 marks) (God folder)

Base case: $n = 1$.

$$\begin{aligned} E(x) &= xp(x) \\ &= \frac{1}{6} \cdot (1 + 2 + 3 + 4 + 5 + 6) \\ &= \frac{21}{6} \end{aligned}$$

$\therefore$ true for $n = 1$.

Assume true for $n = k$, i.e $E(x) = \frac{7n}{2}$. Now for $n = k + 1$:

$$\begin{aligned} E(k + 1) &= E(k) + E(1) \\ &= \frac{7k}{2} + \frac{7}{2} \\ &= \frac{7(k + 1)}{2} \end{aligned}$$

Hence true for $n = k + 1$.

Thus, the statement is true for all integer n

Question 105 (3 marks) (SGHS '21 Q13b)

Base case: $n = 1$

$$\begin{array}{ll} LHS = T_1 & RHS = 3^1 - 2 \\ = 1 & = 1 \end{array}$$

$\therefore$ true for $n = 1$.

Assume true for $n = k, k \geq 1$, i.e

$$T_k = 3^k - 2$$

Prove true for $n = k + 1$, RTP

$$T_{k+1} = 3^{k+1} - 2$$

$$\begin{aligned} LHS &= T_{k+1} \\ &= 3T_k + 4 \\ &= 3(3^k - 2) + 4 \\ &= 3^{k+1} - 6 + 4 \\ &= 3^{k+1} - 2 \\ &= RHS \end{aligned}$$

$\therefore$ true by induction for $n = 1, 2, \dots$



Question 106 (3 marks) (Abbotsleigh '24 Q15c i)Prove true for $n = 2$:

$$\begin{aligned}
P_2(x) &= (x - a_1)(x - a_2) \\
P_2'(x) &= (x - a_1) + (x - a_2) \\
\therefore \frac{P_2'(x)}{P_2(x)} &= \frac{(x - a_1) + (x - a_2)}{(x - a_1)(x - a_2)} \\
&= \frac{x - a_1}{(x - a_1)(x - a_2)} + \frac{x - a_2}{(x - a_1)(x - a_2)} \\
&= \frac{1}{x - a_2} + \frac{1}{x - a_1}
\end{aligned}$$

 $\therefore$ true for $n = 1$.Assume true for $n = k$:

$$\frac{P_k'(x)}{P_k(x)} = \frac{1}{x - a_1} + \frac{1}{x - a_2} + \cdots + \frac{1}{x - a_k}$$

Prove true for $n = k$.

Now,

$$\begin{aligned}
P_{k+1}(x) &= (x - a_1)(x - a_2) \dots (x - a_k)(x - a_{k+1}) \\
&= P_k(x)(x - a_{k+1}) \\
P_{k+1}'(x) &= P_k(k) \cdot 1 + (x - a_{k+1})P_k'(x) \\
\therefore \frac{P_{k+1}'(x)}{P_{k+1}(x)} &= \frac{P_k(k) + (x - a_{k+1})P_k'(x)}{P_{k+1}(x)} \\
&= \frac{P_k(x)}{P_k(x)(x - a_{k+1})} + \frac{(x - a_k)P_k'(x)}{P_k(x)(x - a_{k+1})} \\
&= \frac{1}{x - a_{k+1}} + \frac{P_k'(x)}{P_k(x)} \\
&= \frac{1}{x - a_{k+1}} + \frac{1}{x - a_1} + \frac{1}{x - a_2} + \cdots + \frac{1}{x - a_k} \\
&= RHS
\end{aligned}$$

Question 107 (5 marks) (Caringbah '24 Q16d)

(i)

$$\begin{aligned}
1 + \frac{1}{\varphi} &= 1 + \frac{2}{1 + \sqrt{5}} & 1 + \frac{1}{1 - \varphi} &= 1 + \frac{2}{1 - \sqrt{5}} \\
&= \frac{3 + \sqrt{5}}{1 + \sqrt{5}} \times \frac{1 - \sqrt{5}}{1 - \sqrt{5}} & &= \frac{3 - \sqrt{5}}{1 - \sqrt{5}} \times \frac{1 + \sqrt{5}}{1 + \sqrt{5}} \\
&= \frac{-2 - 2\sqrt{5}}{-4} & &= \frac{-2 + 2\sqrt{5}}{-4} \\
&= \frac{1 + \sqrt{5}}{2} = \varphi & &= \frac{1 - \sqrt{5}}{2} = 1 - \varphi
\end{aligned}$$



(ii) For $n = 0$:

$$\begin{aligned}
 f^0(1) &= \frac{\varphi^2 - (1 - \varphi)^2}{\varphi^1 - (1 - \varphi)^1} \\
 &= \frac{\varphi^2 - 1 + 2\varphi - \varphi^2}{2\varphi - 1} \\
 &= \frac{2\varphi - 1}{2\varphi - 1} = 1 \\
 \therefore LHS &= RHS
 \end{aligned}$$

$\therefore$ the statement is true for $n = 0$.

Assume the statement is true for $n = k$, i.e

$$f^k(1) = \frac{\varphi^{k+2} - (1 - \varphi)^{k+2}}{\varphi^{k+1} - (1 - \varphi)^{k+1}}$$

Proving true for $n = k + 1$:

$$\begin{aligned}
 f^{k+1}(1) &= f \circ f^k(1) \\
 &= 1 + \frac{1}{f^k(1)} \\
 &= 1 + \frac{\varphi^{k+1} - (1 - \varphi)^{k+1}}{\varphi^{k+2} - (1 - \varphi)^{k+2}} \quad \text{from assumption} \\
 &= \frac{\varphi^{k+2} - (1 - \varphi)^{k+2} + \varphi^{k+1} - (1 - \varphi)^{k+1}}{\varphi^{k+2} - (1 - \varphi)^{k+2}} \\
 &= \frac{\varphi^{k+2} \left(1 + \frac{1}{\varphi}\right) - (1 - \varphi)^{k+2} \left(1 + \frac{1}{1 - \varphi}\right)}{\varphi^{k+2} - (1 - \varphi)^{k+2}} \\
 &= \frac{\varphi^{k+2}(\varphi) - (1 - \varphi)^{k+2}(1 - \varphi)}{\varphi^{k+2} - (1 - \varphi)^{k+2}} \\
 &= \frac{\varphi^{k+3} - (1 - \varphi)^{k+3}}{\varphi^{k+2} - (1 - \varphi)^{k+2}} \quad \text{from part (i) as required.}
 \end{aligned}$$

(iii)

$$\begin{aligned}
 \lim_{n \rightarrow \infty} f^n(1) &= \lim_{n \rightarrow \infty} \frac{\varphi^{n+2} - (1 - \varphi)^{n+2}}{\varphi^{n+1} - (1 - \varphi)^{n+1}} \\
 &= \lim_{n \rightarrow \infty} \frac{\varphi^{n+2}}{\varphi^{n+1}} \quad \text{as } |1 - \varphi| < 1, \lim_{k \rightarrow \infty} (1 - \varphi)^k = 0 \\
 &= \lim_{n \rightarrow \infty} \frac{\varphi}{1} \quad \text{dividing top and bottom by } \varphi^{n+1} \\
 &= \varphi
 \end{aligned}$$



Question 108 (3 marks) (Ruse 2020 AT3 Q1e)

Prove true for $n = 3$:

A convex polygon with three vertices is a triangle, which has an angle sum of $180^\circ = 180(3 - 2)^\circ$.

$\therefore$ the statement is true for $n = 3$.

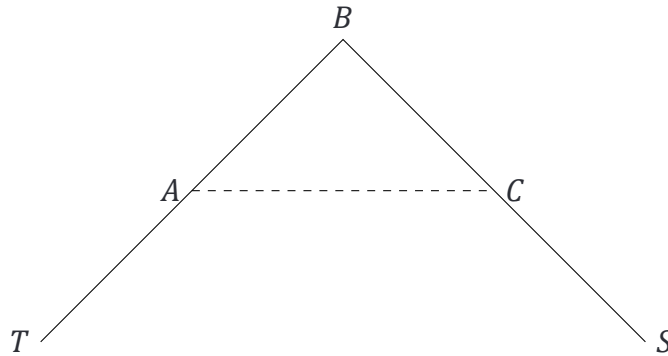
Assume true for $n = k, k \in \mathbb{Z}, k \geq 3$.

i.e, the sum of a convex polygon with k vertices is $180(k - 2)$.

RTP: The sum of interior angles of a complex polygon with $(k + 1)$ vertices is $180((k + 1) - 2)^\circ$.

Let this sum be S_n , where n is the number of vertices.. Suppose B is the corner of a convex polygon with k vertices.

If the sides AB and BC are removed and replaced with side AC , the polygon now has $(k + 1)$ vertices.



$$\begin{aligned} LHS = S_{k+1} &= S_k - \angle ABC + \angle TAC + \angle ACS \\ &= S_k - \angle ABC + (\angle ABC + \angle BCA) + \angle ACS \\ &= S_k + \angle BCA + \angle ACS \\ &= S_k + 180^\circ \\ &= 180(k - 2)^\circ + 180^\circ \\ &= 180((k - 2) + 1)^\circ \\ &= 180((k + 1) - 2)^\circ \\ &= RHS \end{aligned}$$

If the statement is true for $n = k$, it is true for $n = k + 1$.

Hence the statement is true for $n \in \mathbb{Z}, n \geq 3$, by the principle of mathematical induction.



Question 109 (4 marks) (Ruse '24 Q14b)

Base case: $n = 2$

$$\begin{aligned}a_{2-1} &= a_1 \\&= 0 \\&< \frac{1}{2} \\&= a_2\end{aligned}$$

Hence true for $n = 2$.

Assume the statement is true for $n = k, k \in \mathbb{Z}, k > 1$.

$$\therefore a_{k-1} < a_k.$$

$$\begin{aligned}\text{LHS} &= \frac{1 + a_k}{2 + a_k} - \frac{1 + a_{k-1}}{2 + a_{k-1}} \\&= \frac{(1 + a_k)(2 + a_{k-1}) - (1 + a_{k-1})(2 + a_k)}{(2 + a_k)(2 + a_{k-1})} \\&= \frac{(2 + a_{k-1} + 2a_k + a_k a_{k-1}) - (2 + a_k + 2a_{k-1} + a_k a_{k-1})}{(2 + a_k)(2 + a_{k-1})} \\&= \frac{a_k - a_{k-1}}{(2 + a_k)(2 + a_{k-1})} \\&> \frac{a_k - a_k}{(2 + a_k)(2 + a_{k-1})} \quad (\text{by assumption}) \\&= 0 \\&\therefore a_k < a_{k+1}\end{aligned}$$

$\therefore$ if the statement is true for $n = k$, then it is true for $n = k + 1$.

$\therefore$ the statement is true for $n > 1, n \in \mathbb{Z}$ by the principle of mathematical induction.



Question 110 (3 marks) (ST '24 Q14b)

Let $P(n)$ represent the proposition. For $n = 1$:

$$\begin{aligned} LHS &= \frac{d}{dx} (xe^{2x}) & RHS &= 2^0 (2x + 1) e^{2x} \\ &= e^{2x} + x \cdot 2e^{2x} & y &= (2x + 1) e^{2x} \\ &= (2x + 1)e^{2x} & &= LHS \end{aligned}$$

$\therefore$ True for $n = 1$. Assume $P(n)$ is true for $n = k$, i.e. $\frac{d^k}{dx^k} (xe^{2x}) = 2^{k-1}(2x + k)e^{2x}$. Show true for $n = k + 1$, i.e.

$$\begin{aligned} \frac{d^{k+1}}{dx^{k+1}} (xe^{2x}) &= 2^{k+1-1}(2x + k + 1)e^{2x} \\ &= 2^k(2x + k + 1)e^{2x} \end{aligned}$$

$$\begin{aligned} LHS &= \frac{d^{k+1}}{dx^{k+1}} (xe^{2x}) \\ &= \frac{d}{dx} \left(\frac{d^k}{dx^k} \right) \\ &= \frac{d}{dx} (2^{k-1}(2x + k)e^{2x}) && \text{(using assumption)} \\ &= 2^{k-1} \times [(2x + k) \times 2e^{2x} + 2 \times e^{2x}] \\ &= 2^{k-1} \times 2 \times (2x + k + 1) \times e^{2x} \\ &= 2^{k-1+1}(2x + k + 1)e^{2x} \\ &= 2^k(2x + k + 1)e^{2x} \\ &= RHS \end{aligned}$$



Question 111 (3 marks) (SG '24 Q15d)For $n = 1$:

$$\begin{aligned} LHS &= \frac{1^2}{\sqrt{2}} & RHS &= 1^2\sqrt{1} \\ &\approx 0.71 & &= 1 \end{aligned}$$

$\therefore$ Hence true for $n = 1$. Now, assume true for $n = k$. i.e, $\frac{1^2}{\sqrt{2}} + \frac{2^2}{\sqrt{3}} + \dots + \frac{k^2}{\sqrt{k+1}} < k^2\sqrt{k}$ Proving true for $n = k + 1$ i.e RTP $\frac{1^2}{\sqrt{2}} + \frac{2^2}{\sqrt{3}} + \dots + \frac{k^2}{\sqrt{k+1}} + \frac{(k+1)^2}{\sqrt{k+2}} < (k+1)^2\sqrt{k+1}$

$$\begin{aligned} LHS - RHS &= \frac{1^2}{\sqrt{2}} + \frac{2^2}{\sqrt{3}} + \dots + \frac{k^2}{\sqrt{k+1}} + \frac{(k+1)^2}{\sqrt{k+2}} - (k+1)^2\sqrt{k+1} \\ &< k^2\sqrt{k} + \frac{(k+1)^2}{\sqrt{k+2}} - (k+1)^2\sqrt{k+1} \\ &< k^2\sqrt{k} + \frac{(k+1)^2}{\sqrt{k+1}} - (k+1)^2\sqrt{k+1} \\ &= k^2\sqrt{k} + (k+1)\sqrt{k+1} - (k+1)^2\sqrt{k+1} \\ &< k^2\sqrt{k+1} + (k+1)\sqrt{k+1} - (k+1)^2\sqrt{k+1} \\ &= (k^2 + k + 1 - k^2 - 2k - 1)\sqrt{k+1} \\ &= -k\sqrt{k+1} \\ &< 0 \end{aligned}$$

$\therefore LHS < RHS \therefore$ true for $n = k + 1$, $\therefore$ true for all integers $n \geq 1$. **Marker comments:**

- This was a challenging question for many students. Students should carefully examine what is needed to be proven for the inductive step to gain insight on what steps are required to complete the proof.
- Students should read the above solutions and incorporate the techniques into their toolbox of problem solving strategies.
- **1** for correctly showing base case is true
- **1** for correct and substantial progress in the inductive step
- **1** for correctly completing the induction



Question 112 (4 marks) (BBHS '24 Q16 c)

For $n = 2$:

$$LHS = \ln(x_1 + x_2) \quad RHS = \frac{2}{2^n - 1} (\ln x_1 + \ln x_2) = \frac{1}{2} (\ln x_1 + \ln x_2)$$

Since $x_1 + x_2 > \sqrt{x_1 x_2}$ and $\ln x$ is an increasing function,

$$\begin{aligned} \ln(x_1 + x_2) &> \ln(\sqrt{x_1 x_2}) \\ \ln(x_1 + x_2) &> \frac{1}{2} \ln(x_1 x_2) \\ \ln(x_1 + x_2) &> \frac{1}{2} (\ln x_1 + \ln x_2) \\ LHS &> RHS \end{aligned}$$

$\therefore$ this is true for $n = 2$.

Now assuming true for $n = k, k \in \mathbb{Z}^+, \text{ i.e}$

$$\ln(x_1 + x_2 + \cdots + x_k) > \frac{1}{2^{k-1}} (\ln x_1 + \ln x_2 + \cdots + \ln x_k)$$

Proving true for $n = k + 1$

$$\begin{aligned} LHS &= \ln((x_1 + x_2 + \cdots + x_k) + x_{k+1}) \\ \ln((x_1 + x_2 + \cdots + x_k) + x_{k+1}) &> \frac{1}{2} (\ln(x_1 + x_2 + \cdots + x_k) + \ln x_{k+1}) \\ \ln((x_1 + x_2 + \cdots + x_k) + x_{k+1}) &> \frac{1}{2} \left(\frac{1}{2^{k-1}} (\ln x_1 + \ln x_2 + \cdots + \ln x_k) + \ln x_{k+1} \right) \\ &\quad \text{(from assumption)} \\ \ln(x_1 + x_2 + \cdots + x_k + x_{k+1}) &> \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k) + \frac{1}{2} \ln x_{k+1} \end{aligned}$$

Since $\frac{1}{2} > \frac{1}{2^k}$ and $\ln x_{k+1} > 0$, then:

$$\begin{aligned} \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k) + \frac{1}{2} \ln x_{k+1} &> \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k) + \frac{1}{2^k} \ln x_{k+1} \\ \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k) + \frac{1}{2} \ln x_{k+1} &> \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k + \ln x_{k+1}) \\ \therefore \ln(x_1 + x_2 + \cdots + x_k + x_{k+1}) &> \frac{1}{2^k} (\ln x_1 + \ln x_2 + \cdots + \ln x_k + \ln x_{k+1}) \end{aligned}$$

Hence true by mathematical induction for all integers $n \geq 2$.



Question 113 (4 marks) (Ruse '24 Q13 d)

Consider $\frac{1}{2} + \cos \theta + \cos 2\theta + \dots + \cos(n-1)\theta = \frac{\sin\left(\frac{2n-1}{2}\theta\right)}{2 \sin \frac{\theta}{2}}$. Consider $S_2 : \frac{1}{2} + \cos \theta = \frac{\sin \frac{3\theta}{2}}{2 \sin \frac{\theta}{2}}$ for $n \geq 2, n \in \mathbb{Z}, \theta \neq 2k\pi, k \in \mathbb{Z}$.

$$\begin{aligned}
 RHS &= \frac{\sin\left(\theta + \frac{\theta}{2}\right)}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\sin \theta \cos \frac{\theta}{2} + \cos \theta \sin \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\cancel{2 \sin \frac{\theta}{2}} \cos^2 \frac{\theta}{2} + \cancel{\cos \theta \sin \frac{\theta}{2}}}{\cancel{2 \sin \frac{\theta}{2}}} + \frac{\cancel{\cos \theta \sin \frac{\theta}{2}}}{\cancel{2 \sin \frac{\theta}{2}}} \\
 &= \frac{1}{2} (2 \cos^2 \frac{\theta}{2} - 1 + 1) + \frac{\cos \theta}{2} \\
 &= \frac{1}{2} (\cos \theta + 1) + \frac{1}{2} \cos \theta \\
 &= \frac{1}{2} + \cos \theta \\
 &= LHS \\
 \therefore S_2 &\text{ is true}
 \end{aligned}$$

Consider $S_k : \frac{1}{2} + \cos \theta + \cos 2\theta + \dots + \cos(k-1)\theta = \frac{\sin\left(\frac{2k-1}{2}\theta\right)}{2 \sin \frac{\theta}{2}}$ Consider also $S_{k+1} : \frac{1}{2} + \cos \theta + \cos 2\theta + \dots + \cos(k-1)\theta + \cos k\theta = \frac{\sin\left(\frac{2k+1}{2}\theta\right)}{2 \sin \frac{\theta}{2}}$. With S_{k+1} :

$$\begin{aligned}
 LHS &= \frac{1}{2} + \cos \theta + \dots + \cos(k-1)\theta + \cos k\theta \\
 &= \frac{\sin(k\theta - \frac{\theta}{2}) + 2 \cos k\theta \sin \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\sin k\theta \cos \frac{\theta}{2} - \cos k\theta \sin \frac{\theta}{2} + 2 \cos k\theta \sin \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\sin k\theta \cos \frac{\theta}{2} + \cos k\theta \sin \frac{\theta}{2}}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\sin(k\theta + \frac{\theta}{2})}{2 \sin \frac{\theta}{2}} \\
 &= \frac{\sin\left(\frac{2k+1}{2}\theta\right)}{2 \sin \frac{\theta}{2}} \\
 &= RHS
 \end{aligned}$$

$\therefore S_{k+1}$ is true if S_k is true. Since S_2 is true, $S_{2+1} = S_3$ is true, $S_{3+1} = S_4$ is true and so on for all S_n by the principle of mathematical induction.



Question 114 (4 marks) (Manly '24 Q16d)

$$\int_0^{\frac{\pi}{2}} \sin^{2n} x dx = \frac{2\pi(2n)!}{4^{n+1}(1!)^2}$$

For $S(1)$

$$\begin{aligned} LHS &= \int_0^{\frac{\pi}{2}} \sin^2 x dx \\ &= \frac{1}{2} \int_0^{\frac{\pi}{2}} (1 - \cos 2x) dx \\ &= \frac{1}{2} \left[x - \frac{1}{2} \sin 2x \right]_0^{\frac{\pi}{2}} \\ &= \frac{1}{2} \left[\frac{\pi}{2} - \frac{1}{2} \sin \pi \right] - \frac{1}{2} (0 - 0) \\ &= \frac{1}{2} \left[\frac{\pi}{2} - 0 \right] \\ &= \frac{\pi}{4} \\ &= RHS \end{aligned}$$

Therefore $S(1)$ is true. Assume $S(k)$ is true.

$$\int_0^{\frac{\pi}{2}} \sin^{2k} x dx = \frac{2\pi(2k)!}{4^{k+1}(k!)^2}$$

RTP $S(k + 1)$:

$$\int_0^{\frac{\pi}{2}} \sin^{2(k+1)} x dx = \frac{2\pi[2(k+1)]!}{4^{k+2}[(k+1)!]^2}$$

$$\begin{aligned} LHS &= \int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx = \int_0^{\frac{\pi}{2}} \sin^{2k} x \sin^2 x dx \\ &= \int_0^{\frac{\pi}{2}} \sin^{2k} x (1 - \cos^2 x) dx \\ &= \int_0^{\frac{\pi}{2}} \sin^{2k} x dx - \int_0^{\frac{\pi}{2}} \sin^{2k} x \cos^2 x dx \end{aligned}$$

Using IBP for

$$\int_0^{\frac{\pi}{2}} \sin^{2k} x \cos^2 x dx = \int_0^{\frac{\pi}{2}} \cos x \cdot [\cos x \sin^{2k} x] dx$$

Let:

$$\begin{aligned} u &= \cos x & dv &= \cos x (\sin x)^{2k} \\ du &= -\sin x & v &= \frac{\sin^{2k+1} x}{2k+1} \end{aligned}$$



$$\begin{aligned}
\therefore \int_0^{\frac{\pi}{2}} \sin^{2k} x \cos^2 x dx &= \left[\frac{\sin^{2k+1} x \cos x}{2k+1} \right]_0^{\frac{\pi}{2}} - \int_0^{\frac{\pi}{2}} \frac{-\sin^{2k+1} x \sin x}{2k+1} dx \\
&= -[0 - 0] + \int_0^{\frac{\pi}{2}} \frac{\sin^{2k+2} x}{2k+1} dx \\
&= \int_0^{\frac{\pi}{2}} \frac{\sin^{2k+2} x}{2k+1} dx \\
\therefore \int_0^{\frac{\pi}{2}} \sin^{2(k+1)} x dx &= \int_0^{\frac{\pi}{2}} \sin^{2k} x dx - \int_0^{\frac{\pi}{2}} \frac{\sin^{2k+1} x}{2k+1} dx
\end{aligned}$$

$$\begin{aligned}
\left[1 + \frac{1}{2k+1} \right] \int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \int_0^{\frac{\pi}{2}} \sin^{2k} x dx \\
\frac{2k+2}{2k+1} \int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi(2k)!}{4^{k+1}(k!)^2} && \text{(from assumption)} \\
\int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi(2k+1)(2k)!}{4^{k+1}(2k+2)(k!)^2} \\
\int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi(2k+2)!}{4^{k+1}(2k+2)^2(k!)^2} \\
\int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi[2(k+1)]!}{4^{k+1}([2(k+1)](k!))^2} \\
\int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi[2(k+1)]!}{4^{k+1}(4)[(k+1)k!]^2} \\
\int_0^{\frac{\pi}{2}} \sin^{2k+2} x dx &= \frac{2\pi[2(k+1)]!}{4^{k+2}[(k+1)!]^2}
\end{aligned}$$

Therefore $S(k) \Rightarrow S(k+1)$ therefore the statement is true by induction.



Question 115 (4 marks) (SBHS '24 Q13b)

(i)

$$\begin{aligned}u_1 &= \frac{9}{4} - \frac{1}{1} - 2^{-1+1} \\&= \frac{9}{4} - 1 - 2^0 \\&= \frac{1}{4}\end{aligned}$$

Hence true for $n = 1$. Induction Hypothesis: Assume true for $n = k$

$$u_k = \frac{9}{4} - \frac{1}{k} - 2^{-k+1}$$

Inductive Step: Prove true for $n = k + 1$, i.e. prove that

$$\begin{aligned}u_{k+1} &= \frac{9}{4} - \frac{1}{k+1} - 2^{-(k+1)+1} \\&= \frac{9}{4} - \frac{1}{k+1} - 2^{-k}\end{aligned}$$

Using the given expression for u_{n+1} and the expression for u_k from the induction hypothesis

$$\begin{aligned}u_{k+1} &= u_k + \frac{1}{k(k+1)} + 2^{-k} \\&= \frac{9}{4} - \frac{1}{k} - 2^{-k+1} + \frac{1}{k(k+1)} + 2^{-k} \\&= \frac{9}{4} - \left(\frac{1}{k} - \frac{1}{k(k+1)} \right) - ((2 \times 2^{-k}) - 2^{-k}) \\&= \frac{9}{4} - \left(\frac{k+1-1}{k(k+1)} \right) - 2^{-k}(2-1) \\&= \frac{9}{4} - \frac{1}{k+1} - 2^{-k}\end{aligned}$$

Hence, by the principle of mathematical induction:

$$u_n = \frac{9}{4} - \frac{1}{n} - 2^{-n+1}$$

Marker note: generally answered well

(ii) From part (i)

$$\begin{aligned}u_n &= \frac{9}{4} - \frac{1}{n} - 2^{-n+1} \\&= \frac{9}{4} - \left(\frac{1}{n} + 2^{-n+1} \right)\end{aligned}$$



Recognise that $\frac{1}{n} + 2^{-n+1} > 0$ for $n \in \mathbb{Z}^+$, so:

$$\begin{aligned} -\left(\frac{1}{n} + 2^{-n+1}\right) &< 0 \\ \frac{9}{4} - \left(\frac{1}{n} + 2^{-n+1}\right) &< \frac{9}{4} \\ \therefore u_n &< \frac{9}{4} \end{aligned}$$

Common errors:

- Not mentioning the direction that $\frac{1}{n} + 2^{-n+1} > 0$ is approaching 0 from.

Question 116 (3 marks) (NSG '24 Q16a)

Base case:

$$14 = 8 + 2(3)$$

Assume true for $\$n$, where $n \geq 14$. Prove true for $\$(n + 1)$:

Case 1: There is at least one \$8 coin.

Replace one \$ coin by three \$3 coins.

\$8 has become \$9, so \$n has become $\$(n + 1)$.

Case 2: There are no \$8 coins.

As n is at least 14, there are at least five \$3 coins.

Replace five \$3 coins by 2 \$8 coins.

\$15 has become \$16 so \$n has become $\$(n + 1)$

$\therefore$ true for $n + 1$ when true for n . As true for $n = 14$, and true for $n + 1$ when true for n , $\therefore$ by MI, every whole dollar amount of \$14 or more can be achieved by taking a mix of \$3 and \$8 coins.



Question 117 (5 marks) (SGGHS '24 Q15a)

(i)

$$\begin{aligned}
 RHS - LHS &= \frac{2(k+1)(k+2) - 2k(k+3)}{(k+3)(k+2)} \\
 &= \frac{2(k^2 + 3k + 2) - 2k^2 - 6k}{(k+3)(k+2)} \\
 &= \frac{\cancel{2k^2} + \cancel{6k} + 4 - \cancel{2k^2} - \cancel{6k}}{(k+3)(k+2)} \\
 &= \frac{4}{(k+3)(k+2)} \\
 &> 0 \\
 \therefore \frac{2k+2}{2k+3} - \frac{2k}{k+2} &> 0 \\
 \therefore \frac{2k}{k+2} &< \frac{2k+2}{k+3}
 \end{aligned}$$

(ii) For $n = 1$:

$$\begin{aligned}
 LHS &= \frac{1}{3!} & RHS &= \frac{2(1)}{1+2} - \frac{1}{(1+2)!} \\
 &= \frac{1}{6} & &= \frac{2}{3} - \frac{1}{3!} = \frac{1}{2}
 \end{aligned}$$

Now $\frac{1}{6} < \frac{1}{2}$, i.e $LHS < RHS$, $\therefore$ true for $n = 1$.

Assume the statement is true for $n = k, k \in \mathbb{Z}^+, k \geq 1$, i.e

$$\frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \cdots + \frac{k}{(k+2)!} < \frac{2k}{k+2} - \frac{1}{(k+2)!}$$

Prove true for $n = k + 1$, i.e prove

$$\frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \cdots + \frac{k}{(k+2)!} + \frac{k+1}{(k+3)!} < \frac{2(k+1)}{k+3} - \frac{1}{(k+3)!}$$



$$\begin{aligned}
LHS &= \frac{1}{3!} + \frac{2}{4!} + \frac{3}{5!} + \cdots + \frac{k}{(k+2)!} + \frac{k+1}{(k+3)!} \\
&< \frac{2k}{k+2} - \frac{1}{(k+2)!} + \frac{k+1}{(k+3)!} \text{ by } te \\
&= \frac{2k}{k+2} - \frac{1}{(k+2)!} + \frac{k+2-1}{(k+3)!} \\
&= \frac{2k}{k+2} - \frac{1}{(k+2)!} + \frac{k+2}{(k+3)!} - \frac{1}{(k+3)!} \\
&= \frac{2k}{k+2} - \frac{k+3}{(k+2)!} \cdot \frac{k+2}{k+3} + \frac{k+2}{(k+3)!} - \frac{1}{(k+3)!} \\
&= \frac{2k}{k+2} - \frac{k+3}{(k+3)!} - \frac{k+2}{(k+3)!} - \frac{1}{(k+3)!} \\
&= \frac{k+2}{2k} - \left(\frac{k+3}{(k+3)!} - \frac{k+2}{(k+3)!} \right) - \frac{1}{(k+3)!} \\
&= \frac{2k}{k+2} - \frac{k+3-k-2}{(k+3)!} - \frac{1}{(k+3)!} \\
&= \frac{2k}{k+2} - \frac{2}{(k+3)!} \\
&< \frac{2k}{k+2} - \frac{1}{(k+3)!} \\
&< \frac{2(k+1)}{k+3} - \frac{1}{(k+3)!}
\end{aligned}$$

$\therefore$ true for $n = k+1$ when true for $n = k$. By the principle of mathematical induction, the statement is true for all integers $n \geq 1$.



Question 118 (4 marks) (SGGHS '24 Q16a)

Base case: $n = 0$

$$\begin{aligned}T_n &= 2 \cos\left(\frac{\pi}{2^{n+1}}\right) \\T_0 &= 2 \cos\left(\frac{\pi}{2^{0+1}}\right) \\&= 2 \cos\left(\frac{\pi}{2}\right) \\&= 2 \times 0 \\&= 0\end{aligned}$$

$\therefore$ true for $n = 0$.

Assume true for $n = k$, i.e

$$T_k = 2 \cos\left(\frac{\pi}{2^{k+1}}\right)$$

Prove true for $n = k + 1$, i.e

$$T_{k+1} = 2 \cos\left(\frac{\pi}{2^{k+2}}\right)$$

$$\begin{aligned}LHS &= T_{k+1} \\&= \sqrt{T_k + 2} \\&= \sqrt{2 \cos\left(\frac{\pi}{2^{k+1}}\right) + 2} \\&= \sqrt{2 \left[\cos\left(\frac{\pi}{2^{k+1}}\right) + 1 \right]} \\&= \sqrt{2 \left[\cos 2\left(\frac{\pi}{2 \cdot 2^{k+1}}\right) + 1 \right]} \\&= \sqrt{2 \left[\cos 2\left(\frac{\pi}{2^{k+2}}\right) + 1 \right]} \\&= \sqrt{2 \times 2 \cos^2\left(\frac{\pi}{2^{k+2}}\right)} \\&= \sqrt{4 \cos^2\left(\frac{\pi}{2^{k+2}}\right)} \\&= 2 \cos\left(\frac{\pi}{2^{k+2}}\right) \\&= RHS\end{aligned}$$

$\therefore LHS = RHS$, $\therefore$ true for $n = k + 1$ if true for $n = k$. Hence by mathematical induction, it is true for all integer $n \geq 0$.



Question 119 (6 marks) (Baulko '23 Q16c)

(i) For $n = 1$:

$$\begin{aligned} RHS &= \frac{1}{\sqrt{b_1}} B_1 \quad RHS = \sqrt{b_1} \\ &= \frac{1}{\sqrt{b_1}} b_1 \\ &= \sqrt{b_1} \end{aligned}$$

Hence true for $n = 1$

Assume true for $n = k$, i.e

$$\frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 = \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k}$$

For $n = k + 1$ we wish to prove

$$\begin{aligned} &\frac{1}{\sqrt{b_{k+1}}} B_{k+1} + \left(\frac{1}{\sqrt{b_k}} - \frac{1}{\sqrt{b_{k+1}}} \right) B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \\ &= \sqrt{b_1} + \sqrt{b_2} + \dots + \sqrt{b_k} \\ LHS &= \frac{1}{\sqrt{b_{k+1}}} (B_{k+1} + \left(\frac{1}{\sqrt{b_k}} - \frac{1}{\sqrt{b_{k+1}}} \right) B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1) \\ &= \frac{1}{\sqrt{b_{k+1}}} (B_{k+1} - B_k) + \left(\frac{1}{\sqrt{b_k}} B_k + \left(\frac{1}{\sqrt{b_{k-1}}} - \frac{1}{\sqrt{b_k}} \right) B_{k-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \right) \\ &= \frac{1}{\sqrt{b_{k+1}}} (b_{k+1}) + \sum_{r=1}^k \sqrt{b_r} \\ &= \sqrt{b_{k+1}} + \sum_{r=1}^k \sqrt{b_r} \\ &= \sum_{r=1}^{k+1} \sqrt{b_r} \end{aligned}$$

Hence true by mathematical induction.

(ii)

$$\begin{aligned} \sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} &= \frac{1}{\sqrt{b_k}} A_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) A_{n-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) A_1 \\ &\leq \frac{1}{\sqrt{b_k}} B_n + \left(\frac{1}{\sqrt{b_{n-1}}} - \frac{1}{\sqrt{b_n}} \right) B_{n-1} + \dots + \left(\frac{1}{\sqrt{b_1}} - \frac{1}{\sqrt{b_2}} \right) B_1 \quad \text{since } A_i \leq B_i \text{ for all } 0 < i \leq n \\ &= \sum_{r=1}^n \sqrt{b_r} \end{aligned}$$



(iii) Since all $a_r > 0, b_r > 0$ for $0 < r \leq n$, from (ii),

$$\sqrt{a_r} < \frac{1}{2} \left(\frac{a_r}{\sqrt{b_r}} + \sqrt{b_r} \right) \quad 0 < r \leq n$$

Summing over n terms,

$$\begin{aligned} \sum_{r=1}^n \sqrt{a_r} &\leq \sum_{r=1}^n \frac{1}{2} \left(\frac{a_r}{\sqrt{b_r}} + \sqrt{b_r} \right) \\ &= \frac{1}{2} \left(\sum_{r=1}^n \frac{a_r}{\sqrt{b_r}} \right) + \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) \quad (\text{rearranging the sum}) \\ &\leq \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) + \frac{1}{2} \left(\sum_{r=1}^n \sqrt{b_r} \right) \quad (\text{using the result from Q8(c)(ii)}) \\ &= \sum_{r=1}^n \sqrt{b_r} \quad \text{as required} \end{aligned}$$

Question 120 (3 marks) (Killara '24 Q13b)

Base case, let $n = 0$:

$$x^0 - y^0 = 0$$

which is divisible by $(x + y)$. $n = 2$:

$$x^2 - y^2 = (x - y)(x + y) \therefore \text{true for } n = 2$$

Assume true for $n = k$:

$$x^k - y^k = M(x, y)(x + y) \quad (\text{where } M(x, y) \text{ is a polynomial})$$

For $n = k + 2$:

$$\begin{aligned} LHS &= x^{k+2} - y^{k+2} \\ &= x^2 x^k - y^2 y^k \\ &= x^2 x^k - x^2 y^k + x^2 y^k - y^2 y^k \\ &= x^2 (x^k - y^k) + y^k (x^2 - y^2) \\ &= x^2 (x + y) M(x, y) + y^k (x + y)(x - y) \\ &= (x + y) [x^2 M(x, y) + y^k (x - y)] \\ n = k &\Rightarrow n = k + 2 \end{aligned}$$

As $n = 0, n = 2, n = k$ and $n = k + 2$ are all true, the statement is true for all $n \geq 0$ ($n \geq 2$) by the principal of Mathematical Induction.



Question 121 (5 marks) (Girraween '24 Q14a)

(i)

$$k^2 - k - 2 = (k - 2)(k + 1)$$

$$\therefore k = 2, -1$$

(ii) Base case: $n = 2$

$$\begin{array}{ll} LHS = 3^2 & RHS = 2 \times 2^2 - 3 \\ = 9 & = 5 \end{array}$$

$\therefore LHS > RHS$ is true for $n = 2$.

Assume true for $n = k$, i.e

$$3^k > 2k^2 - 3, k \geq 2$$

Prove true for $n = k + 1$, i.e prove

$$3^{k+1} > 2k + 1^2 - 3$$

or

$$3^{k+1} > 2k^2 + 4k - 1, k \geq 2$$

$$LHS = 3^{k+1} = 3 \times 3^k$$

$$> 3[2k^2 - 3]$$

(by assumption)

$$= 6k^2 - 9$$

$$6k^2 - 9 - (2k^2 + 4k - 1) = 4k^2 - 4k - 8$$

$$4(k^2 - k - 2) > 0 \quad k \geq 2$$

$$6k^2 - 9 > 2k^2 + 4k - 1$$

$$3^{k+1} > 6k^2 - 9 > 2k^2 + 4k - 1$$

$$3^{k+1} > 2k^2 + 4k - 1$$

Hence true for $n = k + 1$. $\therefore$ if true for $n = k$, it will be true for $n = k + 1$, and the statement is true for all integers $n \geq 2$ by the principle of mathematical induction.



Question 122 (4 marks) (NSB '24 Q13a)

(i)

$$\begin{aligned}(k-1)(k+1) &< k^2 \\(k-1)k(k+1) &< k^3 \\ \Rightarrow \frac{1}{(k-1)k(k+1)} &> \frac{1}{k^3}\end{aligned}$$

(ii) Let $f(k) = -\frac{1}{2k(k+1)}$

$$\begin{aligned}f(k) - f(k-1) &= \frac{1}{2(k-1)k} - \frac{1}{2k(k+1)} \\&= \frac{(k+1) + (k-1)}{2(k-1)k(k+1)} \\&= \frac{1}{(k-1)k(k+1)} \\ \therefore \frac{1}{k^3} &< \frac{1}{k(k-1)(k+1)} \\ \Rightarrow \frac{1}{k^3} &< \cancel{f(k)} - \cancel{f(k-1)} \\ \Rightarrow \frac{1}{3^3} &< \cancel{f(3)} - \cancel{f(2)} \\ \frac{1}{4^3} &< \cancel{f(4)} - \cancel{f(3)} \\ \frac{1}{5^3} &< \cancel{f(5)} - \cancel{f(4)} \\ &\vdots \\ \frac{1}{(n-2)^3} &< \cancel{f(n-2)} - \cancel{f(n-3)} \\ \frac{1}{(n-1)^3} &< \cancel{f(n-1)} - \cancel{f(n-2)} \\ \frac{1}{n^3} &< f(n) - \cancel{f(n-1)}\end{aligned}$$

Adding all,

$$\begin{aligned}S_n &< f(n) - f(n-1) \\&= -\frac{1}{2n(2n+1)} - \frac{1}{2 \cdot 2(3)} \\&= \frac{1}{12} - \frac{1}{2n(2n+1)} \\&< \frac{1}{12}\end{aligned}$$



Question 123 (4 marks) (Normo '21 AT2 Q2c)

(i)

$$\begin{aligned}(1+a)^p - 1 - a^p &= \binom{p}{0} + \binom{p}{1}a + \binom{p}{2}a^2 + \dots + \binom{p}{p-1}a^{p-1} + \binom{p}{p}a^p - 1 - a^p \\ &= \binom{p}{1}a + \binom{p}{2}a^2 + \dots + \binom{p}{p-1}a^{p-1}\end{aligned}$$

$\therefore (1+a)^p - 1 - a^p$ is divisible by p .

(ii) Let $P(a)$ be the proposition

$$a^p - a \text{ is divisible by } p \quad \forall a \in \mathbb{Z}^+$$

. Base case: $P(1)$:

$$1^p - 1 = 0, \quad \text{which is divisible by } p$$

Hence $P(1)$ is true.

Inductive step: assume $P(k)$ is true, $k \in \mathbb{Z}^+$, i.e

$$k^p - k = pM, \quad \text{where } M \text{ is a polynomial in terms of } k$$

Test $P(k+1)$:

RTP:

$$(k+1)^p - (k+1) \text{ is divisible by } p$$

From (i),

$$\begin{aligned}(1+k)^p - 1 - k^p &= pN, \quad N \in \mathbb{Z}^+ \\ (k+1)^p - (k+1) &= pN + 1 + k^p - k - 1 \\ &= pN + pM \quad \dots \text{by assumption} \\ &= p(N+M) \quad \text{where } N+M \text{ is a polynomial,}\end{aligned}$$

Hence $P(k+1)$ is true.

By mathematical induction, $P(a)$ is true $\forall n \in \mathbb{Z}^+$.



Question 124 (3 marks) (Normo '22 Q16a)

(i)

$$\begin{aligned}\left(\sqrt{a} + \frac{1}{\sqrt{a}}\right)^2 &\geq 0 \\ a - 2 + \frac{1}{a} &\geq 0 \\ a + \frac{1}{a} &\geq 2\end{aligned}$$

(ii) Let the proposition be $P(n)$.

Base case: $P(1)$

$$\begin{aligned}LHS &= a + \frac{1}{a} \\ RHS &= a^0 + \frac{1}{a^0} = 2 \\ a + \frac{1}{a} &\geq 2\end{aligned}\quad \text{(from (i))}$$

$\therefore LHS > RHS$ and $P(1)$ is true.

Inductive step: Assume $P(k)$ is true, $k \in \mathbb{Z}^+$, i.e

$$a^k + \frac{1}{a^k} \geq a^{k-1} + \frac{1}{a^{k-1}}$$

Prove: Examine $P(k+1)$

$$\text{RTP: } a^{k+1} + \frac{1}{a^{k+1}} \geq a^k + \frac{1}{a^k}$$

$$\begin{aligned}\frac{1}{a^{k+1}} &= \left(a + \frac{1}{a}\right) \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from given} \\ LHS &= \left(a + \frac{1}{a}\right) \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \\ &\geq 2 \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from i} \\ &\geq \left(a^k + \frac{1}{a^k}\right) + \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) \quad \text{from i} \\ \left(a^k + \frac{1}{a^k}\right) - \left(a^{k-1} + \frac{1}{a^{k-1}}\right) &\geq 0 \quad \text{from inductive step} \\ LHS &\geq \left(a^k + \frac{1}{a^k}\right) + 0\end{aligned}$$

$\therefore LHS \geq RHS$, and $P(k+1)$ is true if $P(k)$ is true.

Hence by mathematical induction, $P(n)$ is true for all positive integers $n \geq 1$.



Question 125 (5 marks) (NSG '20 Q15a)

(i) For $n = 6$:

$$\begin{aligned} LHS &= \frac{1}{6!} & RHS &= \frac{1}{e^6} \\ &= \frac{1}{720} & &= 0.00247 \dots \\ &= 0.0013 \dots & \therefore LHS &< RHS \end{aligned}$$

$\therefore$ true for $n = 6$.

Assume true for $n = k, k \in \mathbb{Z}^+, k \geq 6$, i.e

$$\frac{1}{k!} < \frac{1}{e^k}$$

Prove true for $n = k + 1$, i.e

$$\text{RTP: } \frac{1}{(k+1)!} < \frac{1}{e^{k+1}}$$

$$\begin{aligned} LHS &= \frac{1}{k+1} \\ &= \frac{1}{(k+1)k!} = \frac{1}{k+1} \times \frac{1}{k!} \\ &< \frac{1}{k+1} \cdot \frac{1}{e^k} \\ &< \frac{1}{e} \cdot \frac{1}{e^k} \\ &= \frac{1}{e^{k+1}} = RHS \end{aligned}$$

$LHS < RHS \therefore$ true for $n = k + 1$ when true for $n = k$, hence the statement is true by mathematical induction.

(ii)

$$\begin{aligned} \frac{1}{6!} &< \frac{1}{e^6} \\ \frac{1}{7!} &< \frac{1}{e^7} \\ \frac{1}{8!} &< \frac{1}{e^8} \\ \vdots & \quad \quad \vdots \end{aligned}$$



Adding, we get:

$$\begin{aligned}
 \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \dots &< \frac{1}{e^6} + \frac{1}{e^7} + \frac{1}{e^8} + \dots \\
 &= \frac{\frac{1}{e^6}}{1 - \frac{1}{e}} \\
 &= \frac{1}{e^6} \times \frac{e}{e-1} \\
 &= \frac{1}{e^6 - e^5} \\
 \therefore \frac{1}{6!} + \frac{1}{7!} + \frac{1}{8!} + \dots &< \frac{1}{e^6 - e^5}
 \end{aligned}$$

Question 126 (3 marks) (Knox '22 Q16a)

For $n = 1$:

$$\begin{aligned}
 LHS &= 3^1 & RHS &= 1 + 3 - 1 \\
 &= 3 & &= 3
 \end{aligned} \tag{2}$$

As $LHS \geq RHS$, the statement is true for $n = 1$.

Assume the statement is true for $n = k$, i.e

$$3^k \geq k^2 + 3k - 1$$

$$3^{k+1} \geq (k+1)^2 + 3(k+1) - 1$$

Prove true for $n = k + 1$, i.e $3^{k+1} \geq k^2 + 2k + 1 + 3k + 3 - 1$ Now,

$$3^{k+1} \geq k^2 + 5k + 3$$

$$3^k \geq k^2 + 3k - 1 \quad \text{from assumption}$$

$$3^{k+1} \geq 3k^2 + 9k - 3$$

Now consider

$$\begin{aligned}
 y &= (3k^2 + 9k - 3) - (k^2 + 5k + 3) \\
 &= 3k^2 + 9k - 3 - k^2 - 5k - 3 \\
 &= 2k^2 + 4k - 6 \\
 &= 2(k+3)(k-1)
 \end{aligned}$$

which is a parabola with a root at $k = 1$, and is positive for $k \geq 1 \Rightarrow y \geq 0$ for any $k \geq 1$, i.e:

$$(3k^2 + 9k - 3) - (k^2 + 5k + 3) \geq 0 \quad k \geq 1 \tag{3}$$

$$3k^2 + 9k - 3 \geq k^2 + 5k + 3 \tag{4}$$

From (1) and (2), it can be deduced that

$$3^{k+1} \geq 3k^2 + 9k - 3 \geq k^2 + 5k + 3$$

. Hence if the statement is true for $n = k$, it is also true for $n = k + 1$, by the principle of mathematical induction. Hence it is true for all positive integers $n \geq 1$.



Question 127 (10 marks) (SGHS '22 Q16c)

(i)

$$\begin{aligned} u &= \frac{x^{2n-1}}{(2n-1)!} & v &= -\cos x \\ u' &= \frac{(2n-1)x^{2n-2}}{(2n-1)!} & v' &= \sin x \\ &= \frac{x^{2n-2}}{(2n-2)!} \end{aligned}$$

$$\begin{aligned} I_n &= \left[\frac{x^{2n-1}}{(2n-1)!} \times -\cos x \right]_0^1 - \int_0^1 \frac{x^{2n-2}}{(2n-2)!} \times -\cos x \, dx \\ &= -\frac{\cos 1}{(2n-1)!} - (-0) + \int_0^1 \frac{x^{2n-2}}{(2n-2)!} \times \cos x \, dx \end{aligned}$$

$$\begin{aligned} u &= \frac{x^{2n-2}}{(2n-2)!} & v &= \sin x \\ u' &= \frac{(2n-2)x^{2n-3}}{(2n-2)!} & v' &= \cos x \\ &= \frac{x^{2n-3}}{(2n-3)!} \end{aligned}$$

$$\begin{aligned} \int_0^1 \frac{x^{2n-2}}{(2n-2)!} \times \cos x \, dx &= \left[\frac{x^{2n-2}}{(2n-2)!} \times \sin x \right]_0^1 - \int_0^1 \frac{x^{2n-3}}{(2n-3)!} \times \sin x \, dx \\ &= \frac{\sin 1}{(2n-2)!} - 0 - \int_0^1 \frac{x^{2(n-1)-1}}{(2(n-1)-1)!} \times \sin x \, dx \\ &= \frac{\sin 1}{(2n-2)!} - I_{n-1} \\ I_n &= -\frac{\cos 1}{(2n-1)!} + \left(\frac{\sin 1}{(2n-2)!} - I_{n-1} \right) \\ &= \frac{\sin 1}{(2n-2)!} - \frac{\cos 1}{(2n-1)!} - I_{n-1} \end{aligned}$$



(ii) Prove true for $n = 1$:

$$\begin{aligned}
 I_1 &= \int_0^1 \frac{x^{2 \times 1 - 1}}{(2 \times 1 - 1)!} dx \\
 &= \int_0^1 x \sin x dx \\
 &= [x \times -\cos x]_0^1 - \int_0^1 1 \times -\cos x dx \\
 &= -\cos 1 - 0 + [\sin x] \\
 &= -\cos 1 + \sin 1 - 0 \\
 &= \frac{1 \times \sin 1 + (-1) \times \cos 1}{(2 \times 1 - 1)!} \\
 &= \frac{a_1 \sin 1 + b_1 \cos 1}{(2 \times 1 - 1)!}
 \end{aligned}$$

where $a_1 = 1, b_1 = -1$
 $\therefore$ statement is true for $n = 1$.
 Assume true for $n = k$, i.e

$$\frac{a_k \sin 1 + b_k \cos 1}{(2k - 1)!} \quad \text{where } a, b, k \in \mathbb{Z}^+$$

Prove true for $n = k + 1$, i.e prove

$$I_{k+1} = \frac{a_{k+1} \sin 1 + b_{k+1} \cos 1}{(2(k + 1) - 1)!} \quad \text{where } a, b, k \in \mathbb{Z}^+$$

$$\begin{aligned}
 LHS &= I_{k+1} \\
 &= \frac{\sin 1}{(2(k + 1) - 2)!} - \frac{\cos 1}{(2(k + 1) - 1)!} - I_{k+1-1} \\
 &= \frac{\sin 1}{(2k + 2 - 2)!} - \frac{\cos 1}{(2k + 2 - 1)!} - I_k \\
 &= \frac{\sin 1}{(2k)!} - \frac{\cos 1}{(2k + 1)!} - \frac{a_k \sin 1 + b_k \cos 1}{(2k - 1)!} \\
 &= \frac{(2k + 1) \sin 1}{(2k + 1)(2k)!} - \frac{\cos 1}{(2k + 1)!} - \frac{(2k + 1) \times 2k \times (a_k \sin 1 + b_k \cos 1)}{(2k + 1) \times 2k \times (2k - 1)!} \\
 &= \frac{(2k + 1) \sin 1 - \cos 1 - 2k(2k + 1)(a_k \sin 1 + b_k \cos 1)}{(2k + 1)!} \\
 &= \frac{((2k + 1) - 2k(2k + 1)a_k) \sin 1 + (-1 - 2k(2k + 1)b_k) \cos 1}{(2(k + 1) - 1)!} \\
 &= \frac{a_{k+1} \sin 1 + b_{k+1} \cos 1}{(2(k + 1) - 1)!} \\
 \therefore LHS &= RHS
 \end{aligned}$$

where:

$$\begin{aligned}
 a_{k+1} &= (2k + 1) - 2k(2k + 1)a_k \\
 b_{k+1} &= -1 - 2k(2k + 1)b_k
 \end{aligned}$$



If true for $n = k$, then true for $n = k + 1$. Since proven true for $n = 1$, statement is proven true by mathematical induction for every positive integer n .

(iii) Since $0 < I_n < \frac{1}{(2n)!}$, $0 < \frac{a_n \sin 1 + b_n \cos 1}{(2n-1)!} < \frac{1}{(2n)!}$ Multiply through by $(2n-1)!$:

$$0 < a_n \sin 1 + b_n \cos 1 < \frac{1}{2n \times (2n-1)!} \times (2n-1)!$$

$$0 < a_n \sin 1 + b_n \cos 1 < \frac{1}{2n}$$

Divide through by $\cos 1$ and considering $\cos 1 > \frac{1}{2} > 0$:

$$0 < \frac{a_n \sin 1}{\cos 1} + \frac{b_n \cos 1}{\cos 1} < \frac{1}{2n} \times \frac{1}{\cos 1}$$

$$0 < a_n \tan 1 + b_n < \frac{1}{2n} \times \frac{1}{\cos 1}$$

Since $\cos 1 > \frac{1}{2}$, $\frac{1}{\cos 1} < 2$ and $\frac{1}{2n} \times \frac{1}{\cos 1} < \frac{1}{2n} \times 2$.

Hence

$$0 < a_n \tan 1 + b_n < \frac{1}{2n} \times \frac{1}{\cos 1} < \frac{1}{2n} \times 2.$$

$$\therefore 0 < a_n \tan 1 + b_n < \frac{1}{n}$$

(iv) Assume $\tan 1$ is rational, i.e assume $\tan 1 = \frac{p}{q}$, where $p, q \in \mathbb{Z}^+$ and $HCF(p, q) = 1$.

From (iii),

$$0 < a_n \times \frac{p}{q} + b_n < \frac{1}{n}$$

As this inequality is true for all positive integer values of n , then it will be true for $n = q$ since q is a positive integer.

Hence using $n = q$:

$$0 < a_q \times \frac{p}{q} + b_q < \frac{1}{q}$$

Multiplying through by q where $q > 0$

$$0 < a_q \times p + b_q \times q < 1$$

Given that a_q, b_q, p and q are all integers, the value of the expression $a_q \times p + b_q \times q$ must also be an integer.

Hence, it cannot have a value between 0 and 1 as stated prior.

$\therefore$ the assumption that $\tan 1$ is rational must be false. Hence, by contradiction, $\tan 1$ is irrational.



Question 128 (3 marks) (NSG '13 Q16c)

For $n = 1$, $LHS = \tan^{-1} \frac{1}{2}$ hence true for $n = 1$.

$$RHS = \frac{\pi}{4} - \tan^{-1} \frac{1}{3}$$

$$\begin{aligned} LHS - RHS &= \tan^{-1} \frac{1}{2} - \frac{\pi}{4} + \tan^{-1} \frac{1}{3} \\ &= \tan^{-1} \frac{\frac{1}{2} + \frac{1}{3}}{1 - \frac{1}{2} \cdot \frac{1}{3}} - \frac{\pi}{4} \\ &= \tan^{-1} 1 - \frac{\pi}{4} \\ &= \frac{\pi}{4} - \frac{\pi}{4} \\ &= 0 \end{aligned}$$

Assume true for $n = k$, i.e. $LHS = RHS$

$$\tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2 \times k^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+1}$$

Prove true for $n = k + 1$, i.e. RTP

$$\tan^{-1} \frac{1}{2 \times 1^2} + \tan^{-1} \frac{1}{2 \times 2^2} + \tan^{-1} \frac{1}{2 \times 3^2} + \dots + \tan^{-1} \frac{1}{2 \times (k+1)^2} = \frac{\pi}{4} - \tan^{-1} \frac{1}{2k+3}$$

$$\begin{aligned} LHS - RHS &= \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{2k+1} \right) + \tan^{-1} \frac{1}{2(k+1)^2} - \left(\frac{\pi}{4} - \tan^{-1} \frac{1}{2k+3} \right) \\ &= \tan^{-1} \frac{1}{2(k+1)^2} + \tan^{-1} \frac{1}{2k+3} - \tan^{-1} \frac{1}{2k+1} \\ &= \tan^{-1} \frac{\frac{1}{2(k+1)^2} + \frac{1}{2k+3}}{1 - \frac{1}{2(k+1)^2} \cdot \frac{1}{2k+3}} \times \frac{2(k+1)^2(2k+3)}{2(k+1)^2(2k+3)} - \tan^{-1} \frac{1}{2k+1} \\ &= \tan^{-1} \frac{(2k+3) + 2(k+1)^2}{2(k+1)^2(2k+3) - 1} - \tan^{-1} \frac{1}{2k+1} \\ &= \tan^{-1} \frac{2k^2 + 6k + 5}{4k^3 + 14k^2 + 16k + 5} - \tan^{-1} \frac{1}{2k+1} \\ &= \tan^{-1} \frac{2k^2 + 6k + 5}{(2k+1)(2k^2 + 6k + 5)} - \tan^{-1} \frac{1}{2k+1} \\ &= \tan^{-1} \frac{1}{2k+1} - \tan^{-1} \frac{1}{2k+1} \\ &= 0 \end{aligned}$$

$$LHS = RHS$$

$\therefore$ true for $n = k + 1$ when true for $n = k$

$\therefore$ by Mathematical induction, true for all positive integers n .



Question 129 (3 marks) (HAHS '24 Q13d)Base case: $n = 1$

$$\begin{aligned} LHS &= x^3 - 1 & RHS &= (x - 1)(x^2 + x + 1) \\ & & &= LHS \end{aligned}$$

Assume true for $n = k, k \in \mathbb{Z}$, i.e

$$x^{3^k} - 1 = (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^{k-1}} + x^{3^{k-1}} + 1)$$

Prove true for $n = k + 1$, i.e

$$x^{3^{k+1}} - 1 = (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^k} + x^{3^k} + 1)$$

$$\begin{aligned} RHS &= (x - 1)(x^2 + x + 1) \dots (x^{2 \times 3^{k-1}} + x^{3^{k-1}} + 1) (x^{2 \times 3^k} + x^{3^k} + 1) \\ &= (x^{3^k} - 1) (x^{2 \times 3^k} + x^{3^k} + 1) \\ &= (x^{3^k})^3 - 1 \\ &= x^{3^{k+1}} - 1 \end{aligned}$$

Question 130 (3 marks) (SGS '21 Q15a)For $n = 1$:

$$\begin{aligned} LHS &= (1 + x)^1 & RHS &= 1 + 1 \times x \\ &= 1 + x & &= 1 + x \end{aligned}$$

Hence true for $n = 1$.Assume true for $n = k, k \in \mathbb{Z}^+$, i.e

$$(1 + x)^n \geq 1 + nx.$$

Proving true for $n = k + 1$, i.e

$$(1 + x)^{k+1} \geq 1 + (k + 1)x$$

$$\begin{aligned} LHS &= (1 + x)^{k+1} \\ &= (1 + x)(1 + x)^k \\ &\geq (1 + x)(1 + kx) && \text{(from assumption)} \\ &= 1 + x + kx + kx^2 \\ &= 1 + (k + 1)x + kx^2 \\ &\geq 1 + (k + 1)x && \text{(since } x^2 \geq 0, k \in \mathbb{Z}^+) \end{aligned}$$

Hence $LHS \geq RHS$, and by the principle of mathematical induction, the inequality is true $\forall n \in \mathbb{Z}^+$ 

Question 131 (5 marks) (Ruse '22 AT3 Q10)

For $n = 1$:

$$\begin{aligned} f_1(\sin^2 \theta) &= \frac{1}{2}(1 - \sqrt{1 - \sin^2 \theta}) \\ &= \frac{1}{2}(1 - \sqrt{\cos^2 \theta}) \\ &= \frac{1}{2}(1 - |\cos \theta|) \\ &= \frac{1}{2}(1 - \cos \theta) && (\text{since } 0 < \theta < \frac{\pi}{2}) \\ &= \frac{1}{2}(1 - (1 - 2 \sin^2 \frac{\theta}{2})) \\ &= \sin^2 \frac{\theta}{2} \end{aligned}$$

$\therefore$ the statement is true for $n = 1$.

Assume the statement is true for $n = k, k \in \mathbb{Z}^+, k \geq 1$, i.e

$$f_k(\sin^2 \theta) = \sin^2 \left(\frac{\theta}{2^k} \right)$$

Proving true for $n = k + 1$:

$$\begin{aligned} LHS &= f_{k+1}(\sin^2 \theta) \\ &= f \circ f_k(\sin^2 \theta) \\ &= f \left(\sin^2 \left(\frac{\theta}{2^k} \right) \right) && (\text{By assumption}) \\ &= \frac{1}{2} \left(1 - \sqrt{1 - \sin^2 \left(\frac{\theta}{2^k} \right)} \right) \\ &= \frac{1}{2} \left(1 - \sqrt{\cos^2 \left(\frac{\theta}{2^k} \right)} \right) \\ &= \frac{1}{2} \left(1 - \left| \cos \left(\frac{\theta}{2^k} \right) \right| \right) \\ &= \frac{1}{2} \left(1 - \cos \left(\frac{\theta}{2^k} \right) \right) && \left(0 < \frac{\theta}{2^k} < \frac{\pi}{2} \right) \\ &= \frac{1}{2} \left(1 - \left(1 - 2 \sin^2 \left(\frac{\theta}{2^{k+1}} \right) \right) \right) \\ &= \frac{1}{2} \left(2 \sin^2 \left(\frac{\theta}{2^{k+1}} \right) \right) \\ &= \sin^2 \left(\frac{\theta}{2^{k+1}} \right) \end{aligned}$$

$\therefore$ If the statement is true for $n = k$, it is true for $n = k + 1$.

$\therefore$ The statement is true by the principle of mathematical induction.



Question 132 (4 marks) (SBHS '21 AT2 Q11d)

Base case: $n = 2$, $LHS = \frac{1}{2+1} + \frac{1}{2+2} + \frac{1}{2+3} \therefore \text{true for } n = 2.$

$$= \frac{47}{60}$$
$$< 1$$

$< RHS$
Assume true for $n = k, k \in \mathbb{Z}^+, k > 1$, i.e

$$\frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \cdots + \frac{1}{2k+1} < 1$$

Prove true for $n = k + 1$, i.e

$$\frac{1}{k+2} + \frac{1}{k+3} + \frac{1}{k+4} + \cdots + \frac{1}{2(k+1)+1} < 1$$

$$\begin{aligned} LHS &= \frac{1}{k+2} + \frac{1}{k+3} + \frac{1}{k+4} + \cdots + \frac{1}{2k+1} + \frac{1}{2k+2} + \frac{1}{2k+3} \\ &= \frac{1}{k+1} - \frac{1}{k+1} + \frac{1}{k+2} + \frac{1}{k+3} + \cdots + \frac{1}{2k+1} + \frac{1}{2k+2} + \frac{1}{2k+3} \\ &< 1 + \frac{1}{2k+2} + \frac{1}{2k+3} - \frac{1}{k+1}, \quad \text{by assumption} \\ &= 1 + \frac{1}{2(k+1)} - \frac{1}{k+1} + \frac{1}{2k+3} \\ &= 1 + \frac{(2k+3) - 2(2k+3) + 2(k+1)}{2(k+1)(2k+3)} \\ &= 1 - \frac{1}{(2k+2)(2k+3)} \\ &< 1 \quad (\text{as } k > 1, 0 < \frac{1}{(2k+2)(2k+3)} < 1) \\ &= RHS \end{aligned}$$

$\therefore$ true for $n = k + 1$

Hence by mathematical induction, the statement is proven true for $n > 1$.



Question 133 (3 marks) (NSB '19 Q15b)

$$x_0 = a^{0+1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^0 + \left(\frac{-1 - \sqrt{5}}{2} \right)^0 \right] = a(1 + 1) = 2a$$

$$x_1 = a^{1+1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^1 + \left(\frac{-1 - \sqrt{5}}{2} \right)^1 \right] = a^2 \left(-\frac{1}{2} - \frac{1}{2} \right) = -a^2$$

Assume true for $n = k - 1$ and $n = k$, i.e

$$\left. \begin{aligned} x_k &= a^k \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^k + \left(\frac{-1 - \sqrt{5}}{2} \right)^k \right] \\ x_{k-1} &= a^{k-1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k-1} + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k-1} \right] \end{aligned} \right\} \text{Inductive hypothesis}$$

Prove true for $n = k + 1$

$$\begin{aligned} x_{k+1} &= -ax_k + a^2x_{k-1} \\ &= -a \cdot a^{k+1} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^k + \left(\frac{-1 - \sqrt{5}}{2} \right)^k \right] + a^2 \cdot a^k \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k-1} + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k-1} \right] \\ &= a^{k+2} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k-1} \left(1 - \frac{-1 + \sqrt{5}}{2} \right) + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k-1} \left(1 - \frac{-1 + \sqrt{5}}{2} \right) \right] \\ &= a^{k+2} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k-1} \left(\frac{3 - \sqrt{5}}{2} \right) + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k-1} \left(\frac{3 - \sqrt{5}}{2} \right) \right] \\ \left(\frac{-1 + \sqrt{5}}{2} \right)^2 &= \frac{1 + 5 - 2\sqrt{5}}{4} = \frac{6 - 2\sqrt{5}}{4} = \frac{3 - \sqrt{5}}{2} \\ \left(\frac{-1 - \sqrt{5}}{2} \right)^2 &= \frac{1 + 5 + 2\sqrt{5}}{4} = \frac{6 + 2\sqrt{5}}{4} = \frac{3 + \sqrt{5}}{2} \\ x_{k+1} &= a^{k+2} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k-1} \left(\frac{-1 + \sqrt{5}}{2} \right)^2 + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k-1} \left(\frac{-1 - \sqrt{5}}{2} \right)^2 \right] \\ &= a^{k+2} \left[\left(\frac{-1 + \sqrt{5}}{2} \right)^{k+1} + \left(\frac{-1 - \sqrt{5}}{2} \right)^{k+1} \right] \end{aligned}$$

Hence by mathematical induction, it is true for all $n \geq 0$.



Question 134 (6 marks) (Normo '22 Q13a)

(i)

$$\begin{aligned} I_n &= \int_1^e (1 - \ln x)^n dx \\ &= [x(1 - \ln x)^n]_1^e - \int_1^e \frac{n(1 - \ln x)^{n-1}}{-x} x dx \\ &= e(1 - \ln e)^n - 1(1 - \ln 1)^n + n \int_1^e (1 - \ln x)^{n-1} dx \\ &= 0 - 1 + nI_{n-1} \\ &= -1 + nI_{n-1} \end{aligned}$$

(ii) For $n = 1$:

$$\begin{array}{ll} LHS = I_1 & RHS = 1!e - 1 - {}^1P_1 \\ = -1 + 1 \cdot I_0 & = e - 1 - 1 \\ = -1 + \int_1^e dx & = e - 2 \\ = -1 + e - 1 & = LHS \\ = -2 & \end{array}$$

$\therefore$ statement is true for $n = 1$.

Now assume the statement is true for $n = k$, i.e

$$I_k = k!e - 1 - \sum_{r=1}^k {}^kP_r$$

RTP:

$$I_{k+1} = (k+1)!e - 1 - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!}$$



$$\begin{aligned}
LHS &= -1 + (k+1)I_k \quad \text{from (i)} \\
&= -1 + (k+1) \left(k!e - 1 - \sum_{r=1}^k \frac{k!}{(k-r)!} \right) \quad \text{from inductive step} \\
&= (k+1)!e - 1 - (k+1) - \sum_{r=1}^k \frac{(k+1)!}{(k-r)!} \\
&= (k+1)!e - 1 - (k+1) - \sum_{r=2}^{k+1} \frac{(k+1)!}{(k-(r-1))!} \\
&= (k+1)!e - 1 - (k+1) - \left(\sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!} - \frac{(k+1)!}{(k+1-1)!} \right) \\
&= (k+1)!e - 1 - (k+1) - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!} + (k+1) \\
&= (k+1)!e - 1 - \sum_{r=1}^{k+1} \frac{(k+1)!}{(k+1-r)!}
\end{aligned}$$

$\therefore LHS = RHS$ and the statement is true for $n = k + 1$ if it is true for n .

Hence by mathematical induction, the statement is true for all positive integers $n \geq 1$.



Question 135 (4 marks) (BoS '24 Q16a)

- (i) When $n = 1$, it is given that w^2 is divisible by 11 so $a_0 - a_1 = 11M$ where $m \in \mathbb{Z}$.

$$\begin{aligned} u_2 &= a_0 + 10a_1 \\ &= 11M + a_1 + 10a_1 \\ \text{Consider } &= 11(M + a_1) \\ &= 11N \quad \text{where } N = M + a_1 \in \mathbb{Z} \end{aligned}$$

The statement is true for $n = 1$.

Assume the statement is true for $n = k$, that is for integer S ,

$$u_{2k+2} = 11S.$$

It is given that $w_{2k} = 11P$ and $w_{2k+2} = 11R$ for some integers P and R so

$$\begin{aligned} w_{2k+2} &= \sum_{j=0}^{2k+1} (-1)^j a_j \\ 11R &= \sum_{j=0}^{2k-1} (-1)^j a_j + a_{2k} - a_{2k+1} \\ &= 11P + a_{2k} - a_{2k+1} \\ a_{2k} - a_{2k+1} &= 11(R - P) \end{aligned}$$

Now consider

$$\begin{aligned} u_{2k+2} &= \sum_{j=0}^{2k+1} 10^j a_j \\ &= \sum_{j=0}^{2k-1} 10^j a_j + 10^{2k} a_{2k} + 10^{2k+1} a_{2k+1} \\ &= 11Q + 10^{2k} a_{2k} + 10^{2k+1} a_{2k+1} \quad \text{by assumption} \\ &= 11Q + 10^{2k} (a_{2k} + 10a_{2k+1}) \\ &= 11Q + 10^{2k} (a_{2k+1} + 11(R - P) + 10a_{2k+1}) \\ &= 11Q + 11 \cdot 10^{2k} (a_{2k+1} + R - P) \\ &= 11[Q + 10^{2k} (a_{2k+1} + R - P)] \\ &= 11S \quad \text{where } S = Q + 10^{2k} (a_{2k+1} + R - P) \in \mathbb{Z} \end{aligned}$$

Since the statement is true for $n = 1$, then by induction it is true for all positive integers n

- (ii) If u_{2n} is palindromic number then $a_0 = a_{2n-1}, a_1 = a_{2n-2}, \dots, a_{n-1} = a_n$. This implies that

$$\begin{aligned} w_{2n} &= (a_0 - a_{2n-1}) + (-a_1 + a_{2n-2}) + (a_2 - a_{2n-3}) + \dots + ((-1)^{n-1} a_{n-1} + (-1)^n a_n) \\ &= 0 \end{aligned}$$

Since w_{2n} is divisible by 11 then from part (i), u_{2n} is also divisible by 11. Hence, all palindromic numbers with an even number of digits is divisible by 11.



Question 136 (4 marks) (HAHS '17 Q16c)

For $n = 1$,

$$\begin{aligned} LHS &= F_0 \\ &= 2^{2^0} + 1 = 3 \end{aligned} \qquad \begin{aligned} RHS &= F_1 - 2 \\ &= 2^{2^1} + 1 - 2 = 3 \end{aligned}$$

Hence the statement is true for $n = 1$.

Now assume the statement is true for $n = k, k \in \mathbb{Z}^+$, i.e

$$F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} = F_k - 2$$

Now for $n = k + 1$

$$\begin{aligned} LHS &= F_0 \times F_1 \times F_2 \times \dots \times F_{n-1} \\ &= F_0 \times F_1 \times F_2 \times \dots \times F_k \\ &= F_0 \times F_1 \times F_2 \times \dots \times F_{k-1} \times F_k \\ &= F_k - 2 \times F_k && \text{(by assumption)} \\ &= (F_k)^2 - 2F_k \\ &= (2^{2^k} + 1)^2 - 2(2^{2^k} + 1) \\ &= 2^{2 \times 2^k} + 2 \times 2^{2^k} + 1 - 2 \times 2^{2^k} - 2 \\ &= 2^{2^{k+1}} + 1 - 2 \\ &= (2^{2^{k+1}} + 1) - 2 \\ &= F_{k+1} - 2 && \text{(as required)} \end{aligned}$$

Hence, by the principle of mathematical induction, the statement is true $\forall n \in \mathbb{Z}^+$.

Question 137 (4 marks) (Normo '21 AT2 Q2b)

Let $P(n)$ be the proposition:

n gold plates on the first pillar can be moved to the last pillar using $2^n - 1$ moves.

Base case: $P(1)$:

$P(1) = 2^1 - 1 = 1$. A single plate can be moved to the last pillar with one move.

Hence $P(1)$ is true.

Inductive step: assume $P(k)$ is true, $k \in \mathbb{Z}^+$:

k plates on the first pillar can be moved to the last pillar using $2^k - 1$ moves.

Test $P(k + 1)$:

RTP: $k + 1$ gold plates on the first pillar can be moved to the last pillar using $2^{k+1} - 1$ moves. Using the assumption, the first k plates on the first pillar can be moved to the second pillar using 2^{k-1} moves.

Then, move the $(k + 1)$ th plate on the first pillar to the last pillar.

Finally, move the k plates on the second pillar to the last pillar on top of the $(k + 1)$ th plate using $2k - 1$ moves.

Thus the total number of moves is $2^k - 1 + 1 + 2^k - 1 = 2 \times 2^k - 1 = 2^{k+1} - 1$.

Hence $P(k + 1)$ is true.

By mathematical induction, $P(k)$ is true $\forall n \in \mathbb{Z}^+$.



Question 138 (5 marks) (Ruse '23 AT3 Q8a)

(i)

$$\begin{aligned}f(0 + 0) &= f(0) \times f(0) \\f(0) &= f(0)^2 \\f(0)(1 - f(0)) &= 0 \\f(0) \neq 0 \quad 1 - f(0) &= 0 \\\therefore f(0) &= 1\end{aligned}$$

$$\begin{aligned}f(x + (-x)) &= f(x) \cdot f(-x) \\f(0) &= f(x) \times f(-x) \\1 &= f(x) \times f(-x) \\f(-x) &= \frac{1}{f(x)}\end{aligned}$$

(ii) For $n = 1$:

$$\begin{aligned}LHS &= f(1 \times x) & RHS &= f(x)^1 \\&= f(x) & &= f(x)\end{aligned}$$

Assume true for $n = k, k \in \mathbb{Z}^+$, i.e

$$f(kx) = [f(x)]^k$$

Prove true for $n = k + 1$, i.e

$$f((k + 1)x) = [f(x)]^{k+1}$$

Now,

$$\begin{aligned}f((k + 1)x) &= f(kx + x) \\&= f(kx) \times f(x)\end{aligned}$$

But $f(kx) = [f(x)]^k$.

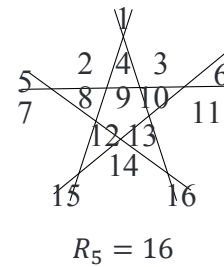
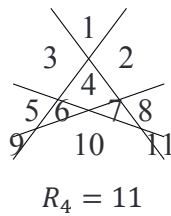
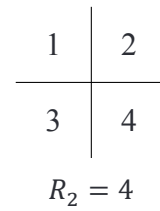
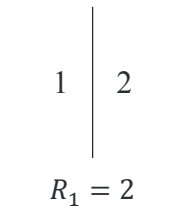
Hence, $f((k + 1)x) = [f(x)]^k \times f(x) = [f(x)]^{k+1}$.

By mathematical induction, $f(nx) = [f(x)]^n$ for all $n \in \mathbb{Z}^+$.



Question 139 (5 marks) (In howard textbook and Lambertnet trial)

(i) this was a pain to make



(ii)

$$\begin{aligned} R_1 &= 2 & R_2 &= R_1 + 2 \\ R_2 &= 4 & R_3 &= R_2 + 3 \\ R_3 &= 7 & R_4 &= R_3 + 4 \\ R_4 &= 11 & R_5 &= R_4 + 5 \\ R_5 &= 16 \end{aligned}$$

$$\begin{aligned} R_1 &= a = 2 \\ R_n &= R_{n-1} + n \\ b &= 1, c = n \end{aligned}$$

(iii) Method 1:

$$\begin{aligned} R_n &= R_{n-1} + n \\ &= n + R_{n-2} + (n - 1) \\ &= n + (n - 1) + (n - 2) + R_{n-3}, \quad \text{and so on} \\ &= n + (n - 1) + (n - 2) + \dots + 2 + R_1 \\ &= n + (n - 1) + (n - 2) + \dots + 2 + 1 + 1 \\ &= \frac{n(n + 1)}{2} + 1 \\ &= \frac{n(n + 1) + 2}{2} \end{aligned}$$



Method 2 – Induction
Show true for $n = 1$:

$$\begin{aligned} LHS &= R_1 = 2 \\ RHS &= \frac{1^1 + 1 + 2}{2} = 2 \end{aligned}$$

$\therefore$ true for $n = 1$.

Assume true for $n = k, k \in \mathbb{Z}^+$, i.e

$$R_k = \frac{k^2 + k + 2}{2}$$

Show true for $n = k + 1$, i.e

$$R_{k+1} = \frac{(k+1)^2 + (k+1) + 2}{2}$$

$$\begin{aligned} LHS &= R_{k+1} \\ &= R_k + (k+1) \\ &= \frac{k^2 + k + 2}{2} + \frac{2(k+1)}{2} \\ &= \frac{k^2 + 2k + 1 + k + 1 + 2}{2} \\ &= \frac{(k+1)^2 + (k+1) + 2}{2} = RHS \end{aligned}$$

$\therefore$ true for $n = k + 1$ if true for $n = k$. Hence true by mathematical induction.



Question 140 (8 marks) (Copium '23 Q14a)

(i) Notice $F_n = F_{n-1} + F_{n-2}$ and $F_{n-3} = F_{n-1} - F_{n-2}$. Thus

$$\begin{aligned}\text{LHS} &= (F_{n-1} + F_{n-2})^2 + (F_{n-1} - F_{n-2})^2 \\ &= F_{n-1}^2 + 2F_{n-1}F_{n-2} + F_{n-2}^2 + F_{n-1}^2 - 2F_{n-1}F_{n-2} + F_{n-2}^2 \\ &= 2(F_{n-1}^2 + F_{n-2}^2) \\ &= \text{RHS}\end{aligned}$$

- **1 mark** Rewriting F_n, F_{n-3} in terms of F_{n-1}, F_{n-2} or vice versa.
- **1 mark** Conclusion.

(ii) A $2 \times (2n - 2)$ rectangle remains to be tiled which is T_{n-1} ways because $2n - 2 = 2(n - 1)$.

- **1 mark** Correct explanation.

(iii) A $2 \times (2n - 4)$ rectangle remains to be tiled, which is T_{n-2} ways because $2n - 4 = 2(n - 2)$.

- **1 mark** Correct explanation.

(iv) From ii. and iii. and the given equation, we add the number of ways to tile the $2 \times 2n$ rectangle based on all cases of covering the leftmost is covered have that $T_n = T_{n-1} + T_{n-2} + 2(T_0 + T_1 + \dots + T_{n-2})$.

$$\begin{aligned}\text{LHS} &= T_n - 2T_{n-1} - 2T_{n-2} + T_{n-3} \\ &= 2(T_0 + T_1 + \dots + T_{n-2}) + T_{n-2} + T_{n-1} - 2T_{n-1} - 2T_{n-2} + T_{n-3} \\ &= 2(T_0 + T_1 + \dots + T_{n-3}) + T_{n-2} - T_{n-1} + T_{n-3} \\ &= 2(T_0 + T_1 + \dots + T_{n-3}) + T_{n-2} \\ &\quad - (2(T_0 + T_1 + \dots + T_{n-3}) + T_{n-3} + T_{n-2}) + T_{n-3} (*) \\ &= 2(T_0 + T_1 + \dots + T_{n-3}) + T_{n-2} \\ &\quad - 2(T_0 + T_1 + \dots + T_{n-3}) - T_{n-3} - T_{n-2} + T_{n-3} \\ &= T_{n-2} - T_{n-3} - T_{n-2} + T_{n-3} \\ &= 0\end{aligned}$$

- **1 mark** Expressing $T_n = T_{n-1} + T_{n-2} + 2(T_0 + T_1 + \dots + T_{n-2})$.
- **1 mark** Substituting and significant progress, up to (*).
- **1 mark** Conclusion.
- **Comment:** Alternative ways of substituting T_n which prove the statement are also awarded marks.



(v) For $n = 0$, there is one way to tile a 2×0 rectangle.

For $n = 1$ we can only tile a 2×2 rectangle with the 2×2 square tetromino. Hence $T_1 = 1 = 1^2 = F_1$.

For $n = 2$, it is given that $T_2 = 4$ so $T_2 = 4 = 2^2 = F_2^2$. Thus it is true for $n = 0, 1, 2$.

Now assume it is true for $n = 0, 1, 2, \dots, k$ where $k \in \mathbb{Z}$ and $k \geq 3$. Thus

$$T_{k-1} = F_{k-1}^2$$

$$T_{k-2} = F_{k-2}^2$$

$$T_{k-3} = F_{k-3}^2$$

and we are required to prove $T_k = F_k^2$. From iv. we have

$$\begin{aligned} T_k &= 2T_{k-1} + 2T_{k-2} - T_{k-3} \\ &= 2F_{k-1}^2 + 2F_{k-2}^2 - F_{k-3}^2 && \text{(inductive assumption)} \\ &= 2(F_{k-1}^2 + F_{k-2}^2) - F_{k-3}^2 \\ &= F_k^2 + F_{k-3}^2 - F_{k-3}^2 && \text{(by part i.)} \\ &= F_k^2 \end{aligned}$$

as needed.

By the principle of mathematical induction, since it is true for $n = 0, 1, 2$ and it is true for $n = k$ if it is true for all $n = 0, 1, \dots, k - 1$, we conclude that $T_n = F_n^2$ for all positive integers n .

- **1 mark** Base case for three integers.
- **1 mark** Correct inductive step and conclusion.
- **Comment:** One can also list out the cases for $n = 3$ and have base cases as $n = 1, 2, 3$ instead, this is a bit longer. Correct quoting of the inductive assumption must be done.



Question 141 (3 marks) ("The -C ult Q2a")

- (i) Let $T(n)$ be the total number of triangles.

Base case: $n = 1$, there is one shaded triangle and no white triangles. Thus $T(1)$ is true for $n = 1$. Assume true for $n = k, k \in \mathbb{Z}^+, k \geq 1$, i.e

$$T(k) = \frac{3^k - 1}{2}$$

RTP for $n = k + 1$, i.e

$$T_n = \frac{3^{k+1} - 1}{2}$$

Each of the shaded triangles at iteration k generates 3 new triangles in the next iteration.

$$\begin{aligned} \therefore T_{k+1} &= T_k + 3^k \\ &= \frac{3^k - 1}{2} + 3^k \\ &= \frac{3^k - 1 + 2 \cdot 3^k}{2} \\ &= \frac{3^k + 2 \cdot 3^k - 1}{2} \\ &= \frac{3^{k+1} - 1}{2} \end{aligned}$$

Hence true by the principle of mathematical induction.

(ii) $\left(\frac{3}{4}\right)^{n-1}$

Question 142 (5 marks) (BoS '18 Q15c)

- (i) When $m = 1$, there are only 2 people playing, and so as the first person removed is P_2 then P_1 must win. Now assume that for $m = k$ for some $k > 0$, that P_1 wins when 2^k people are playing.

Consider the case when $m = k + 1$ so 2^{k+1} people are playing. According to the rules of the game, we remove every person with an even number, so $P_2, P_4, \dots, P_{2^k}$. After removing these 2^k people, there will be half as many people left (i.e. another 2^k people). The first person to be removed will be the person clockwise next to P_1 . However, notice that this is the exact same game as playing with 2^k people (but with relabelling). Hence, by the induction hypothesis, the person who wins will be P_1 . Since it is true for $m = 1$, it is true for all positive integers of m .

- (ii) The idea will be to remove people until the number of people remaining is a power of 2. As 1024 is the closest power of 2 below 2018, this will require 994 people removed. The 994th person to be removed will be the $2 \times 994 = 1988$ th person (labelled P_{1988}). After this, with 1024 people, the next person to be removed will be the 1990th person, and so the 1989th person effectively takes the role of the 1st person in part (i). Hence, the final person remaining will be the 1989th person.



Calculus Proofs & The Squeeze Theorem

Question 143 (12 marks) (God folder)

(i) Comparing upper rectangles,

$$\begin{aligned}\sum_{k=1}^n \frac{1}{k} &> \int_1^n \ln x \, dx \\ &= \ln n \\ \therefore \sum_{k=1}^n \frac{1}{k} &> \ln n\end{aligned}$$

(ii) As $n \rightarrow \infty$, $\ln n \rightarrow \infty$. As $H_n > \ln n$, $H_n \rightarrow \infty$.

(iii) Consider:

$$\begin{aligned}x^{n-1} &\geq 0 \\ x^{n-1} + x^n &\geq x^n \\ x^{n-1}(1+x) &\geq x^n \\ \therefore \frac{x^n}{1+x} &\leq x^{n-1} \quad (\text{since } x \geq 0)\end{aligned}$$

(iv)

$$\begin{aligned}RHS &= \frac{(-x)^n}{1+x} + \frac{(-x)^n - 1}{-x - 1} \quad (\text{sum of GP}) \\ &= \frac{(-x)^n - (-x)^n + 1}{1+x} \\ &= \frac{1}{1+x}\end{aligned}$$

(v) Integrating both sides from 0 to 1:

$$\begin{aligned}\int_0^1 \frac{1}{1+x} \, dx &= \int_0^1 \frac{(-x)^n}{1+x} \, dx + \int_0^1 \sum_{k=0}^{n-1} (-x)^k \, dx \\ \ln 2 &= \int_0^1 \frac{(-x)^n}{1+x} \, dx + \left[x - \frac{1}{2}x^2 + \frac{1}{3}x^3 + \dots + \frac{(-1)^n x^n}{n} \right]_0^1 \\ &= \int_0^1 \frac{(-x)^n}{1+x} \, dx + A_n \\ A_n &= \ln 2 - \int_0^1 \frac{(-x)^n}{1+x} \, dx\end{aligned}$$



(vi) Since $(-1)^n \leq 1$,

$$A_n \geq \ln - \int_0^1 \frac{x^n}{1+x} dx$$

Similarly, $(-1)^n \geq -1$.

$$\text{So } A_n \leq \ln 2 + \int_0^1 \frac{x^n}{1+x} dx$$

$$\therefore \ln 2 - \int_0^1 \frac{x^n}{1+x} dx \leq A_n \leq \ln 2 + \int_0^1 \frac{x^n}{1+x} dx$$

Question 144 (2 marks) (BoS '20 Q15b)

For $t > 0$

$$0 < \frac{1}{t+1} < 1$$

$$0 < \frac{t^2}{t+1} < t^2$$

$$0 < \frac{t^2 - 1 + 1}{t+1} < t^2$$

$$0 < t - 1 + \frac{1}{t+1} < t^2$$

$$0 < \int_0^x \left(t - 1 + \frac{1}{t+1} \right) dx < \int_0^x t^2 dt$$

$$0 < \ln(1+x) - x + \frac{x^2}{2} < \frac{x^3}{3}$$



Question 145 (7 marks) (Abbotsleigh '24 Q16a)

(i)

$$\begin{aligned} x &\geq 0 \\ -x^2 &\leq 0 \leq x \\ 1 - x^2 &\leq 0 \leq 1 + x \end{aligned}$$

Since $x \geq 0, 1 + x > 0, \therefore$ can divide by $1 + x$.

$$\begin{aligned} \frac{1 - x^2}{1 + x} &\leq \frac{1}{1 + x} \leq 1 \\ 1 - x &\leq \frac{1}{1 + x} \leq 1 \end{aligned}$$

(ii) Integrating part (i),

$$\begin{aligned} \int_0^t (1 - x) dx &\leq \int_0^t \frac{1}{1 + x} dx && \leq \int_0^t 1 dx \quad x > 0 \\ \left[x - \frac{x^2}{2} \right]_0^t &\leq [\ln |1 + x|]_0^t && \leq [x]_0^t \\ t - \frac{t^2}{2} &\leq \ln |1 + t| && \leq t \\ 1 - \frac{t}{2} &\leq \frac{1}{t} \ln |1 + t| && \leq 1 \\ 1 - \frac{1}{2} \left(\frac{1}{n} \right) &\leq \frac{1}{\left(\frac{1}{n} \right)} \ln \left| 1 + \frac{1}{n} \right| && \leq 1 \quad \text{(Let } t = \frac{1}{n} \text{)} \end{aligned}$$

Using dummy variable, we have

$$1 - \frac{1}{2x} \leq \ln \left(1 + \frac{1}{x} \right) \leq 1$$

(iii)

$$\begin{aligned} 1 - \frac{1}{2x} &\leq \ln \left(1 + \frac{1}{x} \right)^x \leq 1 \\ e^{1 - \frac{1}{2x}} &\leq e^{\ln \left(1 + \frac{1}{x} \right)^x} \leq e \\ e^{1 - \frac{1}{2x}} &\leq \left(1 + \frac{1}{x} \right)^x \leq e \\ \lim_{x \rightarrow \infty} e^{1 - \frac{1}{2x}} &\leq \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x \leq \lim_{x \rightarrow \infty} e \\ e^1 &\leq \lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x \leq e \end{aligned}$$

Hence, $\lim_{x \rightarrow \infty} \left(1 + \frac{1}{x} \right)^x = e$ by the Squeeze theorem.



Question 146 (5 marks) (BBHS '24 Q16 c)

(i)

$$\begin{aligned}
 f(x) &= \frac{(n+1+x)^{n+1}}{(n+x)^n}, x \geq 0, n \geq 1 \\
 f'(x) &= \frac{(n+x)^n(n+1)(n+1+x)^n - (n+1+x)^{n+1}n(n+x)^{n-1}}{[(n+x)^n]^2} \\
 &= \frac{(n+1+x)^n(n+x)^{n-1}[(n+x)(n+1) - n(n+1+x)]}{(n+x)^{2n}} \\
 &= \frac{(n+1+x)^n[n^2 + n + xn + x - n^2 - n - nx]}{(n+x)^{(2n-n+1)}} \\
 &= \frac{x(n+1+x)^n}{(n+x)^{(n+1)}}
 \end{aligned}$$

Since $x \geq 0, n \geq 1$

$$\begin{aligned}
 (n+1+x)^n &> 0, (n+x)^{n+1} > 0 \\
 \therefore f'(x) &> 0 \\
 \therefore f(x) &\text{ is increasing for } x > 0
 \end{aligned}$$

(ii) Since $f(x)$ is increasing for $x > 0$, then $f(x) > f(0)$

$$\begin{aligned}
 \frac{(n+1+x)^{n+1}}{(n+x)^n} &> \frac{(n+1+0)^{n+1}}{(n+0)^n} \\
 \frac{(n+1+x)^{n+1}}{(n+x)^n} &> \frac{(n+1)^{n+1}}{(n)^n}
 \end{aligned}$$

Since $x > 0, (x+n)^n > 0, (n+1)^{n+1} > 0$

$$\begin{aligned}
 \frac{(n+1+x)^{n+1}}{(n+1)^{n+1}} &> \frac{(n+x)^n}{(n)^n} \\
 \left(\frac{n+1+x}{n+1}\right)^{n+1} &> \left(\frac{n+x}{n}\right)^n \\
 \therefore \left(1 + \frac{x}{n+1}\right)^{n+1} &> \left(1 + \frac{x}{n}\right)^n
 \end{aligned} \tag{1}$$

(iii) Sub $x = 1$ into (1):

$$\begin{aligned}
 \frac{(n+1+1)^{n+1}}{(n+1)^{n+1}} &> \frac{(n+1)^n}{(n)^n} \\
 \frac{(n+2)^{n+1}}{(n+1)^{n+1}} &> \frac{(n+1)^n}{(n)^n}
 \end{aligned}$$

Since $(n+1)^n > 0, (n+1)^{n+1} > 0$

$$\begin{aligned}
 (n+2)^{n+1}n^n &> (n+1)^n(n+1)^{n+1} \\
 \therefore (n+2)^{n+1}n^n &> (n+1)^{2n+1}
 \end{aligned}$$



Question 147 (5 marks) (Ruse '24 Q14d)

(i)

$$\begin{aligned}
 \sum_{k=0}^n (-1)^k x^k &= 1 - x + x^2 - x^3 + \dots + (-1)^n x^n \\
 &= \frac{1(1 - (-x)^{n+1})}{1 - (-x)} \quad (\text{sum of GP}) \\
 &= \frac{(1 - (-x)^{n+1})}{1 + x} \\
 &= \frac{1}{1 + x} - \frac{(-x)^{n+1}}{1 + x} \\
 &= \frac{1}{1 + x} - \frac{(-1)^{n+1} x^{n+1}}{1 + x}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 \frac{x^{n+1}}{1+x} &\geq 0 \text{ for } x \in [0, 1] \\
 \therefore \int_0^1 \frac{x^{n+1}}{1+x} dx &\geq 0
 \end{aligned}$$

LHS proven.

$$\begin{aligned}
 x \in [0, 1] &\Rightarrow 1 + x \in [1, 2] \\
 \therefore \frac{x^{n+1}}{1+x} &\leq x^{n+1} \text{ for } x \in [0, 1] \\
 \int_0^1 \frac{x^{n+1}}{1+x} dx &\leq \int_0^1 x^{n+1} dx \\
 &= \left[\frac{x^{n+2}}{n+2} \right]_0^1 \\
 &= \frac{1}{n+2}
 \end{aligned}$$

RHS proven.

$$\therefore 0 \leq \int_0^1 \frac{x^{n+1}}{1+x} dx \leq \frac{1}{n+2}$$

(iii)

$$\begin{aligned}
 \int_0^1 \sum_{k=0}^n (-1)^k x^k dx &= \int_0^1 \frac{1}{1+x} dx - \int_0^1 \frac{(-1)^{n+1} x^{n+1}}{1+x} dx \\
 \left[\sum_{k=0}^n \frac{(-1)^k x^{k+1}}{k+1} \right]_0^1 &= [\ln |1+x|]_0^1 - (-1)^{n+1} \int_0^1 \frac{x^{n+1}}{1+x} dx \\
 \sum_{k=0}^n \frac{(-1)^k}{k+1} &= \ln 2 - (-1)^{n+1} \int_0^1 \frac{x^{n+1}}{1+x} dx
 \end{aligned}$$



Now,

$$0 \leq \int_0^1 \frac{x^{n+1}}{1+x} dx \leq \frac{1}{n+2} \quad (\text{ii})$$

$$\lim_{n \rightarrow \infty} 0 \leq \lim_{n \rightarrow \infty} \int_0^1 \frac{x^{n+1}}{1+x} dx \leq \lim_{n \rightarrow \infty} \frac{1}{n+2}$$

$$\therefore \lim_{n \rightarrow \infty} \int_0^1 \frac{x^{n+1}}{1+x} dx = 0 \quad (\text{Squeeze theorem})$$

$$\begin{aligned} \therefore \lim_{n \rightarrow \infty} \sum_{k=0}^n \frac{(-1)^{k+1}}{k+1} &= \ln 2 - \lim_{n \rightarrow \infty} (-1)^{n+1} \int_0^1 \frac{x^{n+1}}{1+x} dx \\ &= \ln 2 - 0 \\ &= \ln 2 \end{aligned}$$



Question 148 (5 marks) (Baulko '22 Q13d)

(i)

$$\begin{aligned}
 S_{2n-1} &= 1 - h + h^2 - h^3 + \dots - h^{2n-1} \\
 &= \frac{1 - (-h)^{2n}}{1 - (-h)} && \text{(sum of GP)} \\
 &= \frac{1 - h^{2n}}{1 + h} \\
 &< \frac{1}{1 + h} && \text{(greater numerator)} \\
 S_{2n} &= 1 - h + h^2 - h^3 + \dots + h^{2n} \\
 &= \frac{1 - (-h)^{2n+1}}{1 - (-h)} && \text{(sum of GP)} \\
 &= \frac{1 + h^{2n+1}}{1 + h} \\
 &> \frac{1}{1 + h} && \text{(smaller numerator)} \\
 \therefore S_{2n} &< \frac{1}{1 + h} < S_{2n} \quad \text{as required}
 \end{aligned}$$

(ii)

$$\begin{aligned}
 \int_0^x S_{2n-1} dh &< \int_0^x \frac{1}{1+h} dh < \int_0^x S_{2n} dh \\
 \left[h - \frac{h^2}{2} + \dots - \frac{h^{2n}}{2n} \right]_0^x &< [\ln(1+h)]_0^x < \left[h - \frac{h^2}{2} + \dots + \frac{h^{2n+1}}{2n+1} \right]_0^x \\
 x - \frac{x^2}{2} + \dots - \frac{x^{2n}}{2n} &< \log_e(1+x) < x - \frac{x^2}{2} + \dots + \frac{x^{2n+1}}{2n+1}
 \end{aligned}$$

(iii)

$$\begin{aligned}
 0.25 - \frac{(0.25)^2}{2} + \frac{(0.25)^3}{3} - \frac{(0.25)^4}{4} + \frac{(0.25)^5}{5} - \frac{(0.25)^6}{6} &< \ln(1 + 0.25) < 0.25 - \frac{(0.25)^2}{2} + \frac{(0.25)^3}{3} - \dots \\
 0.223136\dots &< \ln\left(\frac{5}{4}\right) < 0.223145 \\
 \therefore \ln\left(\frac{5}{4}\right) &= 0.2231 \quad \text{to 4 s.f}
 \end{aligned}$$



Question 149 (4 marks) (BBHS '24 Q16 b)

(i) $|f(x)|$ is always positive so $\int |f(x)| dx$ is a positive value. $f(x)$ can cross the x -axis for $a \leq x \leq b$, then $\int_a^b f(x)$ may involve adding negative and positive values. So, $|\int_a^b f(x) dx| < \int_a^b |f(x)| dx$. If $f(x)$ is positive for the domain, then $|\int_a^b f(x) dx| = \int_a^b f(x) dx$. Thus, $|\int_a^b f(x) dx| \leq \int_a^b |f(x)| dx$

(ii)

$$\begin{aligned} \left| \int_0^\pi 27^x \cos x dx \right| &\leq \int_0^\pi |27^x \cos x| dx \\ \left| \int_0^\pi 27^x \cos x dx \right| &\leq \int_0^\pi 27^x dx \quad (|\cos x| \leq 1, 27^x > 0) \\ \left| \int_0^\pi 27^x \cos x dx \right| &\leq \left[\frac{27^x}{\ln 27} \right]_0^\pi \\ \left| \int_0^\pi 27^x \cos x dx \right| &\leq \left[\frac{3^{3x}}{3 \ln 3} \right]_0^\pi \\ \left| \int_0^\pi 27^x \cos x dx \right| &\leq \frac{3^{3\pi}}{3 \ln 3} - \frac{3^{3 \times 0}}{3 \ln 3} \\ \left| \int_0^\pi 27^x \cos x dx \right| &\leq \frac{3^{3\pi} - 1}{3 \ln 3} \end{aligned}$$



Question 150 (5 marks) (NSG '13 Q16d)

(i)

$$f(x) = \log x - x + 1$$

$$f'(x) = \frac{1}{x} - 1$$

$$= \frac{1-x}{x}$$

$\therefore$ stationary point at $x = 1$

$$f''(x) = -\frac{1}{x^2} < 0 \quad \forall x$$

$\therefore$ minimum turning point at $(1,0)$

As domain $x > 0$ (and continuous for all x in domain)

$$\therefore f(x) \leq 0 \quad \forall x > 0$$

(ii)

$$\sum_{r=1}^n \log(np_r) \leq \sum_{r=1}^n (np_r - 1)$$

$$= \sum_{r=1}^n np_r - \sum_{r=1}^n 1$$

$$\sum_{r=1}^n \log(np_r) \leq \sum_{r=1}^n np_r - n$$

(part (i))

(iii) Continuing from part (ii),

$$\sum_{r=1}^n \log np_r \leq n \sum_{r=1}^n p_r - n$$

$$= n \cdot 1 - n$$

$$\sum_{r=1}^n \log np_r \leq 0$$

(iv) Continuing from part (iii)

$$\log np_1 + \log np_2 + \dots + \log np_n \leq 0$$

$$\log(np_1 \cdot np_2 \cdot \dots \cdot np_n) \leq 0$$

$$\log(n^n \cdot p_1 p_2 \dots p_n) \leq 0$$

$$n^n \cdot p_1 p_2 \dots p_n \leq 1$$

Also, since $p_1, p_2, \dots, p_n$ and n are all positive, then $n^n \cdot p_1 p_2 \dots p_n > 0$. $\therefore 0 < n^n \cdot p_1 p_2 \dots p_n \leq 1$



Question 152 (8 marks) (Ruse '22 AT3 Q8b)

(i)

$$\begin{aligned}0 &\leq t \leq \frac{1}{\sqrt{2}} \\0 &\leq t^2 \leq \frac{1}{2} \\-\frac{1}{2} &\leq -t^2 \leq 0 \\\frac{1}{2} &\leq 1 - t^2 \leq 1 \\1 &< \frac{1}{1 - t^2} \leq 2 \\2t^2 &\leq \frac{2t^2}{1 - t^2} < 4t^2 \\0 &\leq \frac{2t^2}{1 - t^2} \leq 4t^2\end{aligned}$$

(ii)

$$\begin{aligned}\frac{2t^2}{1 - t^2} &= \frac{2t^2 - 2 + 2}{1 - t^2} \\&= \frac{2(t^2 - 1)}{1 - t^2} + \frac{2}{1 - t^2} \\&= -2 + \frac{2}{1 - t^2} \\&= -2 + \frac{1}{1 - t} + \frac{1}{1 + t} \\\therefore 0 &\leq \frac{1}{1 + t} + \frac{1}{1 - t} \leq 4t^2\end{aligned}$$

(iii)

$$\begin{aligned}\int_0^x 0 \, dt &\leq \int_0^x \left(\frac{1}{1 + t} + \frac{1}{1 - t} - 2 \right) dt \leq \int_0^x 4t^2 \, dt \\0 &\leq [\ln |1 + t| - \ln |1 - t| - 2t]_0^x \leq \left[\frac{4t^2}{3} \right]_0^x \\0 &\leq \ln \left(\frac{1 + x}{1 - x} \right) - 2x \leq \frac{4x^3}{3} \\e^0 &\leq e^{\ln \left(\frac{1 + x}{1 - x} \right) - 2x} \leq e^{\frac{4x^3}{3}} \\1 &\leq e^{\ln \left(\frac{1 + x}{1 - x} \right)} e^{-2x} \leq e^{\frac{4x^3}{3}} \\1 &\leq \left(\frac{1 + x}{1 - x} \right) e^{-2x} \leq e^{\frac{4x^3}{3}}\end{aligned}$$



Question 153 (7 marks) (Baulko '20 Q16a)

(i)

$$\begin{aligned}\int_n^{2n} \frac{dx}{\sqrt{x}} &= \left[2\sqrt{x} \right]_n^{2n} \\ &= 2\sqrt{2n} - 2\sqrt{n} \\ &= 2\sqrt{n}(\sqrt{2} - 1)\end{aligned}$$

(ii) Area of the lower rectangles < Area under the curve

$$\begin{aligned}\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{n}} &< \int_n^{2n} \frac{dx}{\sqrt{x}} \\ S_n &< 2\sqrt{n}(\sqrt{2} - 1)\end{aligned}$$

Area under the curve < Area of the upper rectangles

$$\begin{aligned}\int_n^{2n} \frac{dx}{\sqrt{x}} &< \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}} \\ 2\sqrt{n}(\sqrt{2} - 1) &< \frac{1}{\sqrt{n}} + \left(\frac{1}{\sqrt{n+1}} + \frac{1}{\sqrt{n+2}} + \dots + \frac{1}{\sqrt{2n-1}} + \frac{1}{\sqrt{2n}} \right) - \frac{1}{\sqrt{2n}} \\ 2\sqrt{n}(\sqrt{2} - 1) - \frac{1}{\sqrt{n}} + \frac{1}{\sqrt{2n}} &< S \\ 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} &< S_n \\ 2\sqrt{n}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2n}} &< S_n < 2\sqrt{n}(\sqrt{2} - 1)\end{aligned}$$

(iii) Let $n = 1\,000\,000$:

$$\begin{aligned}2\sqrt{1000000}(\sqrt{2} - 1) + \frac{1 - \sqrt{2}}{\sqrt{2000000}} &< S_{1000000} < 2\sqrt{1000000}(\sqrt{2} - 1) \\ 828.4268 &< S_{1000000} < 828.4271\end{aligned}$$



Question 154 (6 marks) (Ruse '20 Q13a)

(i) Comparing areas,

$$\frac{1}{m+1} < \int_m^{m+1} \frac{dx}{x} < \frac{1}{m}$$

So

$$\frac{1}{1+m} < \ln(m+1) - \ln m < \frac{1}{m}$$

(ii) Let m successively take on the values $1, 2, \dots, k-1$ and form the sum from $m = 1$ to $m = k-1$:

$$\begin{aligned} \sum_{m=1}^{k-1} \frac{1}{m+1} &< \sum_{m=1}^{k-1} (\ln(m+1) - \ln m) < \sum_{m=1}^{k-1} \frac{1}{m} \\ \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k} &< \ln k - \ln 1 < 1 + \frac{1}{2} + \dots + \frac{1}{k-1} \end{aligned}$$

That is, for $S_k = \frac{1}{2} + \frac{1}{3} + \dots + \frac{1}{k}$,

$$S_k - 1 < \ln k < S_{k-1}$$

From the left inequality,

$$S_k - 1 < \ln k$$

so

$$S_k < 1 + \ln k \tag{1}$$

From the right inequality:

$$\ln k < S_{k-1}$$

Replace k with $k+1$, then

$$\ln(k+1) < S_k \tag{2}$$

Hence by (1) and (2):

$$\ln(k+1) < S_k < 1 + \ln k$$



Question 155 (8 marks) (Normo '23 AT3 Q2)

(a) (i)

$$\begin{aligned} x^4 &< x^6 < x^9 \\ \sqrt[3]{x^2} &< x < \sqrt{x^3} \\ \frac{1}{\sqrt{x^3}} &< \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}} \\ 0 &< \frac{1}{\sqrt{x^3}} < \frac{1}{x} < \frac{1}{\sqrt[3]{x^2}} \end{aligned}$$

(ii)

$$\begin{aligned} \int_1^t \frac{1}{\sqrt{x^3}} dx &\leq \int_1^t \frac{1}{x} dx \leq \int_1^t \frac{1}{\sqrt[3]{x^2}} dx \\ -2 \left[\frac{1}{\sqrt{x}} \right]_1^t &\leq [\ln x]_1^t \leq 3 [\sqrt[3]{x}]_1^t \\ -2 \left(\frac{1}{\sqrt{t}} - 1 \right) &\leq \ln t \leq 3 (\sqrt[3]{t} - 1) \\ \frac{2(\sqrt{t} - 1)}{\sqrt{t}} &\leq \ln t \leq 3 (\sqrt[3]{t} - 1) \end{aligned}$$

(iii) Multiplying by $\frac{1}{\sqrt{t}}$

$$\frac{2}{\sqrt{t}} - \frac{2}{t} \leq \frac{\ln t}{\sqrt{t}} \leq \frac{3}{\sqrt[3]{t}} - \frac{3}{\sqrt{t}}$$

$$\begin{aligned} \lim_{t \rightarrow \infty} \left(\frac{2}{\sqrt{t}} - \frac{2}{t} \right) &= 0 \\ \lim_{t \rightarrow \infty} \left(\frac{3}{\sqrt[3]{t}} - \frac{3}{\sqrt{t}} \right) &= 0 \end{aligned}$$

$$\text{As } t \rightarrow \infty, 0 \leq \frac{\ln t}{\sqrt{t}} \leq 0.$$

$$\therefore \lim_{t \rightarrow \infty} \frac{\ln t}{\sqrt{t}} = 0$$



(b) (i)

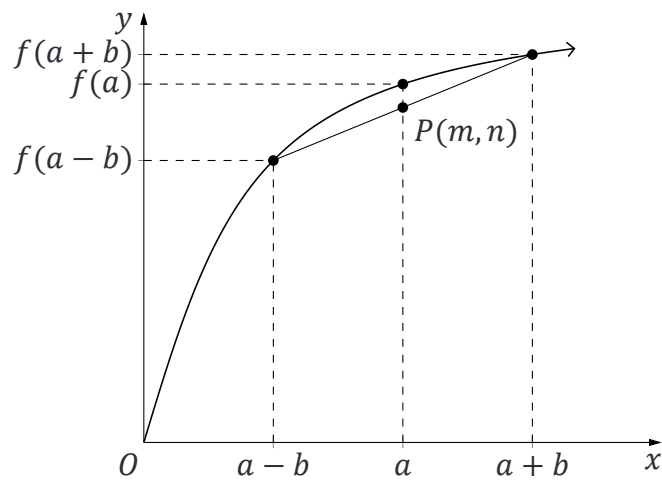
$$f'(x) = \frac{1}{1+x^2}$$

$$f''(x) = -\frac{2x}{(1+x^2)^2}$$

Since $(1+x^2)^2 > 0$ for $x > 0$,

$$\therefore f''(x) < 0 \quad \text{for } x > 0$$

(ii) From part (i), $f(x)$ is concave down and increasing for $x > 0$. Let $P(m, n)$ be the midpoint of the secant.



$$n < f(a)$$

$$\frac{\tan^{-1}(a-b) + \tan^{-1}(a+b)}{2} < \tan^{-1}(a)$$

$$\therefore \tan^{-1}(a-b) + \tan^{-1}(a+b) < 2 \tan^{-1}(a)$$



Question 156 (10 marks) (HGHS '24 Q16b)

(i)

$$\begin{aligned} I_n &= I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx \\ I_{n+2} &= \int_0^{\frac{\pi}{4}} \tan^{n+2} x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \tan^2 x \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x (\sec^2 x - 1) \, dx \\ &= \int_0^{\frac{\pi}{4}} \tan^n x \sec^2 x \, dx - \int_0^{\frac{\pi}{4}} \tan^n x \, dx \\ I_{n+2} &= \left[\frac{\tan^{n+1} x}{n+1} \right]_0^{\frac{\pi}{4}} - I_n \\ I_{n+2} + I_n &= \left[\frac{\tan^{n+1} \left(\frac{\pi}{4} \right)}{n+1} - \frac{\tan^{n+1}(0)}{n+1} \right] \\ &= \left[\frac{(1)^{n+1}}{n+1} - 0 \right] \\ \therefore I_n + I_{n+2} &= \frac{1}{n+1} \end{aligned}$$

(ii)

$$\begin{aligned} I_4 &= \frac{1}{(2)+1} - I_2 \\ I_2 &= \frac{1}{(0)+1} - I_0 \\ I_0 &= \int_0^{\frac{\pi}{4}} \tan^0 x \, dx = \int_0^{\frac{\pi}{4}} x \, dx \\ &= [x]_0^{\frac{\pi}{4}} \\ &= \frac{\pi}{4} \\ \therefore I_2 &= 1 - \frac{\pi}{4} = \frac{4-\pi}{4} \\ I_4 &= \frac{1}{3} - \frac{4-\pi}{4} \\ &= \frac{4-3(4-\pi)}{12} \\ &= \frac{3\pi-8}{12} \text{ or } \frac{\pi}{4} - \frac{2}{3} \end{aligned}$$



(iii)

$$\begin{aligned}
J_n &= (-1)^n I_{2n} \\
J_{n-1} &= (-1)^{n-1} I_{2(n-1)} \\
J_n - J_{n-1} &= (-1)^n I_{2n} - (-1)^{n-1} I_{2(n-1)} \quad (\text{for } n \geq 1) \\
&= (-1)^{n-1} [(-1)I_{2n} - I_{2(n-1)}] \\
&= (-1)^{n-1} (-1) [I_{2n} + I_{2(n-1)}] \\
&= (-1)^n [I_{2n} + I_{2(n-1)}] \\
I_{n+2} + I_n &= \frac{1}{n+1} I_{(2n)+2} + I_{(2n)} = \frac{1}{(2n)+1} \quad (\text{from i}) \\
I_{(2n)} + I_{(2n)-2} &= \frac{1}{(2n-2)+1} \\
\therefore J_n - J_{n-1} &= (-1)^n \left(\frac{1}{2n-1} \right) \quad \text{for } n \geq 1
\end{aligned}$$

(iv)

$$\begin{aligned}
J_n - J_{n-1} &= (-1)^n \left(\frac{1}{2n-1} \right) \\
J_m &= (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3}) + \dots + (J_1 - J_0) + J_0 \\
&= J_0 + (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3}) + \dots + (J_1 - J_0) \\
&= J_0 + (J_m - J_{m-1}) + (J_{m-1} - J_{m-2}) + (J_{m-2} - J_{m-3}) \\
&= J_0 + (-1)^1 \left(\frac{1}{2(1)-1} \right) + (-1)^2 \left(\frac{1}{2(2)-1} \right) + (-1)^3 \left(\frac{1}{2(3)-1} \right) + \dots + (-1)^m \left(\frac{1}{2m-1} \right) \\
&= (-1) \\
&= (-1)^0 I_{2(0)} + \sum_{n=1}^m (-1)^n \left(\frac{1}{2n-1} \right) \\
&= I_0 + \sum_{n=1}^m \frac{(-1)^n}{2n-1} \\
\therefore J_m &= \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1}
\end{aligned}$$

(v)

$$\begin{aligned}
I_n &= \int_0^{\frac{\pi}{4}} \tan^n x \, dx & u &= \tan x \\
&= \int_0^1 u^n \frac{du}{1+u^2} & \frac{du}{dx} &= \sec^2 x = \tan^2 x + 1 \\
&= \int_0^1 \frac{u^n}{1+u^2} du & \frac{dx}{du} &= \frac{1}{\tan^2 x + 1} \\
& & dx &= \frac{1}{u^2 + 1} du \\
& & x = \frac{\pi}{4} &\rightarrow u = 1 \\
& & x = 0 &\rightarrow u = 0
\end{aligned}$$



(vi) From (i), $I_n + I_{n+2} = \frac{1}{n+1}$.

Since $I_n = \int_0^{\frac{\pi}{4}} \tan^n x \, dx > 0$ and $I_{n+2} = \int_0^{\frac{\pi}{4}} \tan^{n+2} x \, dx > 0$

Then $I_n < \frac{1}{n+1}$ and $I_{n+2} < \frac{1}{n+1}$.

Hence $0 < I_n < \frac{1}{n+1}$.

As $\lim_{n \rightarrow \infty} \frac{1}{n+1} = 0$, $\lim_{n \rightarrow \infty} I_n = 0$ and $\lim_{n \rightarrow \infty} I_{2n} = 0$, and $\lim_{n \rightarrow \infty} J_n = \lim_{n \rightarrow \infty} (-1)^n I_{2n}$.

$$\therefore \lim_{n \rightarrow \infty} J_n = 0$$

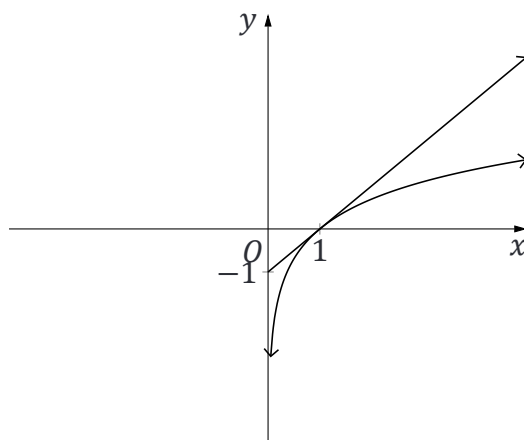
$$\therefore \frac{\pi}{4} + \sum_{n=1}^m \frac{(-1)^n}{2n-1} = 0$$

$$\frac{\pi}{4} = \sum_{n=1}^m \frac{(-1)^n}{2n-1}$$

Question 157 (4 marks) (NSG '22 Q16b ii)

(i)

$$\begin{aligned} y &= \ln x \\ y' &= \frac{1}{x} \\ x &= 1, m = 1 \\ y - \ln 1 &= 1(x - 1) \\ y &= x - 1 \end{aligned}$$



(ii)

$$\begin{aligned}\ln(b_1^{a_1} b_2^{a_2} \dots b_n^{a_n}) &= \ln(b_1^{a_1}) + \ln(b_2^{a_2}) + \dots + \ln(b_n^{a_n}) \\ &= a_1 \ln b_1 + a_2 \ln b_2 + \dots + a_n \ln b_n \\ &< a_1 (b_1 - 1) + a_2 (b_2 - 1) + \dots + a_n (b_n - 1) \\ &= 1 - 1 \\ &= 0 \\ b_1^{a_1} b_2^{a_2} \dots b_n^{a_n} &\leq e^0 = 1\end{aligned}$$



Appendix

1. Several questions have been removed because I forgot where I got them from :D
2. Some questions were excluded (2021 MFIS MCQ) on purpose
3. Question 53 (NSB '24 Q14c): Apparently, nearly no one (if anyone) answered this question correctly.
4. Question 58 (Baulko '21 Q14c): The file this is sourced from is called "Revision 1". The formatting of the paper and solutions mirrors the way Baulkham formats their own trial papers, hence why it is named as such.
5. Question 73 (Baulko '22 Q16c): NSB Q5b) (i)
6. Question 75 (Sydney Tech '22 Q16c): the last part is baulko q16c, so it wasn't added in this section., (iii) was in some '10 paper
7. Question 85 (Ruse '24 Q12d): (ii) is identical to Killara '24 Q12 a) ii)
8. Question 89 (Baulko '23 Q14b): Normo '24 question dupe
9. Question 96 (NSB '22 Q16b v)): Note that this question assumes AM/GM for n variables, with part b being particularly difficult and not the best use of time in the lead up to the HSC.
10. Question 106 (Abbotsleigh '24 Q15c i): By the way, the solutions to this paper aren't on THSC for some reason but you can find them on Studocu.
11. Question 118 (SGGHS '24 Q16a): Ruse '16 Q13a, Ruse 2020 AT3 Q4d
12. Question 119 (Baulko '23 Q16c): Same as Ruse '06 Q8c
13. Question 122 (NSB '24 Q13a): Identical to Ruse '04 8 b) (i), except for mark allocations; (b) is worth 2 less marks here.
14. Question 128 (NSG '13 Q16c): Just a slightly scaffolded version of SGHS's Q16c that same year.
15. Question 134 (Normo '22 Q13a): Caringbah Q16c
16. Question 137 (Normo '21 AT2 Q2b): I did not include the diagram for this question.
17. Question 140 (Copium '23 Q14a): from AMO
18. Question 153 (Baulko '20 Q16a): tikz makes me want to curl up and die, anyways this question was used somewhere else too lol

